Communication Frictions, Managerial Expectations, and Firm Dynamics *

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Haoran Wang †

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Abstract

This paper studies how imperfect communication within firms shapes managerial expectations, input choices, and aggregate productivity. I develop a dynamic model of team production in which capital and labor managers share the firm's objective but observe different information about productivity. The model yields restrictions on forecast-error predictability that jointly identify communication frictions and behavioral distortions in expectations data. Empirically, investment forecasts of U.S. public firms simultaneously underreact to news about hiring and overreact to news about investment. Complementary evidence from U.S. manufacturing establishments shows that managerial forecasts underreact more to news about the other manager's decision when capital and labor decisions are made at different locations. Together, these findings are inconsistent with the common-information benchmark and support a model of firm dynamics with communication frictions. Quantitatively, communication friction generates substantial variations in the capital-labor ratio and large input misallocation. The associated aggregate productivity loss is approximately 5% relative to the frictionless allocation. This productivity loss is moderated by the negative covariance between the marginal revenue products generated by the communication friction.

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†Department of Economics, University of Maryland, College Park. Email: hw2688@umd.edu

1 Introduction

A large empirical literature establishes that management and organizational practices are important drivers of firm productivity (Bloom and Van Reenen, 2007; Bloom et al., 2013, 2019). However, less is known about which aspects of management matter and through what channels they affect firms' decisions and aggregate allocative efficiency. This paper studies a specific channel, linking how firms organize their internal information flows to their input decisions and aggregate productivity.[1] I document two complementary sources of evidence on information sharing within firms. Among U.S. public firms, investment forecasts simultaneously overreact to news about investment but underreact to news about hiring. Among U.S. manufacturing establishments, the investment forecasts of U.S. manufacturing establishments with more segmentation between capital and labor decisions underreact more to hiring. These facts provide evidence against a common-information benchmark and support a model of firm dynamics with communication frictions within firms. I show that the communication friction is quantitatively important: it generates substantial variations in the capital-labor ratio and dispersion in marginal revenue products, leading to a 5% TFP loss relative to the frictionless allocation. The TFP loss is limited relative to the marginal-product dispersion because segmented information induces a negative covariance between the marginal revenue products.

I start the analysis by developing a dynamic team production framework in the spirit of Marschak and Radner (1972), in which each firm is modeled as a team consisting of a capital manager and a labor manager. The managers choose their respective inputs under uncertainty about the firm's productivity but share the objective of maximizing the firm's value. Imperfect communication is modeled by allowing each manager to receive a private signal about the firm's productivity, making the information sets of managers non-nested and segmented within the firm. The managers form expectations and make investment and hiring decisions based on these segmented information sets.

The model yields an empirical test for information segmentation using expectation data. I show that the communication friction leads the rational investment forecast error of the capital manager to be positively predicted by past hiring.[2] The intuition is that when the capital manager makes forecasts under segmented information, he is uncertain about

[1] There is a large literature on imperfect information sharing within an organization (e.g. Mian (2006), Giroud (2013), Impink et al. (2024)). For a review, see Malenko (2023).

[2] The investment forecast error is defined as the difference between actual investment and previously forecast of investment. The rational forecast is the Bayesian expectation conditional on all the information available to the capital manager at the time of the forecsat.

the contemporaneous private signals the labor manager receives and the hiring decision she makes. As a result, his forecasts underreact to current hiring decisions. The actual investment realized next period, on the other hand, will incorporate information revealed by the current period's hiring. The difference between actual and forecasted investment will be positively predicted by past hiring. This "cross-decision" predictability cannot arise under common information, where the labor manager's hiring decision is included in the capital manager's information set.

Because reported forecasts in the data may also reflect behavioral expectations that deviate from rational expectations, cross-decision predictability alone is not sufficient to identify information segmentation. To address this empirical challenge, I decompose the cross-decision predictability into behavioral and rational components analytically. I show that under common information, the cross-decision predictability is solely driven by the behavioral component, which has the same sign as the predictability of investment forecast error by the capital manager's past investment (which I refer to as the "own-decision" predictability). This is because, under common information, if capital and labor are complements in the revenue function, investment and hiring move in the same direction in response to the same news about productivity. As a result, behavioral overreaction or underreaction to news makes the forecast error predictable by past investment and past hiring in the same direction. By contrast, communication frictions can move own- and cross-decision predictability in opposite directions, since the rational component of investment forecast error can be positively predicted by past hiring. Hence, the own- and cross-decision predictability showing opposite signs is inconsistent with the common-information benchmark and consistent with models with communication frictions. I show that the opposite signs cannot be generated by the alternative common-information mechanisms, such as non-convex adjustment costs, time-to-build and non-neutral technologies.

I test the own- and cross-decision predictability using a dataset of U.S. public firms that merges self-reported expectations of capital expenditure from the IBES Guidance dataset with realized outcomes from Compustat. The merged dataset is annual and covers the period between 2003 and 2024. In the data, investment forecasts display exactly this opposite-sign pattern: the investment forecast errors are negatively predicted by past investment but simultaneously positively predicted by past hiring. In other words, investment forecasts simultaneously overreact to investment but underreact to hiring. The evidence therefore supports a model in which capital and labor decisions are based on segmented information.

The establishment-level data from U.S. Census provide further organizational evidence for

the communication frictions. The 2015 wave of Management and Organizational Practices Survey (MOPS) contains not only investment and hiring forecasts but also the location of capital and labor decisions for the manufacturing establishments showing up in Annual Survey of Manufacturers (ASM). I use whether the capital and labor decisions are made in the same location as a proxy for communication friction within the firm between capital and labor decisions. I find that cross-decision predictability of both investment and hiring forecasts are more positive for the establishments whose capital and labor decisions are made in different locations, while own-decision predictability does not vary with this organizational arrangement. This is consistent with the predictions in my model that communication frictions make a manager's forecasts to underreact more (or overreact less) to the other manager's decisions.

I next ask whether communication friction is quantitatively relevant for firm dynamics and allocative efficiency. To do so, I extend the model to incorporate time-to-build in capital, over-extrapolative beliefs, and Hsieh–Klenow-style reduced-form distortions. The model is calibrated to match the own- and cross-decision predictabilities and standard moments of input adjustment: the responsiveness of input choices to productivity, and marginal-product dispersion from the IBES/Compustat sample. I then use the calibrated model to investigate how much capital–labor-ratio and marginal-product dispersion communication friction generates at the estimated parameters, and how the resulting distortions affect aggregate TFP. I show that communication frictions are quantitatively important: when other frictions are muted, communications frictions alone generate about 40 percent of observed dispersion in capital-labor ratio, 28 percent of observed dispersion in marginal revenue product of capital (MRPK), and 20 percent of observed dispersion in marginal revenue product of labor (MRPN). The associated aggregate TFP loss of communication friction is about 5 percent relative to the frictionless allocation.

The quantitative analysis highlights a distinction between communication friction and common information or behavioral frictions. When managers share the same noisy information and behavioral expectations, their input choices respond to a common productivity belief. This common belief changes the scale of inputs without altering the capital–labor ratio, and generates a large correlation between investment and hiring. This makes it akin to a revenue distortion in Hsieh and Klenow (2009) which affects all inputs symmetrically. In contrast, communication frictions can introduce a wedge between the beliefs underlying capital and labor decisions. In the calibrated model, managers place substantially more weight on their private signals than on the common signal. Their productivity beliefs therefore need not move together, weakening the comovement of investment and hiring. This wedge between

capital and labor managers' beliefs generates large variations in the capital–labor ratio. In this sense, communication frictions are akin to input-specific distortions that change the capital-labor ratio of the firm.

An important implication of this property is that communication frictions generate a negative covariance between MRPK and MRPN. It arises from managers' reliance on private information under communication friction. Holding actual productivity and the labor manager's information fixed, a favorable realization of the capital manager's private-signal noise raises capital investment. The additional investment lowers the MRPK due to diminishing returns but raises MRPN because capital and labor are complements, making the marginal revenue product of inputs move in opposite directions. This negative correlation between marginal revenue products significantly moderates the aggregate productivity gain from removing communication frictions. In the calibrated model, communication frictions generate substantially larger MRPK dispersion than convex adjustment costs and about the same level of MRPN dispersion, yet its associated TFP loss is smaller. This is exactly because communication friction generates a negative MRPK–MRPN correlation, whereas convex adjustment costs and common information generate positive correlations. This mechanism suggests that advances in ICT and AI that improve information sharing within firms may yield smaller gains in allocative efficiency than reductions in marginal-product dispersion alone would suggest, because these reductions can be accompanied by higher MRPK–MRPN covariance.

Related Literature This paper is related to multiple strands of literature. First, it contributes to the literature on management and organizational economics. This includes empirical works on the relationship between management and firm productivity (Bloom and Van Reenen, 2007; Bloom et al., 2013, 2019; Guner et al., 2018; Sadun et al., 2025); models with imperfect communication within organizations (Marschak and Radner, 1972; Bolton and Dewatripont, 1994; Garicano, 2000; Dessein, 2002; Dessein and Santos, 2006; Alonso et al., 2008; Calvó-Armengol et al., 2015); and empirical evidence of internal communication frictions from geographical variations (e.g. Mian (2006), Giroud (2013),Giroud and Mueller (2015)) and confidential firm information such as email transcripts and hierarchical details (e.g. Dessein et al. (2022), Impink et al. (2024)). I contribute to this literature by linking management and internal information flows to firms' expectations, which has two advantages. First, firm expectations are increasingly available for a larger number of firms (see, for example, the survey by Bachmann et al. (2023)), making it possible to quantify the aggregate implications of the friction. Additionally, my empirical test retains power for teams whose managers are in the same location and does not rely on the geographic information.

Second, the paper contributes to the burgeoning literature of behavioral macroeconomics. Theoretically, my paper is related to the models of imperfect common knowledge and coordination friction in macroeconomics (e.g. Cooper and John (1988); Woodford (2001); Maćkowiak and Wiederholt (2009); Angeletos and Lian (2016, 2018); Angeletos et al. (2021)). These papers focus on the coordination frictions across firms, while my paper argues that it is also important to model coordination frictions within the firms.[3] Empirically, my strategy of identifying communication frictions builds on the literature of testing Full-Information-Rational-Expectation (FIRE) hypothesis using forecast-error predictability (Coibion and Gorodnichenko, 2012, 2015; Bordalo et al., 2020; Kohlhas and Walther, 2021). I show that the method can be applied to study within-firm frictions, and I document a new fact about managerial expectation: investment forecasts simultaneously overreact to investment and underreact to hiring. Existing empirical works on managerial expectation, such as Gennaioli et al. (2016), Ma et al. (2020) and Barrero (2022), focus on the behavioral and informational frictions in one particular decision, whereas my paper speaks to the frictions across different decisions within the firm.

Lastly, the paper contributes to the literature on the implications of information and behavioral frictions for allocative efficiency. David et al. (2016) and David and Venkateswaran (2019) quantify the contribution of information frictions to misallocation under two internal information structures: either a firm-level noisy signal that is commonly known to the managers, or labor chosen *ex post* so that only capital is subject to the friction. My paper nests both as limiting cases and uses expectation moments to discipline the information parameters. Moreover, I show that communication friction presents distinct properties from common, noisy information and common behavioral distortions, in that it generates dispersions in capital-labor ratio and a negative correlation between marginal revenue products of inputs, which plays an important role in the quantification of TFP losses. Quantitatively, my estimates of misallocation and TFP loss from information frictions are in line with the structural estimates of Ma et al. (2020) and Barrero (2022) and the reduced-form estimates in Gautier et al. (2025) and Ropele et al. (2024).

Layout The remainder of the paper is organized as follows. Section 2 presents the theory. Section 3 describes the empirical test for within-firm communication friction and applies it to U.S. public firms and Census establishments in Section 4. Section 5 quantifies the aggregate implications of communication frictions, discussing parameter identification and calibration

[3] An exception is Lian (2020), which studies the coordination friction between different selves in behavioral economics.

results. Section 6 concludes.

2 Model

In this section, I describe the model of dynamic input choices under communication friction. Throughout the paper, there is no aggregate uncertainty, so that all the aggregate variables are fixed at their steady-state values in equilibrium and are common knowledge to all the agents in the economy.

Household There is a representative household in the economy, who has standard preference on final-good consumption, discounts future consumption utility by a discount factor β and supplies a fixed amount of labor $\bar{N}$ to the firms at a wage rate W_t, and saves in a risk-free bond at a real interest rate r in each period. In a stationary equilibrium, consumption is constant over time, which implies $\beta(1+r)=1$.

Final-Good Producer There is a final good producer who combines intermediate goods $\{Y_{it}\}$ into a final good Y_t by a CES technology

$$Y_t = \left(\int_0^1 (A_{it} Y_{it})^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}} \tag{1}$$

where A_{it} can be interpreted as an idiosyncratic productivity shock on intermediate-good producer i or a demand shock on the product i, and $\epsilon > 1$ is the parameter for elasticity of substitution.

The price of final good is normalized to one. Given price $\{P_{it}\}$, the final good producers choose how much intermediate goods to purchase from intermediate-good producers. This yields a standard demand curve

$$P_{it} = \left(\frac{Y_t}{Y_{it}} \right)^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} \tag{2}$$

Intermediate-Good Producers There is a continuum $i \in [0,1]$ of monopolistically competitive intermediate-good producers in the economy. Each producer i has a constant-return-

to-scale production function to combine capital and labor inputs into intermediate goods:

$$Y_{it} = K_{i,t-1}^{\alpha} N_{it}^{1-\alpha} \tag{3}$$

where $\alpha \in (0,1)$ is the parameter for capital share, and the capital is predetermined to reflect time-to-build in capital. The producer's profit Π_{it} is given by

$$\Pi_{it} = Y_t^{\frac{1}{\epsilon}} A_{it}^{1-\frac{1}{\epsilon}} K_{i,t-1}^{\alpha\left(1-\frac{1}{\epsilon}\right)} N_{it}^{(1-\alpha)\left(1-\frac{1}{\epsilon}\right)} - R_t K_{it} - W_t N_{it} - \Phi^k(K_{i,t-1}, K_{it}) - \Phi^n(N_{i,t-1}, N_{it}) \tag{4}$$

where R_t and W_t denote rental price and wage rate. Φ^k and Φ^n are convex adjustments on capital and labor respectively:

$$\Phi^k(K_{i,t-1}, K_{it}) \equiv \frac{\xi_k}{2}\left(\frac{I_{it}}{K_{i,t-1}} - \delta\right)^2 K_{i,t-1}, \qquad \Phi^n(N_{i,t-1}, N_{it}) \equiv \frac{\xi_n}{2}\left(\frac{H_{it}}{N_{i,t-1}}\right)^2 N_{i,t-1}$$

where $I_{it} \equiv K_{it} - (1-\delta)K_{i,t-1}$ is investment and $H_{it} \equiv N_{it} - N_{i,t-1}$ is hiring, and δ is the depreciation rate of capital.

Information The only source of uncertainty in the model is the idiosyncratic firm fundamentals A_{it}. I assume that the log of A_{it}, denoted as a_{it}, follows an AR(1) process

$$a_{it} = \rho a_{i,t-1} + \mu_{it}, \text{ with } \mu_{it} \overset{iid}{\sim} N(0, \sigma_\mu^2). \tag{5}$$

The firm managers know the data generating process in (5). However, the true value of a_{it} is never fully revealed to the firm and its managers. Hence, when making decisions, the capital and labor manager of firm i need to learn dynamically from the Gaussian noisy signals they receive to form expectations about the "latent variable" a_{it} over time using Kalman filters, which I will specify later in detail.

Communication friction is formalized by allowing each manager's decision to be dependent on different information about productivity. Specifically, I assume that in period t, each manager $m \in \{k, n\}$ is endowed with a private, noisy signal s_{it}^m about a_{it}:

$$s_{it}^m = a_{it} + \epsilon_{it}^m, \text{ with } \epsilon_{it}^m \overset{iid}{\sim} N(0, \sigma_{\epsilon,m}^2) \text{ for } m \in \{k, n\}. \tag{6}$$

Besides, both managers receive a common, noisy signal s_{it}^p:

$$s_{it}^p = a_{it} + \epsilon_{it}^p, \text{ with } \epsilon_{it}^p \overset{iid}{\sim} N(0, \sigma_{\epsilon,p}^2) \tag{7}$$

whose realization is commonly known by the managers. The signal noise $\epsilon_{it}^k, \epsilon_{it}^n, \epsilon_{it}^p$ and the innovation in the firm fundamental μ_{it} are independent to each other, and their standard deviation $\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}$ are commonly known by the managers.

Since the realization of signal s_{it}^m is private to manager m and both managers choose their decisions simultaneously, the action that manager m takes in period t is not known by the other manager m', and vice versa. At the beginning of period $t+1$, the actions taken by the managers in period t, K_{it} and N_{it}, are revealed to both managers and become common knowledge within the firm. This assumption allows the managers to use the other manager's action in the previous period as an additional signal to update their beliefs about the unknown fundamental $a_{i,t+1}$.[4] Formally, denote the managers' information sets as $\mathcal{I}_{it}^k$ and $\mathcal{I}_{it}^n$ respectively. Their law of motion can be expressed as

$$\mathcal{I}_{it}^k = \mathcal{I}_{i,t-1}^k \cup \{s_{it}^p, s_{it}^k, N_{i,t-1}\} \text{ and } \mathcal{I}_{it}^n = \mathcal{I}_{i,t-1}^n \cup \{s_{it}^p, s_{it}^n, K_{i,t-1}\} \tag{8}$$

Equilibrium Under communication friction, the firm faces a dynamic team production problem with incomplete information, as in Marschak and Radner (1972). The firm's input decisions are characterized as a Markov Perfect Bayesian Equilibrium (MPBE), in which

- Given labor manager's policy function $N_{it}(\cdot)$, capital manager's policy function $K_{it} : \mathcal{I}_{it}^k \to \mathbb{R}^+$ solves Bellman equation

$$V^k(K_{i,t-1}, N_{i,t-1}; \mathcal{I}_{it}^k) = \max_K \mathbb{E}\left[\Pi(K, N_{it}; K_{i,t-1}, N_{i,t-1}, A_{it}) + \beta V^k(K, N_{it}; \mathcal{I}_{i,t+1}^k) \middle| \mathcal{I}_{it}^k\right] \tag{9}$$

- Given capital manager's policy function $K_{it}(\cdot)$, labor manager's policy function $N_{it} : \mathcal{I}_{it}^n \to \mathbb{R}^+$ solves Bellman equation

$$V^n(K_{i,t-1}, N_{i,t-1}; \mathcal{I}_{it}^n) = \max_N E_t\left[\Pi(K_{it}, N; K_{i,t-1}, N_{i,t-1}, A_{it}) + \beta V^n(K_{it}, N; \mathcal{I}_{i,t+1}^n) \middle| \mathcal{I}_{it}^n\right] \tag{10}$$

- In period t, each manager updates their belief about a_{it} using their observed signals and the past actions $K_{i,t-1}, N_{i,t-1}$, consistent with the policy function $K_{it}(\cdot), N_{it}(\cdot)$ and the Bayes's rule.

[4] In general, revelation of N_{it} does not fully reveal the true productivity a_{it} to the capital manager (and vice versa), unless the labor decision is not subject to information friction (as in David and Venkateswaran (2019)). However, it does reveal the labor manager's private signal s_{it}^n to the capital manager.

The advantage of using MPBE in (9) and (10) to characterize firm's behavior lies in its flexibility of accommodating essentially any possible internal information structure of the firm. Three classes of extensively studied models in the firm dynamics literature are nested by my model. First, the full-information rational expectation (FIRE) benchmark can be regarded as a special case of (9) and (10), with $\mathcal{I}_{it}^k = \mathcal{I}_{it}^n = \{a_{it}, a_{i,t-1}, \cdots, a_{i0}\}$, i.e. the true firm fundamental is always revealed to both managers. Using our parameterization (6) and (7) above, this full-information benchmark maps to the limit $\sigma_{\epsilon,p} \to 0$ and $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$. Second, the common, noisy information model (as in David, Hopenhayn, and Venkateswaran (2016)) is nested by assuming $\mathcal{I}_{it}^k = \mathcal{I}_{it}^n = \{s_{it}^p, \cdots, s_{i0}^p\}$. Using our parameterization, it maps to the limit $\sigma_{\epsilon,p}$ being finite and $\sigma_{\epsilon,m} \to \infty$ for $m \in \{k, n\}$. In that case, the capital and labor managers do not pay attention to their private signals, and their investment and hiring decisions are driven by the same noisy signal. Lastly, my model can also accommodate the cases in which only capital decisions are subject to information frictions (e.g. David and Venkateswaran (2019)), by assuming the labor managers' private signals to be perfectly precise.

Equation (9) and (10) illustrate that the managers face two types of uncertainty when they are making their own decisions. For the capital manager, apart from the fundamental uncertainty about a_{it} or A_{it}, he faces a strategic uncertainty about the action of the labor manager N_{it}, which is measurable to labor manager's information set only. The source of this strategic uncertainty is the fact that information is not perfectly shared within the firm, so that there is labor-specific information that is only known to the labor manager and not to the capital manager. This strategic uncertainty affects capital manager's decision because labor manager's action affects both the marginal revenue product of capital and the continuation value of the capital manager, and therefore capital manager has an incentive to form expectation about N_{it} to guide his own decision about K_{it}.

Log-Quadratic Approximation An MPBE requires that the belief dynamics of both managers are consistent with the policy function and Bayes's rule. It is a difficult task to analytically characterize or compute the evolution of beliefs since it is an infinite-dimensional object. Here, I get around this technical challenge by doing a log-quadratic approximation on the profit function with respect to $(K_{it}, N_{it}, K_{i,t-1}, N_{i,t-1}, A_{it})$ around their frictionless steady-state values, as in Woodford (2003).[5] I can therefore express the log-quadratic approximated profit function π_{it} as a quadratic function in terms of the log deviations of capital,

[5]By frictionless, I mean the case where there is no information friction and adjustment friction, i.e. a_{it} is fully revealed to the firm, and $\xi_k = \xi_n = 0$.

labor and firm fundamental from their own frictionless steady-state value. Denote these log-deviation values as k_{it}, n_{it} and a_{it} respectively, and define investment (rate) as $\iota_{it} \equiv k_{it} - k_{i,t-1}$ and hiring (rate) as $h_{it} \equiv n_{it} - n_{i,t-1}$. The approximated profit function can be written as

$$\pi(\iota_{it}, h_{it}; x_{it}) = x_{it}' P x_{it} + x_{it}' Q \iota_{it} + x_{it}' R h_{it} + H_\iota \iota_{it}^2 + H_{\iota h} \iota_{it} h_{it} + H_h h_{it}^2 \tag{11}$$

where $x_{it} = [1, k_{i,t-1}, n_{i,t-1}, a_{it}]'$ is the state vector of the firm with law of motion

$$x_{i,t+1} = A x_{it} + B \iota_{it} + C h_{it} + D \mu_{i,t+1} \tag{12}$$

and matrices $A, B, C, D, P, Q, R, H_\iota, H_h$ and $H_{\iota h}$ are constants imputed from the frictionless steady-state values and model parameters. The derivations of (11) are left to Appendix A.1.

The log-quadratic approximation in the profit function Π makes the game defined in (9) and (10) a Linear-Quadratic-Gaussian (LQG) game with incomplete information. As I will show shortly, the value functions (9) and (10) in the MPBE are in quadratic forms, and the policy functions that solve (9) and (10) are linear. This simplifies the computation significantly. Moreover, the Gaussian-distributed productivity and the linearity of policy functions imply that the investment and hiring also follow Gaussian distributions, making it easy for the managers to integrate other manager's past decision in the Kalman filtering process once it is revealed at the beginning of each period.

Characterization of Linear MPBE I first characterize the belief dynamics of the MPBE. Under linear policy functions, investment and hiring are jointly Gaussian with firm fundamentals and signals, and the managers can simply treat the other manager's lagged action as a noisy signal of the firm fundamentals and learn from it in the same manner as from other signals. Since the latent productivity a_{it} is a scalar, the characterization of belief dynamics becomes even simpler. At the beginning of period when the decisions of both managers in the previous period are revealed, each manager's private signal in the previous period becomes common knowledge. The managers can therefore use them to make an additional update on their beliefs about the productivity before they receive the current-period signals. The steady-state beliefs resulting from this Kalman filtering process are characterized by the following lemma.

Lemma 1. *Let $\bar{\Sigma}$ be the fixed point of*

$$\bar{\Sigma} = \left[\left(\rho^2 \bar{\Sigma} + \sigma_\mu^2 \right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2} + \sigma_{\epsilon,n}^{-2} \right]^{-1}$$

and denote capital manager's posterior beliefs as $a_{it}|\mathcal{I}_{it}^k \sim N(\hat{a}_{it}^k, \hat{\Sigma}^k)$ *and labor manager's posterior beliefs as* $a_{it}|\mathcal{I}_{it}^n \sim N(\hat{a}_{it}^n, \hat{\Sigma}^n)$*. At the steady state, the posterior variances of capital and labor managers are given by*

$$\hat{\Sigma}^k \equiv \left((\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2}\right)^{-1}, \hat{\Sigma}^n \equiv \left((\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,n}^{-2}\right)^{-1}$$

and the posterior mean is given by

$$\hat{a}_{it}^k \equiv \mathbb{E}[a_{it}|\mathcal{I}_{it}^k] = \hat{\Sigma}^k \left((\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}\bar{a}_{it} + \sigma_{\epsilon,p}^{-2}s_{it}^p + \sigma_{\epsilon,k}^{-2}s_{it}^k\right), \tag{13}$$

$$\hat{a}_{it}^n \equiv \mathbb{E}[a_{it}|\mathcal{I}_{it}^n] = \hat{\Sigma}^n \left((\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}\bar{a}_{it} + \sigma_{\epsilon,p}^{-2}s_{it}^p + \sigma_{\epsilon,n}^{-2}s_{it}^n\right), \tag{14}$$

where $\bar{a}_{it}$*, the common prior of the period after the past decisions of the managers are revealed, evolves according to*

$$\bar{a}_{it} = \rho\bar{\Sigma}\left((\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}\bar{a}_{i,t-1} + \sigma_{\epsilon,p}^{-2}s_{i,t-1}^p + \sigma_{\epsilon,k}^{-2}s_{i,t-1}^k + \sigma_{\epsilon,n}^{-2}s_{i,t-1}^n\right). \tag{15}$$

Proof. See Appendix A.2. □

Now I characterize the linear MPBE associated with (9) and (10). Let $\zeta_{it} \equiv (\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}\bar{a}_{it} + \sigma_{\epsilon,p}^{-2}s_{it}^p$ denote the common-information component of the posterior mean, and let the state vector of capital and labor manager be $y_{it}^k \equiv [1, k_{i,t-1}, n_{i,t-1}, \zeta_{it}, s_{it}^k]'$ and $y_{it}^n \equiv [1, k_{i,t-1}, n_{i,t-1}, \zeta_{it}, s_{it}^n]'$ respectively. Under log-quadratic approximation (11) and (12), the value functions (9) and (10) are quadratic in y_{it}^k and y_{it}^n with corresponding policy functions linear in y_{it}^k and y_{it}^n. The following lemma characterizes the MPBE.

Lemma 2. *The solution to the MPBE Bellman equations (9) and (10) is characterized by value functions*

$$V^k(y_{it}^k) = (y_{it}^k)'P_k y_{it}^k + Q_k, V^n(y_{it}^n) = (y_{it}^n)'P_n y_{it}^n + Q_n$$

with corresponding policy functions

$$\iota_{it} = F_\iota y_{it}^k, h_{it} = F_h y_{it}^n \tag{16}$$

where $P_k, P_n \in \mathbb{R}^{5\times5}$ *and* $F_\iota, F_h \in \mathbb{R}^{1\times5}$ *jointly solve the system of Riccati equations*

$$F_\iota = -\frac{1}{2}(H_\iota + \beta\mathsf{B}'P_k\mathsf{B})^{-1}(Q'\mathsf{M}_k + H_{\iota h}F_h\mathsf{T}_n + \beta\mathsf{B}'(P_k + P_k')\mathsf{A}_k) \tag{17}$$

$$F_h = -\frac{1}{2}(H_h + \beta\mathsf{C}'P_n\mathsf{C})^{-1}(R'\mathsf{M}_n + H_{\iota h}F_\iota\mathsf{T}_k + \beta\mathsf{C}'(P_n + P_n')\mathsf{A}_n) \tag{18}$$

$$P_k = \mathsf{M}_k' P \mathsf{M}_k + \mathsf{M}_k' R F_h \mathsf{T}_n + \mathsf{T}_n' F_h' H_h F_h \mathsf{T}_n + \beta \mathsf{A}_k' P_k \mathsf{A}_k + \frac{1}{2} F_\iota' (Q' \mathsf{M}_k + H_{\iota h} F_h \mathsf{T}_n + \beta B' (P_k + P_k') \mathsf{A}_k) \tag{19}$$

$$P_n = \mathsf{M}_n' P \mathsf{M}_n + \mathsf{M}_n' Q F_\iota \mathsf{T}_k + \mathsf{T}_k' F_\iota' H_\iota F_\iota \mathsf{T}_k + \beta \mathsf{A}_n' P_n \mathsf{A}_n + \frac{1}{2} F_h' (R' \mathsf{M}_n + H_{\iota h} F_\iota \mathsf{T}_k + \beta \mathsf{C}' (P_n + P_n') \mathsf{A}_n) \tag{20}$$

where the matrices $\mathsf{M}_k, \mathsf{M}_n, \mathsf{T}_k, \mathsf{T}_n, \mathsf{A}_k, \mathsf{A}_n$ *are matrices defined in Appendix A.3, and* $H_h, H_\iota, H_{\iota h}, P, Q, R$ *are the coefficient matrices in the log-quadratic approximation (11).*

Proof. See Appendix A.3. □

Lemma 2 makes it convenient to solve the MPBE conveniently by iterating the system of Riccati equations (19) and (20) together with policy matrices (17) and (18). When the communication friction is muted, the entries of investment and hiring policies on the private signals, F_ι^s and F_h^s, are both equal to 0, and the MPBE degenerates to a Markov Perfect Equilibrium with common noisy signal s_{it}^p. The belief dynamics in Lemma 1 and the policy functions in Lemma 2 characterize a law of motion for the joint distribution of actions, beliefs and productivity. A stationary equilibrium requires this distribution to be at its stationary distribution implied by the law of motion. The details of the computation of this stationary distribution are left to Appendix A.4.

3 Identifying Communication Frictions

The key object that maps theory to data is the managerial expectation of investment or hiring. In this section, I study the implications of communication frictions for managerial expectations and show how to identify them from the expectation data.

3.1 Implications of Communication Friction for Expectations

The belief dynamics and decision rules characterized by Lemma 1 and 2 imply that the capital manager's forecast error, i.e. the difference between the actual investment and the one-period-ahead forecast of investment made by the capital manager, takes the following form:

Lemma 3. *In a MPBE defined in Lemma 2, the investment policy function can be rewritten*

as

$$\iota_{it} = F_0 + F_1 k_{i,t-1} + F_2 n_{i,t-1} + F_3 \bar{a}_{it} + F_4 u^k_{it} + F_5 u^p_{it} \tag{21}$$

where $\bar{a}_{it}$ is the prior of the capital manager in period t after the labor manager's past decision is revealed, and $u^k_{it} \equiv s^k_{it} - \bar{a}_{it}$, $u^p_{it} \equiv s^p_{it} - \bar{a}_{it}$ captures the capital manager's belief updates from prior $\bar{a}_{it}$, using signals s^k_{it} and s^p_{it}. As a result, the rational forecast error of firm i's investment takes the form

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = (F_2 + JF_3)(h_{i,t-1} - \mathbb{E}^k_{i,t-1}[h_{i,t-1}]) + F_4 u^k_{it} + F_5 u^p_{it} \tag{22}$$

where $J > 0$ is the capital manager's Kalman gain on the past hiring of the labor manager at the beginning of period t.

Proof. See Appendix A.5. □

Lemma 3 shows that the rational investment forecast error has three components. The first term speaks directly to the communication friction. It is the gap between actual hiring in period $t-1$ and the capital manager's expectation in period $t-1$ about $h_{i,t-1}$. Absent communication friction, this term will always equal zero. This is because the capital manager knows that the labor manager's decision is based on the same information as his, and therefore he knows the labor manager's decision in $t-1$ when he is forming expectations in $t-1$. On the other hand, if communication is imperfect within the organization, the capital manager does not observe the exact $h_{i,t-1}$ in period $t-1$ because the labor manager's contemporaneous private signal is not observed by the capital manager. Additionally, at the beginning of period t, the revelation of past hiring $h_{i,t-1}$ provides a new signal for the capital manager to update his belief about productivity a_{it}, making the prior $\bar{a}_{it}$ different from the posterior $\mathbb{E}^k_{i,t-1}[\bar{a}_{it}]$ at the end of period $t-1$. As a result, his belief about the labor manager's action in that period generally differs from the labor manager's actual decision. The second and third term of (22) captures the belief updates of the capital manager from new signals in period t, which are orthogonal to variables in $t-1$.

A Test of Common Information The structure of capital manager's investment forecast error in Lemma 3 leads us to consider the regression

$$\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}] = \beta_0 + \beta_h h_{i,t-1} + \nu_{it} \tag{23}$$

The coefficient of interest is β_h, which captures whether the capital manager's investment forecast error can be predicted by past hiring. The following corollary characterizes the sign of β_h implied by the model:

Proposition 1. *The predictability of capital manager's investment forecast error by past hiring β_h can be used as a test of common/segmented information within the firm:*

- *If there is no communication friction, then $\beta_h = 0$;*
- *Under communication friction, $\beta_h > 0$.*

Proof. See Appendix A.6. □

The first part of Proposition 1 suggests that regression (23) is a test of common information under rational expectation: any $\beta_h \neq 0$ indicates that hiring in $t-1$ is not incorporated in the capital manager's information set in $t-1$. This directly follows from the decomposition in Lemma 3: under full-information rational expectation (FIRE) or common, noisy information, there is no information asymmetry between capital and labor managers, so that hiring is fully incorporated in the capital manager's information set. As a result, the first term of the right hand side of (22) disappears, leaving behind only the terms orthogonal to past hiring. Hence, absent communication frictions, the capital manager's investment forecast error cannot be predicted by past hiring.

The second part of the corollary considers the direction of investment forecast error predictability by past hiring under communication frictions. Because of communication frictions, the capital manager's expectation in $t-1$ responds sluggishly to the labor manager's decisions because he cannot tell whether the labor manager's contemporaneous private signal conveys the same message as the signals he receives. At the beginning of period t when the hiring in $t-1$ is revealed to the capital manager, if the labor manager's hiring decision responds positively to her private signal, the capital manager will interpret a high realization of past hiring as a positive signal on the productivity and hence revise his belief upwards in period t. The investment in period t will therefore be higher. As a result, high realizations in past hiring is associated with positive investment forecast errors of the capital manager, making the sign of β_h positive.

The conclusions in Lemma 3 and Proposition 1 are not restricted to the investment forecast error of the capital manager. More generally, the predictability of forecast errors of one

manager's decision by the *other* manager's past decisions, which I refer to as *cross-decision predictability*, tests whether the information is perfectly shared within an organization. For instance, the test (23) can be mirrored to the hiring expectations of the hiring manager if they are available in the data. Specifically, let $\mathbb{E}^n_{it}[\cdot] \equiv \mathbb{E}[\cdot|\mathcal{I}^n_{it}]$ denote the expectation of the hiring manager. A similar regression

$$h_{it} - \mathbb{E}^n_{i,t-1}[h_{it}] = \alpha_0 + \alpha_\iota \iota_{i,t-1} + \mu_{it} \tag{24}$$

tests whether the hiring manager fully incorporates the capital manager's investment decisions in her information set. Note that it is not necessary to have both investment forecasts and hiring forecasts available in the data to run the test: rejecting the null hypothesis of common information requires nonzero predictability in either (23) or (24).

3.2 Distinguishing Communication and Behavioral Frictions

Proposition 1 establishes that communication friction makes the investment forecast error predictable by past hiring under rational expectation. An empirical challenge is that self-reported or survey expectations in the data may deviate from rational expectation, which could generate the same cross-decision predictability in Proposition 1. To see this, let $\mathbb{F}^k_{i,t-1}[\cdot]$ denote the capital manager's one-period-ahead forecast in the data. The observed investment forecast error, defined as the difference between actual investment ι_{it} and investment forecast $\mathbb{F}^k_{i,t-1}[\iota_{it}]$, can be written as

$$\iota_{it} - \mathbb{F}^k_{i,t-1}[\iota_{it}] = \underbrace{\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}]}_{\text{Rational Forecast Error}} + \underbrace{\mathbb{E}^k_{i,t-1}[\iota_{it}] - \mathbb{F}^k_{i,t-1}[\iota_{it}]}_{\text{Behavioral Distortion}}. \tag{25}$$

We have established in Lemma 3 and Proposition 1 that the rational component of investment forecast error, $\iota_{it} - \mathbb{E}^k_{i,t-1}[\iota_{it}]$, cannot be predicted by past hiring under common information. However, if the investment forecast $\mathbb{F}^k_{i,t-1}$ deviates from rational expectation $\mathbb{E}^k_{i,t-1}$, the wedge between them can be correlated with hiring $h_{i,t-1}$. Hence, merely running the regression

$$\iota_{it} - \mathbb{F}^k_{i,t-1}[\iota_{it}] = \tilde{\alpha}_h + \tilde{\beta}_h h_{i,t-1} + \nu_{it} \tag{26}$$

and getting a nonzero coefficient $\tilde{\beta}_h$ cannot distinguish communication frictions and behavioral frictions.

Joint Sign Pattern in the Model To investigate whether the cross-decision predictability is driven by communication friction or behavioral distortion, I consider an alternative regression

$$\iota_{it} - \mathbb{F}^k_{i,t-1}[\iota_{it}] = \tilde{\alpha}_\iota + \tilde{\beta}_\iota \iota_{i,t-1} + \nu_{it} \tag{27}$$

which regresses the capital manager's investment forecast error on his own past decision. The coefficient of interest $\tilde{\beta}_\iota$ is the predictability of the capital manager's investment forecast error by his own past decisions. Since the investment $\iota_{i,t-1}$ is based upon the capital manager's information set in $t-1$, the capital manager should know perfectly his decision $\iota_{i,t-1}$ when he is making forecast in $t-1$, and therefore the rational component of investment forecast error in (25) is orthogonal to $\iota_{i,t-1}$ by definition. Hence, the own-decision predictability can only originate from deviations from rational expectation.

In comparison, the cross-decision predictability $\tilde{\beta}_h$ contains information about both behavioral distortion and how the managers communicate within the firm, through the rational component of the forecast error in (25). Following Proposition 1, this rational component can be positively predicted by past hiring if there is communication friction. Therefore, taking into account the signs of own-decision predictability $\tilde{\beta}_\iota$ and the cross-decision predictability $\tilde{\beta}_h$ can help us separately identify the communication friction and behavioral deviations from rational expectations. Specifically, the sign pattern of $\tilde{\beta}_\iota < 0$ and $\tilde{\beta}_h > 0$ is inconsistent with common information and can only be reproduced with communication friction and behavioral overreaction.

To see this point, let's consider two classes of behavioral distortions. The first is extrapolative beliefs, which is captured by allowing the managers' perceived persistence of productivity $\hat{\rho}$ in Lemma 1 to be different from the actual persistence ρ. The case $\hat{\rho} > \rho$ means that the managers *overextrapolates* today's productivity shocks into the future, leading the managerial beliefs to *overreact* to today's shocks. It is supported by the empirical evidence documented in the behavioral macroeconomics literature, in the context of stock market participants (Greenwood and Shleifer, 2014; Gennaioli et al., 2016), professional forecasters (Angeletos et al., 2021) and firm managers (Barrero, 2022). The case $\hat{\rho} < \rho$ represents *underextrapolation* (Ma et al., 2020) and conceptually similar distortions such as cognitive discounting (Gabaix, 2020) and level-k thinking (Farhi and Werning, 2019). The following proposition characterizes the theoretical own- and cross-decision predictability under belief extrapolation:

Proposition 2. *Suppose both managers hold a subjective persistence $\hat{\rho} \in (0,1)$ that can be different from the actual persistence ρ. The theoretical cross-decision predictability $\tilde{\beta}_h$ in (26)*

and own-predictability $\tilde{\beta}_\iota$ in (27) are given by

$$\tilde{\beta}_\iota = \kappa_\iota(\rho - \hat{\rho}) \tag{28}$$

where $\kappa_\iota > 0$ is a function of deep parameters, and

$$\tilde{\beta}_h = \beta_h + \kappa_h(\rho - \hat{\rho}) \tag{29}$$

where β_h is the predictability of the rational component in (25) by past hiring. It is positive under communication friction and zero under common information. The coefficient κ_h is a function of model parameters and is positive under common information.

Proof. See Appendix A.7. □

Proposition 2 shows that own- and cross-decision predictability convey different messages about expectations. Equation (28) implies that own-decision predictability has the sign of $\rho - \hat{\rho}$ under both common and segmented information. In contrast, the sign of cross-decision decomposition (29) depends not only on the sign of $\rho - \hat{\rho}$ from belief overextrapolation, but also on the predictability of the rational component, captured by β_h. Without communication friction, the rational component of investment forecast error can no longer be predicted by past hiring, so $\beta_h = 0$. As a result, $\tilde{\beta}_\iota$ and $\tilde{\beta}_h$ are both proportional to $\rho - \hat{\rho}$ and therefore have the same sign. In other words, without communication friction, it is impossible for capital manager's beliefs to simultaneously overreact to one decision while underreact to another. The following corollary is immediate:

Corollary 1. *Having a sign pattern of $\tilde{\beta}_\iota < 0$ and $\tilde{\beta}_h > 0$ is inconsistent with common information but consistent with communication friction under belief overextrapolation.*

Following our previous discussion, a negative $\tilde{\beta}_\iota$ indicates that the capital manager's expectation is subject to overextrapolation, so that his expectations overreact to news about productivity. Without communication friction, the capital manager knows that he is overreacting to the same news that drives both investment and hiring. Hence, investment expectations should also overreact to hiring, making $\tilde{\beta}_h < 0$. Given $\tilde{\beta}_\iota < 0$, a positive $\tilde{\beta}_h$ is possible only when the belief overreaction from overextrapolation is overturned by communication friction, which makes the capital manager's investment expectation underreact to hiring.

Apart from extrapolative beliefs, another class of behavioral distortions I consider is non-Bayesian learning. For simplicity, I assume that the capital manager's belief $\mathbb{F}^k_{it}$ is distorted

by

$$\mathbb{F}_{it}^{k}[a_{it}] = \hat{a}_{it}^{k} + \theta(\hat{a}_{it}^{k} - \rho\hat{a}_{i,t-1}^{k}) \tag{30}$$

where the parameter θ captures the deviation of expectation from Bayesian learning. $\theta = 0$ means the expectation $\mathbb{F}_{it}^{k}$ is at the rational benchmark. $\theta < 0$ means that the expectation $\mathbb{F}_{it}^{k}$ underreacts to the new signals received in period t compared to the Bayesian benchmark. The case $\theta > 0$ captures overreaction and are referred to as "diagnostic expectation" in the literature (Bordalo et al., 2020; Bianchi et al., 2024). The following proposition characterizes the theoretical own- and cross-decision predictability under non-Bayesian expectations.

Proposition 3. *Suppose the capital and labor manager's Kalman gains are distorted as in (30). The theoretical cross-decision predictability $\tilde{\beta}_h$ in (26) and own-predictability $\tilde{\beta}_\iota$ in (27) are given by*

$$\tilde{\beta}_\iota = -\omega_\iota\theta \tag{31}$$

where $\omega_\iota > 0$ is a function of deep parameters, and

$$\tilde{\beta}_h = \beta_h - \omega_h\theta \tag{32}$$

where $\beta_h > 0$ is the predictability of the rational component in (25) by past hiring. It is positive under communication friction and zero under common information. The coefficient ω_h is a function of model parameters and is positive under common information.

Proof. See Appendix A.8. □

Proposition 3 shows that non-Bayesian beliefs cannot deliver the sign pattern of $\tilde{\beta}_\iota < 0$ and $\tilde{\beta}_h > 0$ in Corollary 1 by itself. The sign of the own-decision predictability is the same as $-\theta$: when $\theta > 0$, the capital manager's expectation overreacts to the news in productivity with respect to the Bayesian benchmark, making past investment negatively predict the investment forecast error. Without communication friction, the capital manager knows that the labor decisions are driven by the same information as investment, and therefore the investment forecast error should be negatively predicted by past hiring as well, making $\tilde{\beta}_h$ having the same sign as $\tilde{\beta}_\iota$. It is impossible for the investment forecast to be simultaneously overreacting to investment and underreacting to hiring under common information. The following corollary is immediate from Proposition 3.

Corollary 2. *Having a sign pattern of $\tilde{\beta}_\iota < 0$ and $\tilde{\beta}_h > 0$ is inconsistent with common information but consistent with communication friction under non-Bayesian beliefs (30).*

So far, I have discussed the theoretical sign pattern under extrapolative beliefs and non-Bayesian beliefs separately. For more general classes of behavioral expectations, the theoretical own- and cross-decision predictability can hardly be analytically characterized. In Section 5, I show numerically that, for a more general class of behavioral expectations with arbitrary combination of perceived persistence and Kalman gain, models with common information between managers cannot match the sign pattern of $\tilde{\beta}_\iota < 0$ and $\tilde{\beta}_h > 0$. The takeaway is that the joint sign of own- and cross-decision predictability is informative of information sharing within the firm. Communication friction is necessary to match the sign pattern of $\tilde{\beta}_h > 0$ and $\tilde{\beta}_\iota < 0$.

Testing Communication Frictions with Organizational Practices The opposite-sign pattern of own- and cross-decision predictability helps identify the communication friction from behavioral distortions with expectation data only. There are two limitations of this test. First, as shown in Corollary 1 and 2, the test of joint predictability pattern will reject the null hypothesis of common information only when the communication friction is strong enough to flip the negative predictability from behavioral overreaction, making it susceptible to type-II errors. In addition, a recent literature in behavioral economics (e.g. Lian (2020), Giustinelli and Rossi (2026), Rabin and Weizsäcker (2009)) documents evidence of "narrow bracketing" or "narrow thinking": economic agents tend to make incoherent beliefs when faced with multi-dimensional decisions. The opposite-sign pattern of predictability may arise from the incoherence of the capital manager's belief among his multiple decisions rather than communication friction from organizational practices.

The test can be sharpened if the information about management and organizational practices of the firms is available in the dataset. If the cross-decision predictability is indeed driven by communication friction within the firm, then it should vary across the firms with different management practices. For instance, whether the capital and labor decisions are made in the same place can be used as a proxy for communication friction. Given $\tilde{\beta}_\iota < 0$, for the firms or plants whose capital and labor decisions are made in the same location (so that the communication friction is likely small), we should expect that their $\tilde{\beta}_h$ is more negative than those whose capital and labor decisions are made in different locations (so that the communication friction is likely large), according to Corollary 1 and 2. Instead, if $\tilde{\beta}_h$ is driven by manager's behavioral frictions, it is not clear how the cross-decision predictability varies with the location of the decisions.

4 Empirics

I now apply the tests in Section 3 to the data and verify that the empirical patterns are consistent with communication frictions. I use two complementary datasets for the empirical analysis.

4.1 Data

IBES/Compustat First, I apply the joint test of sign pattern to U.S. public firms in a merged IBES/Compustat sample, which contains firms' managerial expectations and realized outcomes between 2003 and 2024. The managerial expectations of U.S. public firms are obtained from IBES Guidance, which contains all the guidance activities (i.e. the firm's public announcement on its own expectations about sales, capital expenditure, earnings per share, among many others, for the current fiscal year) of U.S. public firms since 1992. These guidances can be regarded as the self-reported expectation of the firm on its own financial performances. Since all measurements in the guidance are financial indicators (e.g. sales, capital expenditure, earnings per share, etc.), I assume that the expectations posted in IBES Guidance mainly reflect the capital manager's view about the firm's performance.

For the one-period-ahead investment forecast $\mathbb{F}^k_{i,t-1}[\iota_{it}]$, I use the first capital expenditure guidance posted by the firm in the first quarter of fiscal year t.[6] I then merge the IBES Guidance dataset to CRSP/Compustat Fundamental Annual and get a firm-level dataset with both expectations and realized outcomes. The details of data cleaning and sample construction are laid out in Appendix B.1. The resulting IBES-Compustat merged dataset spans over 2003 and 2024 with 14,804 firm-year observations. Table B1 in Appendix B.4 tabulates the summary statistics. On average, the firms in the merged IBES/Compustat sample is about two times larger than those in the CRSP-Compustat dataset in terms of sales and employment. The sample covers about 21.6% of total observations in the CRSP-Compustat Fundamental Annual Dataset between 2003 and 2024, and the total sales of firms in the merged IBES-Compustat sample take up about 29% of U.S. GDP in the sample period.

For each firm-year observation in the IBES-Compustat merged sample, I take the realized capital expenditure from Compustat and impute the forecast/nowcast error as the

[6] Here, I regard capital expenditure (CPX) as a proxy for investment, although it is not exactly the same as ι_{it} in the model.

arc-percent difference between the actual value and the predicted value, as in Davis and Haltiwanger (1992) and Bloom, Codreanu, and Fletcher (2025):

$$\text{CapEX Forecast Error}_{it} = \frac{\text{CapEX}_{it} - \mathbb{F}^k_{i,t-1}[\text{CapEX}_{it}]}{1/2(\text{CapEX}_{it} + \mathbb{F}^k_{i,t-1}[\text{CapEX}_{it}])}$$

The arc-difference definition of forecast errors has two advantages over the log difference. First, it is bounded between $[-2, 2]$, so that outliers in the realized values and expectations would not affect the forecast/nowcast error and the regression results by much. Additionally, it avoids the problems of taking logs when the actual values or expectations are close to 0. This may not be a big issue for the large-firm samples like IBES or Compustat because there are very few firm-year observations recording a zero capital expenditure or issuing a zero capital expenditure guidance. I verify that the regression results are robust to alternative definitions of forecast/nowcast errors (e.g. the log difference between actual and predicted values).

Census Additionally, I investigate whether or not the cross-decision predictability of U.S. manufacturing establishments varies with locations of capital and labor decisions using the Management and Organizational Practices Survey (MOPS) from U.S. Census Bureau to provide further evidence of communication friction within firms. MOPS survey is a supplement to the Annual Survey of Manufacturers (ASM) in year 2010, 2015 and 2021. From the 2015 wave on, it elicits both the management practices and managerial expectations from the establishments in the ASM sample, making it an ideal dataset to investigate the relationship between management practices and expectation formation within firms and establishments. Each wave of MOPS survey receives responses from approximately 35,000 manufacturing establishments, which is about 70% of the ASM sample (Buffington et al., 2017; Bloom et al., 2019). The detailed data cleaning procedures and summary statistics are relegated to Appendix B.3.

I focus the empirical analysis on the 2015 wave of survey, which was carried out in the second quarter of 2016. [7] In Section D of the 2015 MOPS survey, the plant managers are asked to

[7] The questionnaire of the 2015 MOPS survey can be accessed from the Census website: https://www.census.gov/programs-surveys/mops/technical-documentation/questionnaires.html. The 2010 MOPS survey does not elicit managerial expectations. The 2021 MOPS survey contains managerial expectation for employment and sales, but the corresponding realized outcomes are not yet available to researchers under the new Annual Integrated Economic Survey (AIES) structure, which replaces the old ASM survey from 2023 on. This makes it difficult to construct the forecast errors in the 2021 wave of MOPS.

provide a forecast of sales, capital expenditure, employment and material costs for the year 2016, together with a backcast of the same variables for year 2015. The forecast errors are constructed for each establishment, using the arc difference between the realized outcome in ASM 2016 and the reported values in MOPS. The backcast errors are constructed in the same way as a control in the regression.

In Section B of the MOPS survey, the plant managers report the locations where the establishment's decisions are made. Question 18 asks whether the decisions of hiring full-time employees are determined by the establishment, the headquarter or both, and Question 23 asks the plant manager to report the maximum dollar amount of capital expenditure for the establishment without prior ratification of the headquarter. I use the answers to these questions to construct a dummy D_i that indicates whether the capital and labor decisions are made in the same location ($D_i = 1$) or not ($D_i = 0$) for establishment i. I set $D_i = 1$ if establishment i reports the same location of capital and labor decisions or is at the same location of the headquarter. I use this co-location dummy D_i as a proxy for communication friction between capital and labor decisions for the establishment.

4.2 Empirical Results

IBES/Compustat Results Table 1 tabulates the regression result for the joint test of (26) and (27). The first column shows the result for own-decision regression (27). We can see that $\tilde{\beta}_\iota$ is negative and statistically significant, showing that the investment expectation of U.S. public firms systematically overreact to their own investment. This is consistent with the empirical findings in Barrero (2022) that sales expectations of firm managers in the U.S. present an overreaction pattern. According to Proposition 2 and 3, this behavioral overreaction pattern can originate from over-extrapolation ($\hat{\rho} > \rho$) or over-responding to new signals ($\theta > 0$, as in diagnostic expectation).

The second column shows the result for cross-decision regression (26). We can see that β_h is positive and statistically significant, which implies that the investment expectation of U.S. public firms systematically underreact to their hiring. From the previous discussions in Corollary 1 and 2, a negative own-decision predictability should predict a negative cross-decision predictability under perfect communication. The fact that own-decision and cross-decision predictability are in opposite directions indicates that the communication friction between capital and labor managers is strong enough to offset the channel of belief overreaction and overturn the sign of cross-decision predictability. The third column includes both lagged

hiring and lagged investment in the regression, and these multivariate estimates are similar to the separate estimates in the first two columns. It shows that the correlation between past investment and past hiring does not bias the separate regression results in the first two columns.

Taken together, the results in Table 1 document that the investment forecasts of U.S. public firms simultaneously overreact to investment and underreact to hiring. This opposite-sign pattern rejects rational expectations through the nonzero $\tilde{\beta}_\iota$ and rejects common information within the baseline scalar-fundamental model through the positive $\tilde{\beta}_h$. In Appendix B.2, I show that the opposite-sign pattern is robust to controlling lag nowcast errors and alternative definitions of forecast errors.

Table 1: Predictability of Investment Forecast Error by Past Hiring and Investment

	Dependent variable:		
	$\iota_t - \mathbb{F}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
ι_{t-1}	-0.018^{**}		-0.022^{**}
	(0.009)		(0.009)
h_{t-1}		0.088^{***}	0.094^{***}
		(0.018)	(0.018)
Q1 Forecast Only	Yes	Yes	Yes
Firm Fixed Effect	Yes	Yes	Yes
Sector-year Fixed Effect	Yes	Yes	Yes
Observations	14,804	14,804	14,804
R^2	0.535	0.536	0.536

Notes: This table reports firm-level regressions of the capital-expenditure forecast error on lagged investment and lagged hiring. The forecast error is the arc-percent difference between realized capital expenditures and the firm's forecast. The sample uses the firm's first annual capital-expenditure guidance made in Q1 and requires nonmissing forecast errors and lagged hiring, as well as nonzero lagged capital expenditures. Lagged investment is the logarithm of capital expenditures in the preceding year, and lagged hiring is the preceding year's log change in employment. All specifications include firm and sector-by-year fixed effects, where sectors are defined using three-digit NAICS codes for manufacturing and two-digit NAICS codes otherwise. The sample contains 14,804 firm-year observations from 2,418 firms over 2003–2024. Standard errors clustered by firm are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Census Results I use Census data to test whether or not the own- and cross-decision predictability vary with the proxy for communication friction, i.e. whether or not the capital and labor decisions are made in the same location for the establishment. Beforehand, I repeat the analysis of own-decision and cross-decision predictability for the investment forecast error and hiring forecast error of the plants in the 2015 MOPS/ASM sample. Column (1) of Table 2 and Table 3 show that the plant manager's investment forecast error is negatively predicted by the past investment, and that his hiring forecast error is negatively predicted by the past hiring, which is consistent with behavioral overreaction documented in earlier analysis of expectations in MOPS/ASM establishments (Bloom et al., 2020). Unlike the results in IBES/Compustat, the cross-decision predictability of Census establishments does not follow an opposite direction to the own-decision predictability. The predictability of investment forecast error by past hiring in Table 2 is negative but insignificant, and the predictability of hiring forecast error by past investment in Table 3 is negative.

Next, I separate the MOPS/ASM sample into a same-location group, which contains the establishments whose capital and labor decisions are made in the same location, and a different-location group, which contains the plants whose capital and labor decisions are made in different locations. I run the same regression on the two groups separately and tabluate the results in Column (2) and (3) in Table 2 and Table 3. We can see that the sign patterns of cross-decision predictability are different across the groups. If the plant's capital and labor decisions are made in the same location, both investment and hiring negatively predicts both investment and hiring forecast errors. However, if the plant's capital and labor decisions are made in different locations, the own-decision predictability remains negative, but the cross-decision predictability becomes weakly positive (though statistically insignificant). To systematically test whether or not the own- and cross-decision predictability vary with the co-location dummy of capital and labor decisions, I run the regressions

$$\begin{aligned}\iota_{i,2016} - \mathbb{F}_{i,2015}[\iota_{i,2016}] =& \tilde{\beta}_\iota \iota_{i,2015} + \tilde{\beta}_h h_{i,2015} + \gamma_D D_i \\ &+ \gamma_\iota D_i \times \iota_{i,2015} + \gamma_h D_i \times h_{i,2015} + \Gamma_\iota' X_i + \varepsilon_{i,2016}\end{aligned} \tag{33}$$

and

$$\begin{aligned}h_{i,2016} - \mathbb{F}_{i,2015}[h_{i,2016}] =& \tilde{\alpha}_\iota \iota_{i,2015} + \tilde{\alpha}_h h_{i,2015} + \delta_D D_i \\ &+ \delta_\iota D_i \times \iota_{i,2015} + \delta_h D_i \times h_{i,2015} + \Gamma_h' X_i + \varepsilon_{i,2016}\end{aligned} \tag{34}$$

where the controls X_i include sector fixed effects and backcast errors of investment, employment and sales. The coefficients of interest are γ_ι, γ_h, δ_ι and δ_h, which capture how the

own- and cross-decision predictability of investment and hiring forecast error changes when capital and labor decisions are made in the same location. Corollary 1 and 2 predict that establishments more susceptible to communication frictions between capital and labor decisions ($D_i = 0$) should have more positive cross-decision predictability, and therefore I expect $\gamma_h < 0$ and $\delta_\iota < 0$ under communication frictions. In comparison, own-decision predictability is solely driven by behavioral overreaction, and in the model, it should not vary with the location of capital and labor decisions. Hence, I expect the interaction terms γ_ι and δ_h to be statistically insignificant from the null.

The results of (33) and (34) are tabulated in the fourth column of Table 2 and Table 3. I verify that they are consistent with the predictions of the model with communication frictions. Own-decision predictability, which results from behavioral distortions, does not vary with the proxy for communication frictions: the coefficients for the interaction term on own decisions, γ_ι and δ_h, are statistically insignificant from the null. In comparison, cross-decision predictability is more positive when the capital and labor decisions are organizationally segmented and suscepitble to communication frictions: the coefficients on the interaction terms of cross decisions γ_h and δ_ι are both negative and statistically significant. This is consistent with the takeaway in Proposition 2 and 3 that information segmentation from communication frictions makes the investment and hiring forecasts underreact to the other decisions.

In sum, the results in Table 2 and 3 complements the findings in the IBES/Compustat and indicates the existence of communication frictions between capital and labor decisions *within* establishments. This adds to the existing literature on the communication frictions and spillovers *across* establishments within multi-unit firms (e.g. Giroud (2013); Giroud and Mueller (2015); Giroud et al. (2024)).

4.3 Discussion: Alternative Common-information Explanations

I have showed in Corollary 1 and 2 that the opposite-sign pattern $\tilde{\beta}_\iota < 0 < \tilde{\beta}_h$ is inconsistent with the common-information benchmark under the current model specification with convex adjustment costs and Cobb-Douglas production technology. A crucial identification question is whether deviations from the current model setup, when combined with behavioral overreaction, can cause lagged investment and lagged hiring to predict investment forecast errors in opposite directions even though managers share a common information set. In this subsection, I analyze the possibility of various mechanisms in generating the opposite-sign

Table 2: Capital expenditure forecast error (2016) and lagged investment and hiring

	Dependent variable: Capex forecast error in 2016			
	Pooled (1)	Co-located (2)	Separated (3)	Pooled (4)
Investment rate, 2015	-0.401***	-0.403***	-0.424***	-0.441***
	(0.039)	(0.043)	(0.066)	(0.062)
Hiring, 2015	-0.041	-0.084**	0.016	0.011
	(0.028)	(0.034)	(0.046)	(0.044)
Co-located decisions				0.070***
				(0.019)
Investment rate, 2015 × Co-located (γ_ι)				0.055
				(0.072)
Hiring, 2015 × Co-located (γ_h)				-0.096*
				(0.053)
Industry FE (4-digit NAICS)	Yes	Yes	Yes	Yes
Backcast-error controls	Yes	Yes	Yes	Yes
Sample	Full	Co-located	Separated	Full
Observations	23,500	13,000	10,500	23,500
R^2	0.219	0.223	0.225	0.220

Notes: OLS estimates at the establishment level. The dependent variable is the capex forecast error, 2016: the realized ASM value minus the 2015 MOPS forecast, as a symmetric (arc/DHS) growth deviation. "Investment rate, 2015" is 2015 capital expenditure divided by the establishment's capital stock at the end of 2014 (I_{2015}/K_{2014}), and "Hiring, 2015" is 2014–2015 employment growth (arc/DHS), both from the ASM. "Co-located decisions" D_i equals one when the establishment's capital and labor decisions are made at the same location (headquarters is at the establishment, or both the hiring and the capital-expenditure decision rest with the establishment, or both rest with headquarters), and zero otherwise; columns (2) and (3) split the sample on this indicator and column (4) interacts it with the regressors. All specifications include 4-digit NAICS industry fixed effects and control for the establishment's 2015 backcast errors in sales, capital expenditure, and employment (coefficients not reported). Standard errors, in parentheses, are two-way clustered by firm and 4-digit NAICS industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sample is the set of manufacturing establishments in the 2015 MOPS that report complete 2016 and 2017 forecasts and are observed in the ASM in 2014–2016; all statistics are unweighted. Observation counts are rounded per Census disclosure rules.

Table 3: Employment forecast error (2016) and lagged investment and hiring

	Dependent variable: Employment forecast error in 2016			
	Pooled (1)	Co-located (2)	Separated (3)	Pooled (4)
Investment rate, 2015	-0.023**	-0.044***	0.008	0.004
	(0.010)	(0.012)	(0.015)	(0.015)
Hiring, 2015	-0.125***	-0.129***	-0.119***	-0.112***
	(0.012)	(0.015)	(0.018)	(0.018)
Co-located decisions				0.001
				(0.003)
Investment rate, 2015 × Co-located (δ_ι)				-0.044**
				(0.018)
Hiring, 2015 × Co-located (δ_h)				-0.022
				(0.023)
Industry FE (4-digit NAICS)	Yes	Yes	Yes	Yes
Backcast-error controls	Yes	Yes	Yes	Yes
Sample	Full	Co-located	Separated	Full
Observations	23,500	13,000	10,500	23,500
R^2	0.606	0.584	0.633	0.606
Within R^2	0.604	0.581	0.628	0.604

Notes: The dependent variable is the employment forecast error, 2016: the realized ASM value minus the 2015 MOPS forecast, as a symmetric (arc/DHS) growth deviation. Investment rate, 2015 is 2015 capital expenditure divided by the establishment's capital stock at the end of 2014 (I_{2015}/K_{2014}), and Hiring, 2015 is 2014–2015 employment growth (arc/DHS), both from the ASM. "Co-located decisions" equals one when the establishment's capital and labor decisions are made at the same location (headquarters is at the establishment, or both the hiring and the capital-expenditure decision rest with the establishment, or both rest with headquarters), and zero otherwise; columns (2) and (3) split the sample on this indicator and column (4) interacts it with the regressors. All specifications include 4-digit NAICS industry fixed effects and control for the establishment's 2015 backcast errors in sales, capital expenditure, and employment (coefficients not reported). Standard errors, in parentheses, are two-way clustered by firm and 4-digit NAICS industry. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The sample is the set of manufacturing establishments in the 2015 MOPS that report complete 2016 and 2017 forecasts and are observed in the ASM in 2014–2016; all statistics are unweighted. Observation counts are rounded per Census disclosure rules.

pattern.

Non-neutral Technology The conclusions in Corollary 1 and 2 hinges on the assumption that capital and labor are complements in the revenue function and that the productivity is Hicks-neutral, so that both investment and hiring move in the same direction in response to a productivity shock. Now I slack these assumptions and consider the implications of non-neutral technology for the sign pattern. To isolate this channel and simplify the analysis, I mute the adjustment costs and consider a common-information benchmark in which the firm's gross revenue is

$$Y_{it} = \left[\alpha^{1/\sigma}(A_{k,it}K_{it})^{1-1/\sigma} + (1-\alpha)^{1/\sigma}(A_{n,it}N_{it})^{1-1/\sigma}\right]^{\frac{\sigma(1-1/\epsilon)}{\sigma-1}} \tag{35}$$

where $\epsilon > 1$ is the product-demand elasticity and $\sigma > 0$ is the capital–labor elasticity of substitution. The Cobb-Douglas benchmark is nested by $\sigma = 1$. $A_{k,it}$ and $A_{n,it}$ are capital-augmenting and labor-augmenting technologies, respectively. Their logged values follow two independent AR(1) processes with a common persistence $\rho < 1$. The sign pattern of $(\tilde{\beta}_\iota, \tilde{\beta}_h)$ under non-neutral technology (35) is characterized in the following proposition.

Proposition 4 (Non-neutral technology under common information)**.** *Suppose the managers of the firm have common, noisy information about factor-augmenting technologies* $(A_{k,it}, A_{n,it})$ *and use Bayes-rational Kalman gain but with perceived persistence* $\hat{\rho} > \rho$*. Define the steady-state effective-capital share*

$$\bar{s}_K \equiv \frac{\alpha^{1/\sigma}(\bar{A}_k\bar{K})^{1-1/\sigma}}{\alpha^{1/\sigma}(\bar{A}_k\bar{K})^{1-1/\sigma} + (1-\alpha)^{1/\sigma}(\bar{A}_n\bar{N})^{1-1/\sigma}} \in (0,1). \tag{36}$$

Define the own-technology factor-demand responses

$$\psi_{kk} \equiv \bar{s}_K(\epsilon-1) + (1-\bar{s}_K)(\sigma-1),$$
$$\psi_{nn} \equiv (1-\bar{s}_K)(\epsilon-1) + \bar{s}_K(\sigma-1).$$

If $\psi_{kk} > 0$ *and* $\psi_{nn} > 0$*, then, in the first-order log-linear approximation, the population coefficients in regressions (26) and (27) satisfy*

$$\tilde{\beta}_\iota < 0, \qquad \operatorname{sgn}(\tilde{\beta}_h) = \operatorname{sgn}(\sigma-\epsilon). \tag{37}$$

Consequently,

$$\tilde{\beta}_\iota < 0 < \tilde{\beta}_h \quad \Longleftrightarrow \quad \sigma > \epsilon. \tag{38}$$

Proof. See Appendix A.9. □

The proposition states that, in order to get the opposite-sign pattern, we need strong substitutability between capital and labor inputs in the production function. When $\sigma > \epsilon$, the capital and labor are substitutes in the revenue function, in the sense that an increase in one input will lower the marginal revenue product of the other. In that case, a positive labor-augmenting technology shock increases hiring and decreases investment. When the capital manager's belief is over-extrapolative, he expects the labor-augmenting shock to last longer than the actual and lowers his investment forecast excessively. When the shock mean-reverts, actual investment exceeds the forecast, producing a larger forecast error and making $\tilde{\beta}_h > 0$. In contrast, when $\sigma < \epsilon$, the capital and labor are complements in the revenue function: an increase in one input improves the marginal revenue product of the other input. In this case, a positive labor-augmenting technology shock will encourage both investment and hiring, and the own- and cross-decision predictability should also be in the same direction. The Cobb-Douglas production function I used in the baseline analysis is nested in this category.

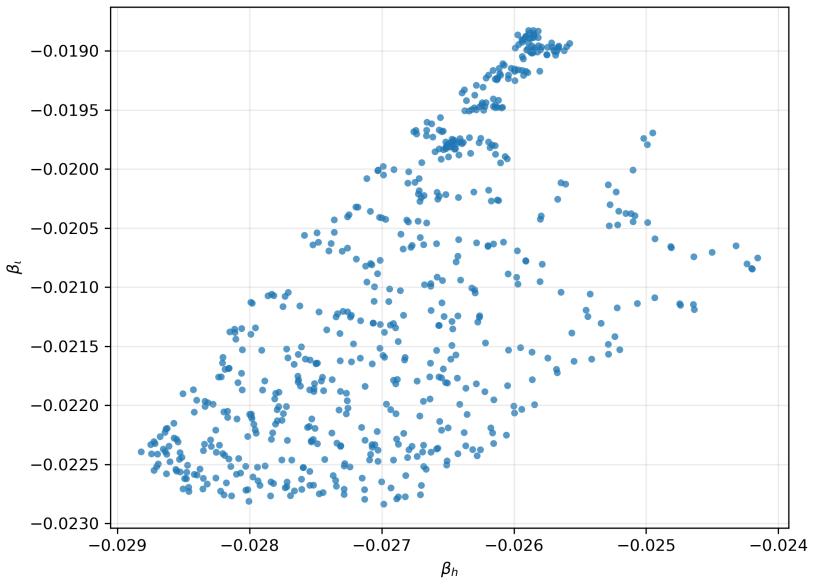
How plausible is $\sigma > \epsilon$? The existing estimates from the empirical literature suggests that it's quite implausible. The estimates of complementarity/substitutability parameter σ are substantially below unity, suggesting that capital and labor are complements rather than substitutes in production: Raval (2019) reports estimates between 0.3 and 0.5, while Oberfield and Raval (2021) reports aggregate estimates between 0.5 and 0.7. In comparison, the demand elasticity parameter ϵ typically lies between 3 and 11 (e.g. Hottman, Redding, and Weinstein (2016)). Hence, non-neutral technology alone is empirically unlikely to account for the sign pattern that communication friction generates.

Lumpiness in Capital and Labor Adjustment Next, I consider whether or not lumpiness in capital and labor adjustment can generate the opposite-sign pattern with belief overreaction. The results established in Proposition 2 and 3 are based on convex adjustment cost, which delivers small, continuous adjustment of capital and labor to the same direction in response to common productivity shocks. Lumpiness generated by non-convex adjustment costs (e.g. Cooper and Haltiwanger (2006), Asker et al. (2014)) can generate asynchronous adjustments between capital and labor and hence break the premises in these propositions.

I evaluate this possibility numerically by augmenting the baseline model with fixed adjustment costs on both capital and labor: if the capital manager makes a nonzero investment, he needs to pay a fixed fraction χ_k of total revenue as fixed costs; similarly, if the labor man-

ager makes a nonzero hiring, she needs to pay a fixed fraction χ_n of total revenue as fixed costs. This setup makes fixed cost scales with firm size, as in Cooper and Haltiwanger (2006) and Bloom (2009). I shut off the information segmentation, let the managers learn from a common, noisy signal, and allow the managers' perceived persistence $\hat{\rho}$ to vary between ρ and 1. I solve the model numerically and simulate a panel of 1,000 firms for 500 periods to obtain the model-implied regression coefficients, with 625 combinations of non-convex costs $(\chi_k, \chi_n) \in [0, 0.01]$ and perceived persistence $\hat{\rho} \in (\rho, 1)$.

Figure 1: Simulated β_ι and β_h under Non-convex Costs & Over-extrapolation



Notes: This figure scatterplots the simulated own- and cross-decision predictability (β_ι, β_h) for 625 combinations of $(\chi_k, \chi_n, \hat{\rho}) \in [0, 0.01] \times [0, 0.01] \times (\rho, 1)$, equally spaced on each dimension, with $\sigma_{\epsilon,p} = 0.3$ and convex costs (ξ_k, ξ_n) fixed at their estimated value in Section 5.

The results are scatter-plotted in Figure 1. We can see that for the parameter combination we consider, both own- and cross-decision predictability are negative. None of them generates the opposite-sign pattern $\beta_\iota < 0$ and $\beta_h > 0$ in the data. The result is not because the simulated model is short of lumpiness: the range of non-convex cost parameters we consider can generate inaction rate up to 50% for capital and labor; in comparison, the inaction rate of capital and labor adjustment in Compustat data is about 10%.[8] Although nonconvex costs can make capital and labor adjust at different dates, they are not sufficient to make capital and labor adjustment respond in opposite directions to the same productivity news. As a result, if the capital manager's belief overreacts to investment, it should also overreact to hiring.

[8]Here, I define inaction as investment or hiring rate below 10 percent, both in the data and in the model.

Time-to-build, Distortions and Private Benefits Lastly, I consider the sign patterns of regression (26) and (27) under three alternative mechanisms: time-to-build in capital, Hsieh-Klenow type of distortions, and managers' private benefits. Each of these mechanisms can be accommodated by an extended version of our baseline model. As Appendix A.10.1, A.10.2 and A.10.3 shows, it turns out that, after taking log-quadratic approximation of the managers' payoff functions, the Markov Perfect Equilibrium under these frictions take the same form as in Lemma 2. This means that the belief dynamics in Lemma 1 and the MPBE characterization in Lemma 2 remain true. These frictions affect investment and hiring by changing the policy matrix F_ι and F_h, and as long as the primitives of these frictions are common knowledge within the firm, the same-sign pattern in Corollary 1 and 2 still holds for these alternative mechanisms under common information: without communication frictions, the cross-decision predictability $\tilde{\beta}_h$ inherits the sign of the own-decision predictability $\tilde{\beta}_\iota$. In other words, these frictions cannot reproduce opposite-sign pattern by themselves.

Taken together, these alternative mechanisms under common information do not readily reproduce the opposite-sign pattern. Frictions that preserve the common linear belief structure retain the same-sign restriction; non-convex costs do not overturn it numerically even with substantial lumpiness; and non-neutral technology requires $\sigma > \epsilon$, contrary to conventional elasticity estimates.

5 Quantitative Analysis

In this section, I incorporate the expectation tests in Section 3 and 4 into a larger quantitative model to structurally estimate the information parameters governing the communication frictions. With the quantitative model, I further rule out the confounders for the sign test and quantify the aggregate effect of communication frictions.

5.1 Calibration

An Extended Model for Quantification To obtain a credible quantification of the aggregate effect of communication friction, I extend our baseline model so that it incorporates Hsieh-Klenow-style reduced-form distortions and time-to-build in capital. The firm's profit

function becomes

$$\Pi_{it} = (1-\tau_{it}^y)R(K_{i,t-1}, N_{it}; A_{it}) - (1+\tau_{it}^k)R_t K_{it} - W_t N_{it} \\ - \frac{\xi_k}{2}\left(\frac{K_{it}}{K_{i,t-1}} - 1\right)^2 K_{i,t-1} - \frac{\xi_n}{2}\left(\frac{N_{it}}{N_{i,t-1}} - 1\right)^2 N_{i,t-1} \tag{39}$$

where $R(K_{i,t-1}, N_{it}; A_{it})$ is the revenue function with capital $K_{i,t-1}$ predetermined to capture time-to-build, and τ_{it}^y and τ_{it}^k are reduced-form firm-specific distortions as in Hsieh and Klenow (2009). τ_{it}^y captures the mechanisms that distort all factors symmetrically without changing the capital-labor ratio (such as price markup), while τ_{it}^k captures the capital-specific distortions (such as financial frictions) that generate variations in the capital-labor ratio. Following David and Venkateswaran (2019), I allow these reduced-form distortions to have a rich correlation structure:

$$\tau_{it}^y = \gamma^y a_{it} + \delta_i^y + \eta_{it}^y \tag{40}$$
$$\tau_{it}^k = \gamma^k a_{it} + \delta_i^k + \eta_{it}^k \tag{41}$$

where $\gamma^y, \gamma^k \in (0,1)$ captures correlation of distortions with productivity. They are referred to as "correlated distortion" as in Bartelsman et al. (2013) and capture size-dependent distortions (such as antitrust regulations). $\eta_{it}^y \sim N(0, \sigma_{\eta,y}^2)$ and $\eta_{it}^k \sim N(0, \sigma_{\eta,k}^2)$ are the transitory distortions uncorrelated to productivity, as in Hsieh and Klenow (2009). Lastly, $\delta_i^y \sim N(0, \sigma_{\delta,y}^2)$ and $\delta_i^k \sim N(0, \sigma_{\delta,k}^2)$ are firm characteristics that do not change over time. I assume that the correlated distortion γ^y, γ^k, firm characteristic δ_i^y, δ_i^k and the uncorrelated distortions η_{it}^y, η_{it}^k are common knowledge within the firm, so that the only source of uncertainty about τ^y and τ^k is the uncertainty about productivity a_{it}.

The information structure within the firm remains the same as in the baseline model: in each period, the capital manager and the labor manager receive a noisy, private signal about productivity a_{it} respectively, and they both receive a firm-level common signal. When they form expectations in each period, they do not know the other manager's decision in that period. To match the behavioral overreaction in own-decision predictability, I allow the expectations of capital manager to be over-extrapolative, i.e. the perceived persistence of productivity $\hat{\rho}$ is larger than the actual persistence ρ.

Calibration In total, there are 12 parameters to be calibrated in the extended model: convex adjustment costs ξ_k and ξ_n, the behavioral/information parameters $(\sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}, \hat{\rho})$, and the parameters characterizing the reduced-form distortions $(\gamma^k, \gamma^y, \sigma_{\delta,k}, \sigma_{\delta,y}, \sigma_{\eta,k}, \sigma_{\eta,y})$.

I pick 12 empirical moments as targets in the calibration. In particular, I include the own- and cross-decision predictability $\tilde{\beta}_\iota$ and $\tilde{\beta}_h$ as targeted moments. As I have discussed in the previous section, the predictability of investment forecast errors by past investment and by past hiring are indicative of communication friction and belief extrapolation and hence help us identify the information and behavioral friction parameters. To match $\tilde{\beta}_h > 0$ and $\tilde{\beta}_\iota < 0$, we expect $\hat{\rho} > \rho$ and $\sigma_{\epsilon,k}, \sigma_{\epsilon,n} < \infty$, so that the private signals receive weights in belief updating. The rest of the targeted moments are standard in the firm dynamics literature. The serial correlation of input adjustments ($\text{corr}(\iota, \iota_{-1})$ and $\text{corr}(h, h_{-1})$) and the volatility of input adjustments ($var(\iota)$ and $var(h)$) are included to jointly discipline adjustment costs and uncorrelated distortions. The correlation between marginal revenue products and productivity, $\text{corr}(\text{MRPK}, a)$ and $\text{corr}(\text{MRPN}, a)$, is connected with the correlated distortions, and the dispersion in MRPK and MPRN allows the across-firm variation by the firm characteristic terms to absorb the proportion of observed MRPK and MRPN dispersion not captured by other variations.

The extended model is calibrated by minimum distance using analytical stationary moments. No finite panel is simulated in the calibration. For each candidate parameter vector, I proceed as follows:

1. For each $\Theta = (\xi_k, \xi_n, \hat{\rho}, \sigma_{\epsilon,p}, \sigma_{\epsilon,k}, \sigma_{\epsilon,n}, \gamma^k, \gamma^y, \sigma_{\delta,k}, \sigma_{\delta,y}, \sigma_{\eta,k}, \sigma_{\eta,y})$, solve the MPBE according to Lemmas 1 and 2, obtaining the linear policy rules for investment and hiring and the corresponding belief dynamics;

2. Stack the policy rules and belief dynamics into a linear state-space system. I compute the stationary covariance matrix of its dynamic component by solving the associated discrete Lyapunov equation and then add the cross-sectional covariance generated by the permanent distortions δ_i^k and δ_i^y analytically;

3. Construct the population model moments $\mathbf{m}(\Theta)$ directly from the resulting contemporaneous and lagged covariance matrices. In particular, variances, covariances, and correlations are computed from these matrices, and $\tilde{\beta}_h$ and $\tilde{\beta}_\iota$ are the corresponding population projection coefficients.

I choose the parameter vector that minimizes the unweighted quadratic distance from the data moments $\mathbf{m}^{\text{data}}$:

$$\hat{\Theta} = \arg\min_{\Theta} \left[\mathbf{m}(\Theta) - \mathbf{m}^{\text{data}}\right]' \left[\mathbf{m}(\Theta) - \mathbf{m}^{\text{data}}\right].$$

Table 4: Estimated Parameter Values and Targets

Description		Value	Source/Target
Elasticity of substitution	ε	4	Standard Value
Capital share	α	0.47	Cost share in IBES/Compustat
Persistence of Productivity	ρ	0.94	Estimates of $a_{it} = \rho a_{i,t-1} + \mu_{it}$.
Std.dev. of Productivity shock	σ_μ	0.24	
Capital convex adj. cost	ξ_k	0.36	$\text{corr}(\iota, \iota_{-1})$
Labor convex adj. cost	ξ_n	3.75	$\text{corr}(h, h_{-1})$
Noise on the common signal	$\sigma_{\epsilon,p}$	1.40	Reg. Coef. $\tilde{\beta}_\iota$
Noise on K manager's signal	$\sigma_{\epsilon,k}$	0.32	Reg. Coef. $\tilde{\beta}_h$
Noise on N manager's signal	$\sigma_{\epsilon,n}$	0.07	$\text{corr}(h, \Delta a)$
Degree of Extrapolation	$\hat{\rho}/\rho$	1.06	$\text{corr}(\iota, \Delta a)$
Correlated Distortion on revenue	γ_y	0.17	corr(MRPN, a)
Correlated Distortion on capital	γ_k	0.00	corr(MRPK, a)
Transitory Distortion η^y	$\sigma_{\eta,y}$	0.98	$\text{var}(\iota)$
Transitory Distortion η^k	$\sigma_{\eta,k}$	0.41	$\text{var}(h)$
Permanent Distortion δ^y	$\sigma_{\delta,y}$	1.06	var(MRPK)
Permanent Distortion δ^k	$\sigma_{\delta,k}$	1.36	var(MRPN)

Because the model moments are computed analytically, this criterion contains no model-simulation error. Appendix A.4 provides the details of the stationary-covariance calculation. The targeted moments and calibrated parameter values are summarized in Table 4.

Data The data moments are computed from a quantitative sample constructed by merging annual CRSP–Compustat records with IBES Guidance spanning from 2003 to 2024. From IBES, I obtain the first annual capital-expenditure guidance within the fiscal year with a forecast horizon below four quarters and compute the forecast error of investment by combining it with the actual capital expenditure in Compustat. From the Compustat sample, I take the book value of property, plant and equipment (PPE) as the measure of capital stock. Investment rate ι and hiring rate h are obtained by first-differencing the logged capital and employment series respectively. Value added is sales minus material expenditure.[9] I then construct the total revenue factor productivity (TFPR) or sales Solow residual as the firm

[9]Material expenditure is defined as Cost of Goods Sold (COGS) + Selling, General & Administrative Expenses (XSGA) − Depreciation (DP) − Wage Bill (XLR), following Keller and Yeaple (2009) and Flynn and Sastry (2020). For observations with missing Compustat labor expense, I impute the wage bill as employment multiplied by the corresponding industry-year wage from the Census Bureau County Business Patterns (CBP). See Appendix B.4 for details.

productivity a_{it} using time-to-build timing

$$a_{it} = \log(\text{Value Added}_{it}) - (1-\alpha)(1-1/\epsilon)n_{it} - \alpha(1-1/\epsilon)k_{i,t-1}$$

where the capital share α is chosen to match the cost share of labor in the sample, and the elasticity of substitution $\epsilon = 4$ is taken from the literature (Hsieh and Klenow (2009), Hottman et al. (2016)). The corresponding marginal-revenue-product proxies are $\text{MRPK}_{it} = \log(\text{Value Added}_{it}) - k_{i,t-1}$ and $\text{MRPN}_{it} = \log(\text{Value Added}_{it}) - n_{it}$.

Following David and Venkateswaran (2019), I remove industry-by-calendar-year fixed effects from investment, hiring, productivity growth and levels, MRPK, MRPN, and the capital–labor ratio $k_{i,t-1} - n_{it}$. Industries are defined by three-digit NAICS codes for manufacturing and two-digit NAICS codes otherwise; fiscal years ending from January through May are assigned to the preceding calendar year. Before trimming, I construct exact one-year changes in the residualized investment and hiring series. I then jointly retain observations between the first and ninety-ninth percentiles of the capital-expenditure forecast error, lagged capital expenditure, lagged hiring, those investment and hiring changes, and the residualized productivity, marginal-revenue-product, and capital–labor-ratio series. The resulting calibration sample contains 13,194 firm-year observations from 2,086 firms. Appendix B.4 provides the complete construction procedure and summary statistics of the sample.

Table 5 reports the model-implied and empirical targeted moments in the calibration. We can see that our model can match most of the data moments closely, although it overstates the variance of investment and hiring.

5.2 Results

Matching the Own- and Cross-decision Predictability Table 5 shows that the model-implied own-decision predictability $\tilde{\beta}_\iota$ and cross-decision predictability $\tilde{\beta}_h$ closely match the data moments with over-extrapolation combined with a noisy firm-level common signal and more precise manager-level signals. Can models under common, noisy information match the sign pattern observed in the data? To answer to question, I mute communication frictions (i.e. fix $\sigma_{\epsilon,k}, \sigma_{\epsilon,n} \to \infty$) and recalibrate models with alternative specifications of expectations to the same set of targets. The best fits of these models to the own- and cross-decision predictability are tabulated in Table 6.

The first two rows of Table 6 show that, without communication frictions, the own- and cross-

Table 5: Targeted Moments, Model vs Data

Moment	Data moment	Model-implied moment
$\text{corr}(\iota, \iota_{-1})$	0.305	0.306
$\text{corr}(h, h_{-1})$	0.164	0.168
$\text{corr}(\iota, \Delta a)$	0.160	0.159
$\text{corr}(h, \Delta a)$	0.302	0.303
β_h	0.112	0.108
β_ι	-0.026	-0.024
$\text{var}(\iota)$	0.019	0.036
$\text{corr}(\text{MRPK}, a)$	0.465	0.468
$\text{var}(\text{MRPK})$	0.455	0.456
$\text{var}(h)$	0.022	0.044
$\text{corr}(\text{MRPN}, a)$	0.465	0.468
$\text{var}(\text{MRPN})$	0.227	0.231

decision predictability tends to have the same sign, governed by the behavioral expectation. Specifically, in the second row, I consider a more general class of behavioral expectations, with the belief updating rule given by

$$\mathbb{F}_{it}[a_{it}] = \hat{\rho}\mathbb{F}_{i,t-1}[a_{i,t-1}] + G(s^p_{it} - \hat{\rho}\mathbb{F}_{i,t-1}[a_{i,t-1}]). \tag{42}$$

where the managers' perceived persistence $\hat{\rho}$ and the Kalman gain G can vary freely between 0 and 1, as in Angeletos et al. (2021) and Kohlhas and Walther (2021). Calibrating the $\hat{\rho}$ and G to the data gives us positive own- and cross-decision predictability ($\tilde{\beta}_h \approx 0.046, \tilde{\beta}_\iota \approx 0.045$), failing to match the opposite-sign pattern in the data. This is consistent with the takeaway from the theoretical results established in Section 3.The third row of Table 6 shows that behavioral overreaction is also necessary: if I keep communication frictions only and use rational expectation in the calibration (by fixing $\hat{\rho} = \rho$), I am not able to get a nonzero own-decision predictability $\tilde{\beta}_\iota$. The takeaway is that both communication frictions and behavioral overreaction are necessary to match the own- and cross-decision predictability.

Sources of Learning The estimated parameters shed light on the learning patterns of the capital and labor managers. Table 7 shows how much the managers learn from multiple information sources. Repeatedly learning from noisy signals leads to a substantial reduction in uncertainty for both managers: posterior uncertainty is less than 10 percent of prior uncertainty for both the capital and labor manager. In this baseline calibration, managers' private signals play an important role in decision making. Both managers put a large weight

Table 6: Communication frictions are necessary to match the observed $\tilde{\beta}_\iota$ and $\tilde{\beta}_h$

Recalibrated Model	$\tilde{\beta}_h$	$\tilde{\beta}_\iota$
Noisy, Common Info + Extrapolation	0.001	0.001
Noisy, Common Info + Behavioral Expectation (42)	0.046	0.045
Communication Friction + RE	0.098	0
Communication Friction + Extrapolation	**0.108**	**-0.024**
Data	0.112	-0.026

Notes: The upper panel reports the recalibrated values of the own- and cross-decision predictability under alternative specifications of information and behavioral frictions. The lower panel reports the observed own- and cross-decision predictability in the data.

on their private signals, so that the model can make investment forecast underreact to hiring and generate a positive cross-decision predictability. These private signals generate different posterior beliefs at the time investment and hiring decisions are made and break the perfect correlation between capital and labor managers' beliefs under perfect communication. In contrast, firm-level common signals are not major sources of learning for the managers: the Kalman gains on the common signals are below 2% for both managers. This finding is consistent with the literature: David et al. (2016) finds that firm-level public information, such as stock price, is not a major source of learning.

Table 7: Learning Patterns of the Managers

	Uncertainty		Source of Learning		
Manager	Posterior	Posterior/Prior	Common	Private	Prior
Capital Manager	0.035	7.85%	1.81%	34.33%	63.86%
Labor Manager	0.005	1.11%	0.26%	90.71%	9.03%

Notes: "Posterior" reports the posterior variance $\hat{\Sigma}^m$ for manager $m \in \{k, n\}$. "Posterior/Prior" reports $\hat{\Sigma}^m/\sigma_a^2$, the posterior variance relative to the unconditional variance of productivity. The Common, Private, and Prior columns report the weights in the manager's posterior mean on the firm-level common signal s_{it}^p, the manager-specific private signal s_{it}^m, and the common prior $\bar{a}_{it}$, respectively.

Communication Friction as Input-specific Distortions Communication friction plays an important role in two untargeted moments: the investment-hiring correlation and the dispersion of capital-labor ratio. These two moments show that communication frictions affect firm dynamics in a significantly different manner from other behavioral and informational frictions.

Low investment-hiring correlation is an important feature in the US firm data, and our model provides a decent match as an untargeted moment. Which mechanisms are responsible for

it, and which one is the most prominent? In the first column of Table 8, I report the contribution of each friction to the investment-hiring correlation. The numbers in the table are computed under the assumption that only one friction of interest is active (i.e. the relevant parameters are fixed at the estimated level) while all other frictions are muted (i.e. the relevant parameters are set to the frictionless level). From Table 8, we can see that the low investment-hiring correlation is primarily driven by communication friction and time-to-build, with communication friction being the strongest force in the model: communication friction alone generates a 0.33 investment-hiring correlation, while none of the other frictions alone lowers the correlation effectively from 1.

The decomposition of investment-hiring correlation highlights a crucial difference between firm-level, common information/behavioral frictions and within-firm segmented information. We can see that if we remove the information segmentation and allow the managers to learn from all three signals received in each period, the investment-hiring correlation is equal to 1. Manager-level private signals, on the other hand, lower the correlation significantly. This can be seen from the expressions of capital and labor choices. Without other frictions, capital and labor are both proportional to the managers' posterior beliefs about productivity, i.e. $k_{it} \propto \mathbb{E}^k_{it}[a_{it}]$ and $n_{it} \propto \mathbb{E}^n_{it}[a_{it}]$. Absent communication friction, managers share the same information, so that $\mathbb{E}^k_{it} = \mathbb{E}^n_{it} = \mathbb{E}_{it}$, and firm-level common noisy information attenuates the responsiveness of both inputs by the same proportion, without breaking the perfect correlation between them. The existence of private signals breaks this pattern as the managers' expectations are no longer perfectly correlated in general.

The distinction between firm-level common information and manager-level private information is akin to the difference between revenue distortion τ^y and capital distortion τ^k. Common, noisy information and behavioral expectations act like the revenue distortion τ^y as they both affect all the inputs symmetrically, while the information segmentation introduced by communication friction acts like the capital-specific distortion τ^k, as they both affect different inputs disproportionately and generate variations in capital-labor ratio. Communication frictions should therefore be categorized as input-specific distortions, while common noisy information and behavioral expectations belong to the class of revenue distortion.

To make the point clearer, I investigate the role that communication friction plays in the dispersion of capital-labor ratio. As we know from Hsieh and Klenow (2009), revenue distortions cannot distort capital-labor ratio while input-specific distortions can. The second column of Table 8 reports the capital-labor ratio dispersion that each mechanism can generate alone. We can see that in our calibrated model, communication friction is an important source of

the capital-labor ratio variability. Communication friction alone generates a capital-labor ratio dispersion of 0.302, which accounts for approximately 39% of the observed dispersion in the capital-labor ratio and about 38% of the model-implied dispersion. In contrast, revenue distortion τ^y, common noisy information and behavioral expectations cannot distort the capital-labor ratio by themselves. These results show that modeling information and behavioral friction as commonly shared within the firm can be misleading. It ignores an important channel for imperfect information to generate input-specific distortions, which may lead us to underestimate the role of information and behavioral friction in firm dynamics and overestimate the role of other input-specific distortions (e.g. financial frictions, non-neutral technologies) in driving the variations of capital-labor ratio.

Table 8: Sources of input comovement and capital–labor ratio variation

Mechanism	$\text{corr}(\iota_t, h_t)$	$\text{var}(k_{t-1} - n_t)$
Data	0.557	0.774
Model, all mechanisms	0.445	0.802
Time to build only	0.631	0.085
Convex costs only	0.993	0.006
Communication friction only	0.326	0.302
Overextrapolation + common information	1.000	0.000
Capital distortion τ^k only	0.972	2.032
Revenue distortion τ^y only	1.000	0.000

Notes: "Only" rows activate the named mechanism at its estimated value and mute all other mechanisms. The τ^k-only row keeps γ_k, $\sigma_{\delta,k}$, and $\sigma_{\eta,k}$ active; the τ^y-only row keeps γ_y, $\sigma_{\delta,y}$, and $\sigma_{\eta,y}$ active. The common-information counterfactual combines the three estimated signal precisions into one common signal.

Allocative Efficiency What is the implication of communication frictions for input misallocation and allocative efficiency? In Table 9, I tabulate the contribution of each friction on the capital and labor misallocation and quantify the aggregate TFP loss generated by each friction. These numbers are computed under the assumption that only one friction is active each time. We can see that communication friction alone can generate approximately 27 percent of observed MRPK dispersion and 20% of MRPN dispersion. In comparison, if I mute communication friction within the firm, the MRPK and MRPN dispersion generated by behavioral expectations and common noisy information are much smaller. Hence, the information-driven misallocation is primarily generated by communication friction. Compared with other mechanisms, the MRPK and MRPN dispersion driven by communication

Table 9: Misallocation accounting

Mechanism	var(MRPK)	var(MRPN)	$\log A - \log A^*$
Data	0.455 (100.00%)	0.227 (100.00%)	–
Model, all mechanisms	0.456 (100.36%)	0.231 (101.70%)	-37.54%
Time to build only	0.085 (18.79%)	0.000 (0.00%)	-4.91%
Convex costs only	0.029 (6.45%)	0.058 (25.69%)	-8.68%
Communication friction only	0.123 (27.10%)	0.044 (19.28%)	-4.99%
Overextrapolation + common information	0.003 (0.59%)	0.003 (1.18%)	-0.54%
Correlated distortions only	0.013 (2.89%)	0.013 (5.75%)	-2.62%
Transitory distortions η	0.012 (2.56%)	0.018 (7.86%)	-1.01%
Permanent distortions δ	0.316 (69.37%)	0.152 (67.16%)	-18.69%

Notes: Dispersion shares in parentheses use the corresponding data MRPK and MRPN dispersion as the denominator. "Only" rows activate the named mechanism at its estimated value and mute the others. Permanent and transitory distortion rows report the full-model value minus the counterfactual that removes that distortion block. Aggregate TFP loss entries are log TFP gaps relative to the frictionless allocation.

frictions are quantitatively significant: neither adjustment costs nor correlated distortions can generate a significantly larger marginal revenue product dispersion than communication frictions by themselves.

Despite the fact that communication friction can generate a large portion of MRPK and MRPN dispersion, it is not the largest source of aggregate TFP loss. Communication friction by itself generates a sizable 5% of aggregate TFP loss from the frictionless benchmark, but its magnitude is smaller than adjustment costs, which explains much less MRPK dispersion than communication frictions. This is because the MRPK and MRPN dispersion are not sufficient statistics for aggregate TFP loss by themselves. Analytically, the aggregate TFP loss can be expressed as

$$\log(A_t) - \log(A^*) = -\frac{1}{2}\frac{\epsilon^2}{\epsilon - 1}\Big[\hat{\alpha}_1(1-\hat{\alpha}_2)\text{var(MRPK)} + \hat{\alpha}_2(1-\hat{\alpha}_1)\text{var(MRPN)} + 2\hat{\alpha}_1\hat{\alpha}_2 \cdot \text{cov(MRPK, MRPN)}\Big] \tag{43}$$

where $\hat{\alpha}_1$ and $\hat{\alpha}_2$ are the capital and labor elasticities in the revenue function. The covariance term captures how the correlation between capital and labor distortions affect aggregate productivity. Intuitively, if capital distortion and labor distortion comove with each other, the distorted capital choice would distort the labor choice even further due to the complementarity between capital and labor, which leads to larger aggregate TFP loss. If the capital distortion and labor distortion are negatively correlated, this reinforcement effect disappears,

making the resulting aggregate TFP loss smaller.

Table 10: Comovement between marginal revenue products

Mechanism	corr(MRPK, MRPN)
Data	-0.144
Model, all mechanisms	-0.177
Time to build only	-
Convex costs only	0.987
Communication friction only	-0.919
Overextrapolation + common information	1.000
Capital distortion τ^k only	-
Revenue distortion τ^y only	1.000

Notes: Mechanism-only rows activate the named mechanism at its estimated value and mute all other mechanisms. The correlation is undefined for time-to-build-only and τ^k-only because these counterfactuals generate no MRPN dispersion. The τ^k-only row keeps γ_k, $\sigma_{\delta,k}$, and $\sigma_{\eta,k}$ active; the τ^y-only row keeps γ_y, $\sigma_{\delta,y}$, and $\sigma_{\eta,y}$ active. The common-information counterfactual combines the three estimated signal precisions into one common signal.

We can see from Table 10 that the mechanisms in the calibrated model present different patterns of marginal revenue product correlation. Specifically, communication friction generates a negative correlation between marginal revenue product, time-to-build and capital distortion generate a zero correlation, and other mechanisms generate a positive correlation. This explains why communication frictions imply a smaller aggregate TFP loss than convex adjustment costs even though the latter explains a smaller proportion of marginal revenue product dispersion.

But why are MRPK and MRPN negatively correlated under communication friction? It's because under communication friction, the comovement of MRPK and MRPN is primarily driven by manager-level private signals. Suppose there is a favorable realization of the capital manager's private signal while the productivity of the firm remains the same. Upon observing the private signal, the capital manager updates his beliefs and invests in capital, but since the productivity does not change, the marginal revenue product of capital decreases due to diminishing marginal return. Meanwhile, a higher capital improves the marginal revenue product of labor due to complementarity. Therefore, a favorable realization of the capital manager's private signal leads to a low MRPK and a high MRPN. The same mechanism holds for the labor manager's private signal as well. Since both capital and labor managers rely heavily upon their own private signals as in Table 7, the MRPK and MRPN are negatively correlated.

In contrast, other mechanisms considered in the model cannot generate a negative correlation between MRPK and MRPN by themselves. Under time-to-build or capital distortion τ^k, labor decisions are frictionless, and hence the MRPN dispersion is zero. Under common information, behavioral expectation or revenue distortion τ^y, both MRPK and MRPN are affected symmetrically, and the correlation is perfect without other frictions. Under adjustment costs, both investment and hiring respond sluggishly to the same productivity shock, and hence the marginal revenue products tend to comove in the same direction. Hence, communication friction appears to be an important force to make the model-implied MRP correlation negative and close to the untargeted data moment.

In sum, although communication friction generates a large share of observed MRPK and MRPN dispersion alone, the aggregate TFP loss from this mechanism is mitigated by the negative correlation between marginal revenue product of capital and labor. More generally, this shows the importance of investigating the nature of distortions in misallocation and aggregate productivity accounting: for the frictions that move MRPK and MRPN in opposite directions, such as heterogeneous technology in David and Venkateswaran (2019) and communication friction, large dispersions in marginal revenue products do not automatically imply large losses in TFP.

5.3 Discussion

Comparison with the literature My quantification of information-driven misallocation is comparable to the efficiency loss suggested by the literature. David et al. (2016) estimate the US aggregate TFP loss from information friction to be between 4% (in the case of frictionless labor) and 40% (when labor is subject to the same noisy information as capital). David and Venkateswaran (2019)'s estimate is around 7%. My quantification is within the range and closer to the lower bound. Compared with the papers that treat information friction as a revenue distortion, such as Gautier et al. (2025) and Ropele et al. (2024), my quantification is larger.

The takeaways for other frictions are largely consistent with the literature. As in David and Venkateswaran (2019), permanent distortions δ^k and δ^y remain the largest contributors to the capital and labor misallocation and aggregate TFP loss. My current framework focuses on within-firm friction and assumes that the internal information structure is same across firms. Since management and organizational practices are highly sticky over time and heterogeneous across firms (Bloom et al. (2019)), they can be candidates of these firm-level

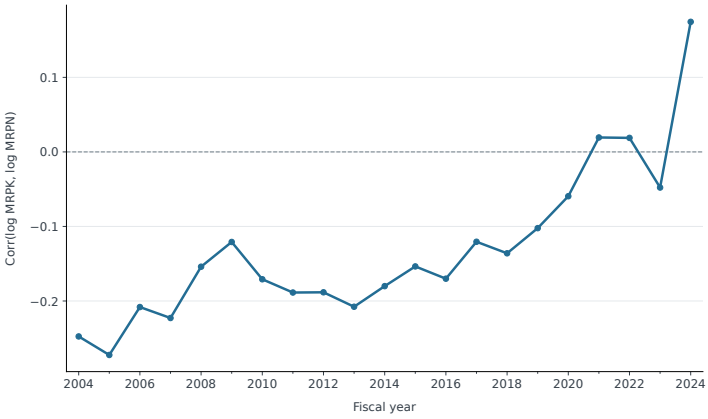
permanent distortions, which are identified as the key driver of marginal revenue product dispersion in David and Venkateswaran (2019). A promising direction for future research is to allow for heterogeneous practices of internal information sharing across firms and evaluate its quantitative plausibility as a candidate of permanent distortions.

Economic consequences of ICT and AI Although the analysis of the paper so far focuses on the steady state and assumes the information parameters to be fixed over time, the mechanism of how communication frictions affect firm dynamics may shed light on several puzzling trend of the U.S. economy in recent decades of information technology revolution. In the past three decades, the U.S. experienced a rapid development in Information and Communication Technologies (ICTs). However, the same episode also features declining business dynamism, low allocative efficiency and slow productivity growth (e.g. Decker et al. (2020)), which is puzzling because improved ICTs should make firms more responsive to information. Leading explanations include increasing adjustment frictions (Decker et al., 2020), imperfect spillover of new technologies (Akcigit and Ates, 2021, 2023; Goldin et al., 2024), rising markups (De Loecker et al., 2020) that may be partially driven by improving ICTs (Farboodi and Veldkamp, 2026; Eeckhout and Veldkamp, 2024).

My paper provides a complementary angle to this puzzle. While better ICTs mitigate the dispersions in MRPK and MRPN from information frictions, improved communication technology also increases the covariance between marginal revenue products, so that the frictions on one input are more likely to influence the other input. This increase in MRPK-MRPN covariance offsets the aggregate TFP gain from better ICTs. In fact, there is indeed an upward trend of MRPK-MRPN correlation in this episode, as illustrated in Figure 2, which echoes with the finding in Blackwood et al. (2026) that the revenue distortions and input-specific distortions are more strongly correlated in the past few decades, a key contributor to the decline in allocative efficiency. Smaller communication frictions from the improved ICTs provide a potential mechanism to account for these trends.

With the recent development of Artificial Intelligence (AI), one may wonder if information and communication frictions would remain relevant in firms' decisions. The framework developed in this paper is helpful to this discussion. First, whether AI would alleviate or exacerbate the communication frictions within firms largely depends on the organizational practices of firms. Although AI is incredibly good at detecting inconsistencies in large-scale projects completed by multiple people, giving managers decentralized access to a capable AI model does not automatically give them access to one another's information: AI may im-

Figure 2: Trend of MRPK-MRPN Correlation



Notes: This is computed from the IBES/Compustat quantitative sample, between 2004 and 2024.

prove the precision of managers' private information while leaving information segmentation in place. Additionally, even if AI is deployed properly so that the communication frictions within firms are alleviated, the gain in aggregate productivity from this channel might be moderated, for exactly the same reasons I've discussed for ICTs: as AI reduces information segmentation, it can lower MRPK and MRPN dispersion while raising their covariance. Consequently, observing more coordinated decisions and narrower marginal-product distributions need not imply a proportionately large TFP gain.

Micro-foundation of communication frictions In the current framework, the firm's internal information structure is exogenously given, and information segmentation is strictly dominated by perfect communication: making the managers' private signals common knowledge within the firm will always improve the value of the firm and generate a net TFP gain in aggregate. In reality, the internal information flow might be endogenously chosen by the firm. For instance, if each manager has limited attention and the complementarity of tasks between managers is weak, then it might be optimal for the firm to take the attention constraint into account and assign segmented information to managers (Dessein et al. (2016)). This might change the welfare implications of this paper: having segmented information within the firm may not be interpreted as a "misallocation" of information but a second-best outcome. I leave this to a companion paper that studies information sharing across plants within multi-unit firms.

6 Conclusion

This paper studies how imperfect information sharing inside firms affects managerial expectations, input choices, and allocative efficiency. I develop a dynamic team-production framework in which capital and labor managers pursue the same objective but make decisions using different information. The framework produces a novel empirical test based on investment forecast errors. Under common information, behavioral overreaction or underreaction makes forecast errors predictable by past investment and past hiring in the same direction. Communication friction breaks this restriction because hiring contains information observed by the labor manager but not fully incorporated into the capital manager's forecast. Applying the test to the data on U.S. public firms, I find that investment forecasts simultaneously overreact to news about investment and underreact to news about hiring. Complementary Census evidence connects forecast-error predictability to organizational practices: for both investment and employment forecasts, cross-decision coefficients are more positive when capital and labor decisions are made at different locations, while own-decision coefficients show no statistically detectable differences. Together, these findings support an organizational source of information segmentation alongside behavioral overreaction.

The quantitative results show why this distinction matters for firm dynamics. Manager-specific information substantially lowers investment–hiring comovement and acts as an input-specific distortion, unlike common noisy information and behavioral expectations, which affect capital and labor symmetrically. Communication frictions alone can generate approximately 40% of observed capital–labor-ratio dispersion, 27% of MRPK dispersion, and 20% of MRPN dispersion. They imply an aggregate TFP loss of about 5% relative to the frictionless allocation. The negative MRPK–MRPN correlation generated by manager-specific private information attenuates the associated TFP loss relative to what the MRPK and MRPN dispersions alone would imply.

The results in this paper highlights the importance of further investigation into within-firm communication frictions. A natural extension of the paper is to study communication across establishments within multi-establishment firms. Applying the framework and forecast-error tests to this setting could help assess whether information segmentation explains the substantial dispersion in marginal revenue products across establishments within the same firm (Kehrig and Vincent (2026), Bloom et al. (2019), Giroud and Mueller (2015)). Another related question is how firms choose their internal information structure. Limited managerial attention may make some information segmentation optimal, so a welfare evaluation in

communication requires accounting for the costs of acquiring and sharing information. Understanding these organizational choices may also help explain the boundaries of firms: as firms expand across establishments and businesses, the benefits of expansion may be offset by increasing difficulties in sharing information and coordinating decisions. These important questions are left for future research.

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A Theory Appendix

A.1 Second-order Approximation of the Profit Function

Denote the frictionless steady state value of variable X_t as $\bar{X}$, and denote the log deviation of X_t from $\bar{X}$ as x_t.

Frictionless Steady State Fix $\bar{A} = 1$. The profit maximization problem without information and adjustment frictions is to choose K and N to maximize

$$\Pi(K, N) = Y^{1/\epsilon} A^{1-1/\epsilon} (K^{\alpha} N^{1-\alpha})^{1-1/\epsilon} - RK - WN$$

FOC:

$$Y^{1/\epsilon} A^{1-1/\epsilon} K^{\alpha(1-1/\epsilon)-1} N^{(1-\alpha)(1-1/\epsilon)} \alpha(1-1/\epsilon) = R$$

$$Y^{1/\epsilon} A^{1-1/\epsilon} K^{\alpha(1-1/\epsilon)} N^{(1-\alpha)(1-1/\epsilon)-1} (1-\alpha)(1-1/\epsilon) = W$$

In the quantitative exercise, $\alpha, \epsilon, \rho, \sigma_u, \delta, \beta$ are calibrated outside the model. Given these parameters, the frictionless steady-state values for rental rate, capital, labor and wage can be expressed analytically. First, rental price $\bar{R}$ is

$$\bar{R} = 1/\gamma - 1 + \delta = \alpha(1-1/\epsilon) \bar{K}^{\alpha(1-1/\epsilon)-1} N^{(1-\alpha)(1-1/\epsilon)}$$

Normalize $\bar{Y} = 1, \bar{A} = 1$, we get

$$\bar{K}^{\alpha(1-1/\epsilon)} \bar{N}^{(1-\alpha)(1-1/\epsilon)} = 1$$

We can use the above two equations, together with FOCs, to back out steady state $\bar{K}, \bar{N}, \bar{W}$:

$$\bar{K} = \frac{\alpha(1-1/\epsilon)}{1/\gamma - 1 + \delta}$$

$$\bar{N} = \left(\frac{1/\gamma - 1 + \delta}{\alpha(1-1/\epsilon)} \right)^{\alpha/(1-\alpha)}$$

$$\bar{W} = \bar{K}^{\alpha} \bar{N}^{-\alpha} (1-\alpha)(1-1/\epsilon)$$

Second-order Approximation We have

$$
\begin{aligned}
\Pi \approx\ & \bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)} + \bar{R}\bar{K}k_t + \bar{W}\bar{N}n_t \\
& + \frac{1}{2}\bar{R}\bar{K}(1-1+\alpha(1-1/\epsilon))k_t^2 + \frac{1}{2}\bar{W}\bar{N}(1-1+(1-\alpha)(1-1/\epsilon))n_t^2 \\
& + \bar{A}^{1-1/\epsilon}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}k_t n_t \\
& - \bar{R}\bar{K} - \bar{R}\bar{K}k_t - \bar{R}\bar{K}\frac{1}{2}k_t^2 \\
& - \bar{W}\bar{N} - \bar{W}\bar{N}n_t - \bar{W}\bar{N}\frac{1}{2}n_t^2 \\
& - \frac{\xi_k}{2}\bar{K}(k_t-k_{t-1})^2 - \frac{\xi_n}{2}\bar{N}(n_t-n_{t-1})^2 \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}(a_t+\frac{1}{2}a_t^2) - \frac{1}{2}\frac{\epsilon-1}{\epsilon^2}\bar{A}^{1-1/\epsilon}\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}a_t^2 \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}\alpha(1-1/\epsilon)\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}k_t a_t \\
& + \frac{\epsilon-1}{\epsilon}\bar{A}^{1-1/\epsilon}(1-\alpha)(1-1/\epsilon)\bar{K}^{\alpha(1-1/\epsilon)}\bar{N}^{(1-\alpha)(1-1/\epsilon)}n_t a_t
\end{aligned}
$$

where the sixth line is an approximation of adjustment cost for capital:

$$
\begin{aligned}
\frac{\xi_k}{2}\left(\frac{K_t-(1-\delta)K_{t-1}}{K_{t-1}}-\delta\right)^2 K_{t-1} &= \frac{\xi_k}{2}\left(\exp(k_t-k_{t-1})-1\right)^2\bar{K}\exp(k_{t-1}) \\
&= \frac{\xi_k\bar{K}}{2}(k_t-k_{t-1})^2(1+k_{t-1}+...) = \frac{\xi_k\bar{K}}{2}(k_t-k_{t-1})^2
\end{aligned}
$$

Define investment $\iota_t \equiv k_t - k_{t-1}$ and hiring $h_t \equiv n_t - n_{t-1}$. Replace $k_t = k_{t-1} + \iota_t$ and $n_t = h_t + n_{t-1}$, we get:

Hence

$$
\begin{aligned}
\pi = & \frac{1}{\epsilon}\bar{Y} + \frac{\epsilon-1}{\epsilon}\bar{Y}a_t + \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y}a_t^2 \\
& + \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1)\left(k_{t-1}^2 + 2k_{t-1}\iota_t + \iota_t^2\right) \\
& + \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1)(n_{t-1}^2 + 2n_{t-1}h_t + h_t^2) \\
& + \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}(k_{t-1}n_{t-1} + \iota_t n_{t-1} + k_{t-1}h_t + \iota_t h_t) \\
& + \frac{\epsilon-1}{\epsilon}\alpha(1-1/\epsilon)\bar{Y}(k_{t-1}a_t + \iota_t a_t) + \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y}(n_{t-1}a_t + h_t a_t) \\
& - \frac{\xi_k}{2}\bar{K}\iota_t^2 - \frac{\xi_n}{2}\bar{N}h_t^2
\end{aligned}
\tag{A1}
$$

Hence the log-quadratic profit function takes the form

$$x_t' P x_t + x_t' Q \iota_t + x_t' R h_t + H_\iota \iota_t^2 + H_{\iota h} \iota_t h_t + H_h h_t^2 \tag{A2}$$

where state vector $x_t = [1, k_{t-1}, n_{t-1}, a_t]$, and the matrices are

$$P = \begin{bmatrix} \frac{1}{\epsilon}\bar{Y} & 0 & 0 & \frac{\epsilon-1}{2\epsilon}\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) & \frac{1}{2}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} & \frac{\epsilon-1}{2\epsilon}\alpha(1-1/\epsilon)\bar{Y} \\ * & * & \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) & \frac{\epsilon-1}{2\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y} \\ * & * & * & \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y} \end{bmatrix}$$

$$Q = \begin{bmatrix} 0 \\ \bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \frac{\epsilon-1}{\epsilon}\alpha(1-1/\epsilon)\bar{Y} \end{bmatrix}$$

$$R = \begin{bmatrix} 0 \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) \\ \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-1/\epsilon)\bar{Y} \end{bmatrix}$$

and $H_\iota = \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) - \frac{1}{2}\xi_k\bar{K}, H_h = \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) - \frac{1}{2}\xi_n\bar{N}, H_{\iota h} = \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}$.

The law of motion for the state vector x_t is given by

$$x_{t+1} \equiv \begin{bmatrix} 1 \\ k_{t-1} + \iota_t \\ n_{t-1} + h_t \\ \rho a_t + \mu_{t+1} \end{bmatrix} = Ax_t + B\iota_t + Ch_t + D\mu_{t+1}$$

where

$$A \equiv \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \rho \end{bmatrix}, B \equiv \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, C \equiv \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, D \equiv \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

A.2 Proof of Lemma 1

Let

$$\Sigma^{-} \equiv \rho^2 \bar{\Sigma} + \sigma_\mu^2.$$

At the beginning of period t, before the current signals arrive, the period-(t-1) actions are revealed at the end of period $t-1$. At this point, the information sets of the two managers are common, which I denote as $\mathcal{I}_{it}^{-}$. Hence their common prior is

$$a_{it} \mid \mathcal{I}_{it}^{-} \sim N\left(\bar{a}_{it}, \Sigma^{-}\right),$$

We first derive the managers' first-order beliefs. Conditional on $\mathcal{I}_{it}^{-}$, the prior and all signal noises are Gaussian and independent. The capital manager observes s_{it}^{p} and s_{it}^{k}. Normal–normal updating therefore gives

$$\begin{aligned} \left(\hat{\Sigma}^{k}\right)^{-1} &= \left(\Sigma^{-}\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2}, \\ \hat{a}_{it}^{k} &= \hat{\Sigma}^{k}\left[\left(\Sigma^{-}\right)^{-1}\bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^{p} + \sigma_{\epsilon,k}^{-2} s_{it}^{k}\right]. \end{aligned}$$

The same calculation for the labor manager yields

$$\begin{aligned} \left(\hat{\Sigma}^{n}\right)^{-1} &= \left(\Sigma^{-}\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,n}^{-2}, \\ \hat{a}_{it}^{n} &= \hat{\Sigma}^{n}\left[\left(\Sigma^{-}\right)^{-1}\bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^{p} + \sigma_{\epsilon,n}^{-2} s_{it}^{n}\right]. \end{aligned}$$

Substituting the definition of Σ^{-} gives equations (13) and (14).

Next consider the capital manager's expectation of the labor manager's posterior. The only term in $\hat{a}_{it}^{n}$ that is not measurable with respect to $\mathcal{I}_{it}^{k}$ is s_{it}^{n}. Since $s_{it}^{n} = a_{it} + \epsilon_{it}^{n}$, with ϵ_{it}^{n} independent of $\mathcal{I}_{it}^{k}$,

$$\mathbb{E}\left[s_{it}^{n} \mid \mathcal{I}_{it}^{k}\right] = \mathbb{E}\left[a_{it} \mid \mathcal{I}_{it}^{k}\right] = \hat{a}_{it}^{k}.$$

Taking the conditional expectation of the labor manager's posterior formula therefore gives

$$\hat{a}_{it}^{(k,n)} \equiv \mathbb{E}[\hat{a}_{it}^{n} | \mathcal{I}_{it}^{k}] = \hat{\Sigma}^{n} \left[\left(\Sigma^{-}\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^{p} + \sigma_{\epsilon,n}^{-2} \hat{a}_{it}^{k} \right],$$

Similarly, the labor manager's belief of the capital manager's posterior is given by

$$\hat{a}_{it}^{(n,k)} \equiv \mathbb{E}[\hat{a}_{it}^{k} | \mathcal{I}_{it}^{n}] = \hat{\Sigma}^{k} \left[\left(\Sigma^{-}\right)^{-1} \bar{a}_{it} + \sigma_{\epsilon,p}^{-2} s_{it}^{p} + \sigma_{\epsilon,k}^{-2} \hat{a}_{it}^{n} \right],$$

The corresponding conditional variances are

$$\hat{\Sigma}^{(k,n)} = \left(\hat{\Sigma}^{n} \sigma_{\epsilon,n}^{-2}\right)^{2} \left(\hat{\Sigma}^{k} + \sigma_{\epsilon,n}^{2}\right),$$
$$\hat{\Sigma}^{(n,k)} = \left(\hat{\Sigma}^{k} \sigma_{\epsilon,k}^{-2}\right)^{2} \left(\hat{\Sigma}^{n} + \sigma_{\epsilon,k}^{2}\right).$$

Thus the second-order beliefs are obtained directly from the first-order posteriors; no recursive covariance state is required.

It remains to establish the common-prior recursion. As the actions in period $t-1$ are revealed, both posterior means in $t-1$ become common knowledge. The posterior formulas above can then be inverted to recover both private signals:

$$s_{i,t-1}^{k} = \sigma_{\epsilon,k}^{2} \left[\left(\hat{\Sigma}^{k}\right)^{-1} \hat{a}_{i,t-1}^{k} - \left(\Sigma^{-}\right)^{-1} \bar{a}_{i,t-1} - \sigma_{\epsilon,p}^{-2} s_{i,t-1}^{p} \right],$$
$$s_{i,t-1}^{n} = \sigma_{\epsilon,n}^{2} \left[\left(\hat{\Sigma}^{n}\right)^{-1} \hat{a}_{i,t-1}^{n} - \left(\Sigma^{-}\right)^{-1} \bar{a}_{i,t-1} - \sigma_{\epsilon,p}^{-2} s_{i,t-1}^{p} \right].$$

After pooling the public and private signals, the common end-of-period posterior mean is

$$\tilde{a}_{i,t-1} = \bar{\Sigma} \left[\left(\Sigma^{-}\right)^{-1} \bar{a}_{i,t-1} + \sigma_{\epsilon,p}^{-2} s_{i,t-1}^{p} + \sigma_{\epsilon,k}^{-2} s_{i,t-1}^{k} + \sigma_{\epsilon,n}^{-2} s_{i,t-1}^{n} \right],$$

and its variance is

$$\bar{\Sigma} = \left[\left(\Sigma^{-}\right)^{-1} + \sigma_{\epsilon,p}^{-2} + \sigma_{\epsilon,k}^{-2} + \sigma_{\epsilon,n}^{-2} \right]^{-1}.$$

Because $a_{it} = \rho a_{i,t-1} + \mu_{it}$, the next common prior has mean $\bar{a}_{it} = \rho \tilde{a}_{i,t-1}$ and variance $\rho^{2}\bar{\Sigma} + \sigma_{\mu}^{2} = \Sigma^{-}$. This gives equation (15), and the fixed-point equation for $\bar{\Sigma}$ stated in the lemma.

A.3 Proof of Lemma 2

To save some notations, I drop the firm subscript i in the proof. Denote the common-information part of the belief as ζ_t:

$$\zeta_t \equiv (\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}\bar{a}_t + \sigma_{\epsilon,p}^{-2}s_t^p$$

Guess the decision rules of capital and labor managers follow

$$\iota_t = F_\iota y_t^k, h_t = F_h y_t^n$$

where $y_t^k \equiv [1, k_{t-1}, n_{t-1}, \zeta_t, s_t^k]'$ and $y_t^n \equiv [1, k_{t-1}, n_{t-1}, \zeta_t, s_t^n]'$. The value function takes the form

$$V^k(y_t^k) = (y_t^k)'P_k y_t^k + Q_k, V^n(y_t^n) = (y_t^n)'P_n y_t^n + Q_n$$

Our goal is to jointly solve $F_\iota, F_h, P_k, Q_k, P_n, Q_n$ so that they are consistent with the MPBE value functions (9) and (10).

First, let's solve for P_k, Q_k, F_ι from the Bellman equation for V^k

$$V^k(y_t^k) = \max_{\iota_t} \mathbb{E}\left[x_t'Px_t + x_t'Q\iota_t + x_t'Rh_t + H_\iota\iota_t^2 + H_{\iota h}\iota_t h_t + H_h h_t^2 + \beta V^k(y_{t+1}^k)\middle|\mathcal{I}_t^k\right] \quad \text{(A3)}$$

Since

$$\hat{x}_t^k \equiv \mathbb{E}[x_t|\mathcal{I}_t^k] = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \hat{\Sigma}^k & \hat{\Sigma}^k\sigma_{\epsilon,k}^{-2} \end{bmatrix}}_{\equiv \mathsf{M}_k} y_t^k \quad \text{(A4)}$$

and

$$\hat{y}_t^{(k,n)} \equiv \mathbb{E}[y_t^n|\mathcal{I}_t^k] = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \hat{\Sigma}^k & \hat{\Sigma}^k\sigma_{\epsilon,k}^{-2} \end{bmatrix}}_{\equiv \mathsf{T}_n} y_t^k \quad \text{(A5)}$$

Each term on the RHS can be expressed in terms of y_t^k.

$$\mathbb{E}\left[x_t'Px_t|\mathcal{I}_t^k\right] = (\hat{x}_t^k)'P\hat{x}_t^k + tr(P\hat{\Sigma}_x^k) = (y_t^k)'\mathsf{M}_k'P\mathsf{M}_k y_t^k + P_{aa}\hat{\Sigma}^k \quad \text{(A6)}$$

$$\mathbb{E}\left[x_t' Q \iota_t | \mathcal{I}_t^k\right] = (\hat{x}_t^k)' Q \iota_t = (y_t^k)' \mathsf{M}_k' Q \iota_t \tag{A7}$$

$$\mathbb{E}\left[x_t' R h_t | \mathcal{I}_t^k\right] = \mathbb{E}[x_t' R F_h y_t^n | \mathcal{I}_t^k] = (y_t^k)' \mathsf{M}_k' R F_h \mathsf{T}_n y_t^k + tr(R F_h \Sigma_{y^n,x}) = (y_t^k)' \mathsf{M}_k' R F_h \mathsf{T}_n y_t^k + R_a F_h^s \hat{\Sigma}^k \tag{A8}$$

$$\mathbb{E}\left[H_{\iota h} \iota_t h_t | \mathcal{I}_t^k\right] = H_{\iota h} \iota_t F_h \mathsf{T}_n y_t^k \tag{A9}$$

$$\mathbb{E}[H_h h_t^2 | \mathcal{I}_t^k] = (y_t^k)' \mathsf{T}_n' F_h' H_h F_h \mathsf{T}_n y_t^k + H_h (F_h^s)^2 (\hat{\Sigma}^k + \sigma_{\epsilon,n}^2) \tag{A10}$$

where P_{aa} denotes the entry in the P matrix for productivity a and F_h^s denotes the entry in F_h vector for the private signal s.

For the last term $\mathbb{E}[V^k(y_{t+1}^k) | \mathcal{I}_t^k] = \mathbb{E}[(y_{t+1}^k)' P_k y_{t+1}^k + Q_k | \mathcal{I}_t^k]$, we need to compute $\mathbb{E}[y_{t+1}^k | \mathcal{I}_t^k]$:

$$\mathbb{E}[k_t | \mathcal{I}_t^k] = k_{t-1} + \iota_t, \mathbb{E}[n_t | \mathcal{I}_t^k] = n_{t-1} + \mathbb{E}[h_t | \mathcal{I}_t^k] = n_{t-1} + F_h \mathsf{T}_n y_t^k \tag{A11}$$

$$\mathbb{E}[\zeta_{t+1} | \mathcal{I}_t^k] = ((\rho^2 \bar{\Sigma} + \sigma_\mu^2)^{-1} + \sigma_{\epsilon,p}^{-2}) \rho \hat{\Sigma}^k (\zeta_t + \sigma_{\epsilon,k}^{-2} s_t^k) \tag{A12}$$

$$\mathbb{E}[s_{t+1}^k | \mathcal{I}_t^k] = \rho \hat{a}_t^k = \rho \hat{\Sigma}^k (\zeta_t + \sigma_{\epsilon,k}^{-2} s_t^k) \tag{A13}$$

Hence

$$\mathbb{E}[y_{t+1}^k | \mathcal{I}_t^k] = \mathsf{A}_k y_t^k + \mathsf{B} \iota_t \tag{A14}$$

where

$$\mathsf{A}_k \equiv \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & ((\rho^2 \bar{\Sigma} + \sigma_\mu^2)^{-1} + \sigma_{\epsilon,p}^{-2}) \rho \hat{\Sigma}^k & ((\rho^2 \bar{\Sigma} + \sigma_\mu^2)^{-1} + \sigma_{\epsilon,p}^{-2}) \rho \hat{\Sigma}^k \sigma_{\epsilon,k}^{-2} \\ 0 & 0 & 0 & \rho \hat{\Sigma}^k & \rho \hat{\Sigma}^k \sigma_{\epsilon,k}^{-2} \end{bmatrix} + \mathsf{C} F_h \mathsf{T}_n$$

and $\mathsf{B} \equiv [0,1,0,0,0]', \mathsf{C} \equiv [0,0,1,0,0]'$.

Hence the optimization problem on the RHS of (A3) becomes

$$\max_{\iota_t} (y_t^k)' \mathsf{M}_k' P \mathsf{M}_k y_t^k + (y_t^k)' \mathsf{M}_k' Q \iota_t + (y_t^k)' \mathsf{M}_k' R F_h \mathsf{T}_n y_t^k + \cdots + H_\iota \iota_t^2 + \iota_t H_{\iota h} F_h \mathsf{T}_n y_t^k + H_h (y_t^k)' \mathsf{T}_n' F_h' F_h \mathsf{T}_n y_t^k$$

$$+\cdots+\beta\left[(y_t^k)'\mathsf{A}_k'P_k\mathsf{A}_k y_t^k+\iota_t\mathsf{B}'(P_k+P_k')\mathsf{A}_k y_t^k+\iota_t^2\mathsf{B}'P_k\mathsf{B}\right] \tag{A15}$$

Take FOC on ι_t:

$$(y_t^k)'\mathsf{M}_k'Q+2H_\iota\iota_t+H_{\iota h}F_h\mathsf{T}_n y_t^k+\beta(\mathsf{B}'(P_k+P_k')\mathsf{A}_k y_t^k+2\mathsf{B}'P_k\mathsf{B}\iota_t)=0$$

and hence

$$\iota_t=-\frac{1}{2}(H_\iota+\beta\mathsf{B}'P_k\mathsf{B})^{-1}(Q'\mathsf{M}_k+H_{\iota h}F_h\mathsf{T}_n+\beta\mathsf{B}'(P_k+P_k')\mathsf{A}_k)y_t^k \tag{A16}$$

Hence, given P_k, the policy matrix F_ι is given by

$$F_\iota=-\frac{1}{2}(H_\iota+\beta\mathsf{B}'P_k\mathsf{B})^{-1}(Q'\mathsf{M}_k+H_{\iota h}F_h\mathsf{T}_n+\beta\mathsf{B}'(P_k+P_k')\mathsf{A}_k) \tag{A17}$$

Plug the policy function (A16) into the (A15), we get

$$\begin{aligned}&(y_t^k)'P_k y_t^k\\&=(y_t^k)'\Big[\mathsf{M}_k'P\mathsf{M}_k+\mathsf{M}_k'QF_\iota+\mathsf{M}_k'RF_h\mathsf{T}_n+F_\iota'H_\iota F_\iota+F_\iota'H_{\iota h}F_h\mathsf{T}_n+H_h\mathsf{T}_n'F_h'F_h\mathsf{T}_n\\&+\beta\mathsf{A}_k'P_k\mathsf{A}_k+\beta F_\iota'\mathsf{B}'(P_k+P_k')\mathsf{A}_k+\beta F_\iota'\mathsf{B}'P_k\mathsf{B}F_\iota\Big]y_t^k\end{aligned} \tag{A18}$$

and hence the Riccati equation for P_k is given by

$$P_k=\mathsf{M}_k'P\mathsf{M}_k+\mathsf{M}_k'RF_h\mathsf{T}_n+H_h\mathsf{T}_n'F_h'F_h\mathsf{T}_n+\beta\mathsf{A}_k'P_k\mathsf{A}_k+\frac{1}{2}F_\iota'(Q'\mathsf{M}_k+H_{\iota h}F_h\mathsf{T}_n+\beta\mathsf{B}'(P_k+P_k')\mathsf{A}_k) \tag{A19}$$

Lastly, the constant Q_k satisfies

$$Q_k=P_{aa}\hat{\Sigma}^k+R_aF_h^s\hat{\Sigma}^k+H_h(F_h^s)^2(\hat{\Sigma}^k+\sigma_{\epsilon,n}^2)+\beta Q_k+\beta\cdot tr(P_k\hat{\Sigma}_{t+1|t}^k) \tag{A20}$$

where $\hat{\Sigma}_{t+1|t}^k\equiv\mathbb{E}[(y_{t+1}-\mathbb{E}[y_{t+1}|\mathcal{I}_t^k])^2|\mathcal{I}_t^k]$.

Similarly, we can solve for P_n and F_n from the Bellman equation for V^n:

$$V^n(y_t^n)=\max_{h_t}\mathbb{E}\left[x_t'Px_t+x_t'Q\iota_t+x_t'Rh_t+H_\iota\iota_t^2+H_{\iota h}\iota_t h_t+H_h h_t^2+\beta V^n(y_{t+1}^n)\big|\mathcal{I}_t^n\right] \tag{A21}$$

and we have

$$\hat{x}_t^n \equiv \mathbb{E}[x_t|\mathcal{I}_t^n] = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & \hat{\Sigma}^n & \hat{\Sigma}^n\sigma_{\epsilon,n}^{-2} \end{bmatrix}}_{\equiv \mathsf{M}_n} y_t^n \tag{A22}$$

$$\hat{y}_t^{(n,k)} \equiv \mathbb{E}[y_t^k|\mathcal{I}_t^n] = \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \hat{\Sigma}^n & \hat{\Sigma}^n\sigma_{\epsilon,n}^{-2} \end{bmatrix}}_{\equiv \mathsf{T}_k} y_t^n \tag{A23}$$

$$\mathbb{E}\left[x_t' P x_t|\mathcal{I}_t^n\right] = (\hat{x}_t^n)' P \hat{x}_t^n + tr(P\hat{\Sigma}_x^n) = (y_t^n)'\mathsf{M}_n' P \mathsf{M}_n y_t^n + P_{aa}\hat{\Sigma}^n \tag{A24}$$

$$\mathbb{E}\left[x_t' Q \iota_t|\mathcal{I}_t^n\right] = (\hat{x}_t^n)' Q \mathbb{E}[\iota_t|\mathcal{I}_t^n] + tr(Q F_\iota \Sigma_{y^k,x}) = (y_t^n)'\mathsf{M}_n' Q F_\iota \mathsf{T}_k y_t^n + Q_a F_\iota \hat{\Sigma}^n \tag{A25}$$

$$\mathbb{E}\left[x_t' R h_t|\mathcal{I}_t^n\right] = (y_t^n)'\mathsf{M}_n' R h_t \tag{A26}$$

$$\mathbb{E}\left[H_{\iota h}\iota_t h_t|\mathcal{I}_t^n\right] = (y_t^n)'\mathsf{T}_k' F_\iota' H_{\iota h} h_t \tag{A27}$$

$$\mathbb{E}[H_h h_t^2|\mathcal{I}_t^n] = H_h h_t^2 \tag{A28}$$

For the last term $\mathbb{E}[V^n(y_{t+1}^n)|\mathcal{I}_t^n] = \mathbb{E}[(y_{t+1}^n)' P_n y_{t+1}^n + Q_n|\mathcal{I}_t^n]$, we need to compute $\mathbb{E}[y_{t+1}^n|\mathcal{I}_t^n]$:

$$\mathbb{E}[k_t|\mathcal{I}_t^n] = k_{t-1} + \mathbb{E}[\iota_t|\mathcal{I}_t^n] = k_{t-1} + F_\iota \mathsf{T}_k y_t^n, \mathbb{E}[n_t|\mathcal{I}_t^n] = n_{t-1} + h_t \tag{A29}$$

$$\mathbb{E}[\zeta_{t+1}|\mathcal{I}_t^n] = ((\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1} + \sigma_{\epsilon,p}^{-2})\rho\hat{\Sigma}^n(\zeta_t + \sigma_{\epsilon,n}^{-2} s_t^n) \tag{A30}$$

$$\mathbb{E}[s_{t+1}^n|\mathcal{I}_t^n] = \rho\hat{a}_t^n = \rho\hat{\Sigma}^n(\zeta_t + \sigma_{\epsilon,n}^{-2} s_t^n) \tag{A31}$$

Hence

$$\mathbb{E}[y_{t+1}^n|\mathcal{I}_t^n] = \mathsf{A}_n y_t^n + \mathsf{C} h_t \tag{A32}$$

where

$$\mathsf{A}_n \equiv \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & ((\rho^2\bar{\Sigma}+\sigma_\mu^2)^{-1}+\sigma_{\epsilon,p}^{-2})\rho\hat{\Sigma}^n & ((\rho^2\bar{\Sigma}+\sigma_\mu^2)^{-1}+\sigma_{\epsilon,p}^{-2})\rho\hat{\Sigma}^n\sigma_{\epsilon,n}^{-2} \\ 0 & 0 & 0 & \rho\hat{\Sigma}^n & \rho\hat{\Sigma}^n\sigma_{\epsilon,n}^{-2} \end{bmatrix} + \mathsf{B}F_\iota\mathsf{T}_k$$

The optimization problem on the RHS of (A21) becomes

$$\max_{h_t}(y_t^n)'\mathsf{M}_n'P\mathsf{M}_n y_t^n + (y_t^n)'\mathsf{M}_n'QF_\iota\mathsf{T}_k y_t^n + (y_t^n)'\mathsf{M}_n'Rh_t + \cdots + H_h h_t^2 + h_t H_{\iota h}F_\iota\mathsf{T}_k y_t^n + H_\iota(y_t^n)'\mathsf{T}_k'F_\iota'F_\iota\mathsf{T}_k y_t^n$$
$$+ \cdots + \beta\left[(y_t^n)'\mathsf{A}_n'P_n\mathsf{A}_n y_t^n + 2h_t\mathsf{C}'P_n\mathsf{A}_n y_t^n + h_t^2\mathsf{C}'P_n\mathsf{C}\right] \tag{A33}$$

Take FOC on h_t:

$$(y_t^n)'\mathsf{M}_n'R + 2H_h h_t + H_{\iota h}F_\iota\mathsf{T}_k y_t^n + \beta(\mathsf{C}'(P_n+P_n')\mathsf{A}_n y_t^n + 2\mathsf{C}'P_n\mathsf{C}h_t) = 0$$

and hence

$$h_t = -\frac{1}{2}(H_h+\beta\mathsf{C}'P_n\mathsf{C})^{-1}(R'\mathsf{M}_n + H_{\iota h}F_\iota\mathsf{T}_k + \beta\mathsf{C}'(P_n+P_n')\mathsf{A}_n)y_t^n \tag{A34}$$

Hence, given P_n, the policy matrix F_h is given by

$$F_h = -\frac{1}{2}(H_h+\beta\mathsf{C}'P_n\mathsf{C})^{-1}(R'\mathsf{M}_n + H_{\iota h}F_\iota\mathsf{T}_k + \beta\mathsf{C}'(P_n+P_n')\mathsf{A}_n) \tag{A35}$$

Plug the policy function (A34) into the (A33), we get

$$\begin{aligned}&(y_t^n)'P_n y_t^n \\ &= (y_t^n)'\Big[\mathsf{M}_n'P\mathsf{M}_n + \mathsf{M}_n'QF_\iota\mathsf{T}_k + \mathsf{M}_n'RF_h + \mathsf{T}_k'F_\iota'H_\iota F_\iota\mathsf{T}_k + \mathsf{T}_k'F_\iota'H_{\iota h}F_h + F_h'H_hF_h \\ &+ \beta\mathsf{A}_n'P_n\mathsf{A}_n + \beta F_h'\mathsf{C}'(P_n+P_n')\mathsf{A}_n + \beta F_h'\mathsf{C}'P_n\mathsf{C}F_h\Big]y_t^n\end{aligned} \tag{A36}$$

and hence the Riccati equation for P_n is given by

$$P_n = \mathsf{M}_n'P\mathsf{M}_n + \mathsf{M}_n'QF_\iota\mathsf{T}_k + \mathsf{T}_k'F_\iota'H_\iota F_\iota\mathsf{T}_k + \beta\mathsf{A}_n'P_n\mathsf{A}_n + \frac{1}{2}F_h'(R'\mathsf{M}_n + H_{\iota h}F_\iota\mathsf{T}_k + \beta\mathsf{C}'(P_n+P_n')\mathsf{A}_n) \tag{A37}$$

Lastly, the coefficient Q_n is defined as

$$Q_n = P_{aa}\hat{\Sigma}^n + Q_a F_\iota^s \hat{\Sigma}^n + H_\iota (F_\iota^s)^2 (\hat{\Sigma}^n + \sigma^2_{\epsilon,k}) + \beta Q_n + \beta \cdot tr(P_n \hat{\Sigma}^n_{t+1|t}) \tag{A38}$$

where $\hat{\Sigma}^n_{t+1|t} \equiv \mathbb{E}[(y_{t+1} - \mathbb{E}[y_{t+1}|\mathcal{I}^n_t])^2|\mathcal{I}^n_t]$.

A.4 Computation of the Stationary Distribution and Model Moments

For a candidate parameter vector Θ, the linear-Gaussian structure of the model allows the population moments to be computed without simulating a finite panel. The computation follows the steps below:

1. For a given Θ, solve the fixed point for $\bar{\Sigma}$ and the posterior variances $\hat{\Sigma}^k, \hat{\Sigma}^n$ in Lemma 1, and the policy vectors F_ι, F_h and value matrices P_k, P_n in Lemma 2, with $\hat{\rho}$ in place of ρ wherever the persistence enters.

2. Define the state vector

   $$z_{it} \equiv [k_{it},\ n_{it},\ \zeta_{it},\ s^k_{it},\ s^n_{it},\ a_{it}]'$$

   and the innovation vector $u_{it} \equiv (\epsilon^p_{it}, \epsilon^k_{it}, \epsilon^n_{it}, \mu_{it})'$. The law of motion for the state vector can be written in the form of

   $$\mathcal{B} z_{it} = \mathcal{C} z_{i,t-1} + \mathcal{D} u_{it}$$

   where matrices $\mathcal{B}, \mathcal{C}, \mathcal{D}$ are

$$\mathcal{B} \equiv \begin{bmatrix} 1 & 0 & -F_\iota^\zeta & -F_\iota^s & 0 & 0 \\ 0 & 1 & -F_h^\zeta & 0 & -F_h^s & 0 \\ 0 & 0 & 1 & 0 & 0 & -\sigma^{-2}_{\epsilon,p} \\ 0 & 0 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \mathcal{D} \equiv \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sigma^{-2}_{\epsilon,p} & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathcal{C} \equiv \begin{bmatrix} 1+F_\iota^k & F_\iota^n & 0 & 0 & 0 & 0 \\ F_h^k & 1+F_h^n & 0 & 0 & 0 & 0 \\ 0 & 0 & (\Sigma^-)^{-1}\rho\bar{\Sigma} & (\Sigma^-)^{-1}\rho\bar{\Sigma}\sigma_{\epsilon,k}^{-2} & (\Sigma^-)^{-1}\rho\bar{\Sigma}\sigma_{\epsilon,n}^{-2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho \end{bmatrix}$$

Hence the law of motion can be written as

$$z_{it} = \underbrace{\mathcal{B}^{-1}\mathcal{C}}_{\equiv\tilde{\mathcal{C}}(\Theta)} z_{i,t-1} + \underbrace{\mathcal{B}^{-1}\mathcal{D}}_{\tilde{\mathcal{D}}(\tilde{\Theta})} u_{it} \tag{A39}$$

3. Let $\Sigma_u \equiv \text{var}(u_{it})$. When $\tilde{\mathcal{C}}(\Theta)$ is stable, the stationary covariance solves the Lyapunov equation

$$\Omega(\Theta) = \tilde{\mathcal{C}}(\Theta)\Omega(\Theta)\tilde{\mathcal{C}}(\Theta)' + \tilde{\mathcal{D}}(\Theta)\Sigma_u\tilde{\mathcal{D}}(\Theta)', \tag{A40}$$

Matrix $\Omega(\Theta)$ characterizes the stationary equilibrium of the state vector z_{it}.

In the quantitative analysis of Section 5, all the targeted moments can be expressed analytically as functions of $\Omega(\Theta)$. The differences are

- The inclusion of over-extrapolation $\hat{\rho}$ changes the law of motion for beliefs in Lemma 1, and the ρ's in the third row of matrix $\mathcal{C}$ should be replaced by $\hat{\rho}$, without changing the last row of $\mathcal{C}$;
- The inclusion of time-to-build and correlated distortion in the full model changes the constant matrices from log-quadratic approximation in (11);
- The inclusion of transitory uncorrelated distortions η^k and η^y should be augmented to the innovation vector u_{it};
- The inclusion of firm characteristics δ^k and δ^y adds a constant term δ_i to the right hand side of the law of motion (A39) for z_{it}.

A.5 Proof of Lemma 3

First, since

$$\iota_t = F_\iota^0 + F_\iota^k k_{t-1} + F_\iota^n n_{t-1} + F_\iota^\zeta \zeta_t + F_\iota^s s_t^k$$

and $\zeta_t = (\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}\bar{a}_t + \sigma_{\epsilon,p}^{-2}s_t^p$, we have

$$\iota_t = \underbrace{F_\iota^0}_{\equiv F_0} + \underbrace{F_\iota^k}_{\equiv F_1} k_{t-1} + \underbrace{F_\iota^n}_{\equiv F_2} n_{t-1} + \underbrace{F_\iota^\zeta(\rho^2\bar{\Sigma} + \sigma_\mu^2)^{-1}}_{\equiv F_3} \bar{a}_t + \underbrace{F_\iota^\zeta\sigma_{\epsilon,p}^{-2}}_{\equiv F_5} s_t^p + \underbrace{F_\iota^s}_{\equiv F_4} s_t^k$$

Let F_h^s denote the entry of the labor manager's policy matrix F_h on the private signal s_t^n. Under communication friction, $F_h^s \neq 0$. Lemma 2 implies that

$$h_{t-1} - \mathbb{E}_{t-1}^k[h_{t-1}] = F_h^s\left(s_{t-1}^n - \hat{a}_{t-1}^k\right) \tag{A41}$$

Using $(\hat{\Sigma}^k)^{-1} = \bar{\Sigma}^{-1} - \sigma_{\epsilon,n}^{-2}$, equation (15) can therefore be written as

$$\begin{aligned}\bar{a}_t &= \rho\bar{\Sigma}\left[(\hat{\Sigma}^k)^{-1}\hat{a}_{t-1}^k + \sigma_{\epsilon,n}^{-2}s_{t-1}^n\right] \\ &= \rho\left[\hat{a}_{t-1}^k + \frac{\bar{\Sigma}\sigma_{\epsilon,n}^{-2}}{F_h^s}\left(h_{t-1} - \mathbb{E}_{t-1}^k[h_{t-1}]\right)\right].\end{aligned} \tag{A42}$$

Hence

$$\bar{a}_t - \mathbb{E}[\bar{a}_t|\mathcal{I}_{t-1}^k] = \underbrace{\frac{\rho\bar{\Sigma}\sigma_{\epsilon,n}^{-2}}{F_h^s}}_{\equiv J}(h_{t-1} - \mathbb{E}_{t-1}^k[h_{t-1}]) \tag{A43}$$

and therefore

$$\begin{aligned}\iota_t - \mathbb{E}_{t-1}^k[\iota_t] &= F_2\left(n_{t-1} - \mathbb{E}_{t-1}^k[n_{t-1}]\right) + F_3(\bar{a}_t - \mathbb{E}_{t-1}^k[\bar{a}_t]) + F_4u_t^k + F_5u_t^p \\ &= F_2(h_{t-1} - \mathbb{E}_{t-1}^k[h_{t-1}]) + F_3J\left(h_{t-1} - \mathbb{E}_{t-1}^k[h_{t-1}]\right) + F_4u_t^k + F_5u_t^p \\ &= (F_2 + F_3J)\left(h_{t-1} - \mathbb{E}_{t-1}^k[h_{t-1}]\right) + F_4u_t^k + F_5u_t^p\end{aligned} \tag{A44}$$

Under common information, $F_h^s = 0$, and hence $h_{t-1} = \mathbb{E}_{t-1}^k[h_{t-1}]$. The investment forecast error degenerates to $F_5u_t^p$.

A.6 Proof of Proposition 1

From Proposition **??**, we get

$$\operatorname{Cov}\left(\iota_t - \mathbb{E}^k_{t-1}[\iota_t], h_{t-1}\right) = \beta_1 \operatorname{Cov}(a_{t-1} - \mathbb{E}^k_{t-1}[a_{t-1}], h_{t-1}) + \beta_2 \operatorname{Cov}(h_{t-1} - \mathbb{E}^k_{t-1}[h_{t-1}], h_{t-1}). \tag{A45}$$

We have

$$\operatorname{Cov}(h_{t-1} - \mathbb{E}^k_{t-1}[h_{t-1}], h_{t-1}) = \operatorname{Var}(h_{t-1} - \mathbb{E}^k_{t-1}[h_{t-1}]) > 0, \tag{A46}$$

and

$$\begin{aligned}
&\operatorname{Cov}(a_{t-1} - \mathbb{E}^k_{t-1}[a_{t-1}], h_{t-1}) \\
&= \operatorname{Cov}(a_{t-1} - \mathbb{E}^k_{t-1}[a_{t-1}], h_{t-1} - \mathbb{E}^k_{t-1}[h_{t-1}]) \\
&= F^s_h \operatorname{Cov}\left(a_{t-1} - \mathbb{E}^k_{t-1}[a_{t-1}], s^n_{t-1} - \hat{E}^k_{t-1}[a_{t-1}]\right) \\
&= F^s_h \hat{\Sigma}^k \geq 0.
\end{aligned} \tag{A47}$$

And the coefficients β_1 and β_2 are both positive:

$$\begin{aligned}
\beta_1 &= \left[F^\zeta_\iota \sigma^{-2}_{\epsilon,p} + F^s_\iota\right] \rho > 0, \\
\beta_2 &= F^n_\iota + F^\zeta_\iota (\Sigma^-)^{-1} \frac{\rho \bar{\Sigma} \sigma^{-2}_{\epsilon,n}}{F^s_h} > 0.
\end{aligned}$$

Substituting these signs together with (A46) and (A47) into (A45) gives

$$\operatorname{Cov}\left(\iota_t - \mathbb{E}^k_{t-1}[\iota_t], h_{t-1}\right) > 0.$$

Since $\operatorname{Var}(h_{t-1}) > 0$, dividing by that variance proves $\beta_h > 0$.

A.7 Proof of Proposition 2

The idea of the proof is to express investment, hiring and investment forecast errors by their moving-average representations and compute the theoretical predictability directly from these MA representations.

Filtering and the policy conditions. Let $\tau_j = \sigma_{\epsilon,j}^{-2} \quad (j = p, k, n)$ be the precision of signals, and $\tau \equiv \tau_p + \tau_k + \tau_n$. Define

$$G = \frac{\tau}{(\Sigma_{\hat{\rho}}^-)^{-1} + \tau}, \qquad \phi = (1 - G)\hat{\rho}, \qquad \omega = \frac{\tau_n}{(\Sigma_{\hat{\rho}}^-)^{-1} + \tau}$$

thus $0 < \phi < r < 1$ and $0 \leq \omega \leq G < 1$. Define a pooled signal $\bar{s}_t$ as

$$\bar{s}_t = \frac{\tau_p s_t^p + \tau_k s_t^k + \tau_n s_t^n}{\tau} = a_t + \bar{\epsilon}_t \tag{A48}$$

where the noise $\bar{\epsilon}_t$ is

$$\bar{\epsilon}_t = \frac{\tau_p \epsilon_t^p + \tau_k \epsilon_t^k + \tau_n \epsilon_t^n}{\tau}, \qquad \mathrm{Var}(\bar{\epsilon}_t) = \tau^{-1} \tag{A49}$$

Let $\tilde{a}_t$ denote the subjective posterior after disclosure of both current private signals. The filter and common statistic satisfy

$$\tilde{a}_t = \phi \tilde{a}_{t-1} + G\bar{s}_t, \qquad \bar{a}_t^{(\hat{\rho})} = \hat{\rho}\tilde{a}_{t-1}, \qquad \zeta_t = (\Sigma_{\hat{\rho}}^-)^{-1}\hat{\rho}\tilde{a}_{t-1} + \tau_p s_t^p. \tag{A50}$$

Since $\bar{\Sigma}_{\hat{\rho}} = \tau^{-1}G$ and $\Sigma_{\hat{\rho}}^- = \tau^{-1}G/(1-G)$, the perceived variance recursion $\Sigma_{\hat{\rho}}^- = \hat{\rho}^2\bar{\Sigma}_{\hat{\rho}} + \sigma_\mu^2$ implies

$$\sigma_\mu^2(1 - G) = \tau^{-1}G(1 - \hat{\rho}\phi). \tag{A51}$$

Write $X_t = (k_t, n_t)'$ and define

$$L = \begin{pmatrix} 1 + F_\iota^k & F_\iota^n \\ F_h^k & 1 + F_h^n \end{pmatrix}, \qquad t = \begin{pmatrix} F_\iota^s \\ F_h^s \end{pmatrix}, \qquad z = \begin{pmatrix} F_\iota^\zeta \\ F_h^\zeta \end{pmatrix},$$

$$v = \tau_p z + t, \qquad d = \hat{\rho}(\Sigma_{\hat{\rho}}^-)^{-1}z, \qquad f = ((\Sigma_{\hat{\rho}}^-)^{-1} + \tau_p)z + t, \qquad J = v_\iota. \tag{A52}$$

An MPBE under the current setup necessarily satisfies

$$L \geq 0, \qquad \rho(L) < 1, \qquad z \geq 0, \qquad t \geq 0, \qquad J > 0. \tag{A53}$$

The moving-average representation of decisions. Substituting the signals into the stock policies gives

$$X_t = LX_{t-1} + d\tilde{a}_{t-1} + va_t + \varepsilon_t^X, \qquad \varepsilon_t^X = \begin{pmatrix} \tau_p F_\iota^\zeta \epsilon_t^p + t_\iota \epsilon_t^k \\ \tau_p F_h^\zeta \epsilon_t^p + t_h \epsilon_t^n \end{pmatrix}. \tag{A54}$$

These are combinations of the original signal noises, not additional shocks. Define

$$w_t = \varepsilon_t^X - v\bar{\epsilon}_t, \qquad \eta_{n,t} = \epsilon_t^n - \bar{\epsilon}_t.$$

so that (A54) can be rewritten as

$$X_t = LX_{t-1} + d\tilde{a}_{t-1} + v\bar{s}_t + w_t \tag{A55}$$

Precision weighting and independence of the primitive noises imply

$$\begin{aligned} &\mathrm{Cov}(\varepsilon_t^X, \bar{\epsilon}_t) = \tau^{-1}v, && \mathrm{Cov}(w_t, \bar{\epsilon}_t) = 0, && \mathrm{Cov}(\eta_{n,t}, \bar{\epsilon}_t) = 0, \\ &\mathrm{Cov}(\eta_{n,t}, w_{\iota,t}) = -\tau^{-1}J, && \mathrm{Cov}(\eta_{n,t}, w_{h,t}) = \frac{t_h}{\tau_n} - \tau^{-1}v_h. \end{aligned} \tag{A56}$$

Equations (A55) give the stationary MA representation

$$X_t = \sum_{\ell=0}^{\infty} Q_\ell \bar{s}_{t-\ell} + \sum_{\ell=0}^{\infty} L^\ell w_{t-\ell} \tag{A57}$$

where

$$Q_0 = v, \qquad Q_\ell = L^\ell v + G\sum_{j=0}^{\ell-1} L^{\ell-1-j} d\,\phi^j \quad (\ell \geq 1). \tag{A58}$$

All Q_ℓ are componentwise nonnegative by (A52). Let $e_\iota = (1,0)'$, $e_h = (0,1)'$, and $P_\ell = e_\iota' Q_\ell$, $R_\ell = e_h' Q_\ell$. Since investment and hiring are changes in stocks,

$$\iota_t = \sum_{\ell=0}^{\infty} P_\ell \Delta\bar{s}_{t-\ell} + \sum_{\ell=0}^{\infty} e_\iota' L^\ell \Delta w_{t-\ell}, \tag{A59}$$

$$h_t = \sum_{\ell=0}^{\infty} R_\ell \Delta\bar{s}_{t-\ell} + \sum_{\ell=0}^{\infty} e_h' L^\ell \Delta w_{t-\ell}. \tag{A60}$$

In particular, $P_0 = J$ and $P_1 = (Lv)_\iota + Gd_\iota$.

The investment forecast error. Let $m_t = \mathbb{F}_t^k[a_t]$. Pooling the other manager's signal gives $\tilde{a}_t = (1-\omega)m_t + \omega s_t^n$, and hence $\mathbb{F}_t^k[\tilde{a}_t] = m_t$. Define

$$\xi_t = \bar{s}_t - \hat{\rho}\tilde{a}_{t-1}, \qquad H = L_{12}t_h + \omega d_\iota, \qquad K = \frac{H + \hat{\rho}\omega J}{1-\omega} = \frac{L_{12}t_h + \hat{\rho}\omega f_\iota}{1-\omega}, \qquad B = (1-G)K. \tag{A61}$$

The private-signal surprise in the current labor stock and the next-period signal surprises imply

$$\begin{aligned} e_{t+1} &\equiv \iota_{t+1} - \mathbb{F}_t^k[\iota_{t+1}] \\ &= H(s_t^n - m_t) + J(a_{t+1} - \hat{\rho} m_t) + \varepsilon_{\iota,t+1}^X \\ &= J(a_{t+1} - \hat{\rho}\tilde{a}_t) + K(s_t^n - \tilde{a}_t) + \varepsilon_{\iota,t+1}^X. \end{aligned}$$

Using $s_t^n - \tilde{a}_t = (1-G)\xi_t + \eta_{n,t}$ yields the complete error representation

$$e_{t+1} = J\xi_{t+1} + B\xi_t + K\eta_{n,t} + w_{\iota,t+1}. \tag{A62}$$

The last term contains fresh noises and is orthogonal to date-t decisions.

A scalar covariance calculation under the actual law. The pooled innovation has the MA representation

$$\xi_t = \sum_{j=0}^{\infty} b_j \bar{s}_{t-j}, \qquad b_0 = 1, \qquad b_j = (\phi - \hat{\rho})\phi^{j-1} \quad (j \geq 1).$$

For $n \geq 0$, let $C_n = \mathrm{Cov}(\xi_t, \Delta\bar{s}_{t-n})$. The actual autocovariance of the pooled signal is

$$\gamma_{\bar{s}}(j) = \frac{\sigma_\mu^2}{1-\rho^2}\rho^{|j|} + \tau^{-1}\mathbf{1}_{\{j=0\}}, \qquad C_n = \sum_{j=0}^{\infty} b_j[\gamma_{\bar{s}}(n-j) - \gamma_{\bar{s}}(n+1-j)].$$

For clarity, the fundamental contribution to this sum is

$$C_n^a = \frac{\sigma_\mu^2}{\rho - \phi}\left[\frac{(\rho - \hat{\rho})\rho^n}{1+\rho} + \frac{(\hat{\rho} - \phi)(1-\phi)\phi^n}{1-\rho\phi}\right].$$

This follows by expanding $(1 - r\mathcal{L})/[(1-\phi\mathcal{L})(1-\rho\mathcal{L})]$ in fundamental innovations, where $\mathcal{L}$ is the lag operator. The signal-noise contribution is $\tau^{-1}(b_n - b_{n+1})$. By (A51),

$$\tau^{-1}(b_n - b_{n+1}) = -\frac{\sigma_\mu^2(1-\phi)\phi^n}{1-r\phi} + \frac{\tau^{-1}}{1-G}\mathbf{1}_{\{n=0\}}.$$

Adding the two contributions and collecting $\rho - \hat{\rho}$ gives

$$C_n = (\rho - \hat{\rho})\frac{\sigma_\mu^2}{1-\rho\phi}A_n + \frac{\tau^{-1}}{1-G}\mathbf{1}_{\{n=0\}}, \tag{A63}$$

where

$$A_n = (1-\rho\phi)\frac{\rho^n/(1+\rho) - \phi^n/(1+\phi)}{\rho-\phi} + \frac{\phi^{n+2}(\hat{\rho}-\phi)}{(1-\hat{\rho}\phi)(1+\phi)}. \tag{A64}$$

At $\rho = \phi$, the divided difference is its derivative limit. Since $x^n/(1+x)$ is strictly increasing for $x > 0$ and $n \geq 1$, we have

$$A_n > 0 \quad (n \geq 1), \qquad A_0 + A_1 = \frac{\phi^2(\hat{\rho}-\phi)}{1-\hat{\rho}\phi} > 0. \tag{A65}$$

Note that A_0 need not be positive.

Own-decision predictability. Combining (A59), (A62), and (A56) gives

$$\mathrm{Cov}(e_{t+1}, \iota_t) = \sum_{\ell=0}^{\infty} P_\ell [JC_{\ell+1} + BC_\ell] - \tau^{-1}KJ.$$

The $n = 0$ term in (A63) contributes $P_0 BV\tau^{-1}/(1-G) = KJ\tau^{-1}$ and cancels the last term exactly. Thus

$$\mathrm{Cov}(e_{t+1}, \iota_t) = (\rho - \hat{\rho})\frac{\sigma_\mu^2}{1-\rho\phi}\mathcal{S}_\iota, \qquad \mathcal{S}_\iota = \sum_{\ell=0}^{\infty} P_\ell [JA_{\ell+1} + BA_\ell]. \tag{A66}$$

All terms are nonnegative except possibly JBA_0, but it is not sufficient to make $S_\iota < 0$. To see this point, let $c \equiv (1-G)/(1-\omega) \leq 1$. Because $v_h \geq t_h$, $d_\iota \geq 0$, and $G \geq \omega$,

$$P_1 = (Lv)_\iota + Gd_\iota \geq L_{12}t_h + \omega d_\iota = \frac{B}{c} - \omega\hat{\rho}J. \tag{A67}$$

(A65) implies $A_0 > -A_1$, and if $B > 0$, we have

$$\begin{aligned}
\mathcal{S}_\iota &\geq J^2A_1 + JBA_0 + BP_1A_1 \\
&> A_1\left[J^2 + \frac{B^2}{c} - (1+\hat{\rho}\omega)JB\right] \\
&\geq A_1\left[(J-B)^2 + (1-\hat{\rho}\omega)JB\right] > 0.
\end{aligned}$$

For $B = 0$, $\mathcal{S}_\iota \geq J^2A_1 > 0$. Dividing (A66) by the positive investment variance proves

$$\tilde{\beta}_\iota = (\rho - \hat{\rho})\kappa_\iota, \qquad \kappa_\iota = \frac{\sigma_\mu^2 \mathcal{S}_\iota}{(1-\rho\phi)\,\mathrm{Var}(\iota_t)} > 0. \tag{A68}$$

Cross-decision predictability and its rational component. The same calculation using (A60) gives

$$\mathrm{Cov}(e_{t+1}, h_t) = \frac{Kt_h}{\tau_n} + (\rho - \hat{\rho})\Gamma_h, \qquad \Gamma_h = \frac{\sigma_\mu^2}{1-\rho\phi}\sum_{\ell=0}^{\infty} R_\ell[JA_{\ell+1} + BA_\ell]. \tag{A69}$$

To define that rational component, write

$$b_t = \mathbb{E}_t^k[a_t], \qquad S_k = \mathrm{Var}(a_t \mid \mathcal{I}_t^k), \qquad \widehat{S}_k = ((\Sigma_{\hat{\rho}}^-)^{-1} + \tau_p + \tau_k)^{-1}.$$

S_k is the objective posterior variance, while $\widehat{S}_k$ is the perceived posterior variance. The rational investment forecast error is

$$\iota_{t+1} - \mathbb{E}_t^k[\iota_{t+1}] = H(s_t^n - b_t) + J(a_{t+1} - \rho b_t) + \varepsilon_{\iota,t+1}^X.$$

Only $t_h s_t^n$ in current hiring is unobserved by the capital manager. It follows that the rational projection component at the behavioral allocation is

$$\beta_h \equiv \beta_h^R \equiv \frac{\mathrm{Cov}(\iota_{t+1} - \mathbb{E}_t^k[\iota_{t+1}], h_t)}{\mathrm{Var}(h_t)} = \frac{t_h[H/\tau_n + (H+\rho J)S_k]}{\mathrm{Var}(h_t)}. \tag{A70}$$

This is positive under communication frictions and equal to 0 under common information, as in Proposition 1. Using

$$\widehat{S}_k = \frac{\omega}{\tau_n(1-\omega)}, \qquad \frac{K}{\tau_n} = \frac{H}{\tau_n} + (H + rJ)\widehat{S}_k,$$

the decomposition relative to this rational component, for $\rho \neq \hat{\rho}$, is

$$\begin{aligned} \tilde{\beta}_h &= \beta_h + (\rho - \hat{\rho})\kappa_h, \\ \kappa_h &= \frac{1}{\mathrm{Var}(h_t)}\left\{\Gamma_h - t_h\left[J\widehat{S}_k + (H+\rho J)\frac{S_k - \widehat{S}_k}{\rho - \hat{\rho}}\right]\right\}. \end{aligned} \tag{A71}$$

Under common information the private-signal terms are absent: $t_\iota = t_h = \omega = K = B = 0$. Hiring is measurable to the capital manager, so $\beta_h = \beta_h^R = 0$. Directly applying the same MA calculation, without the private-noise terms, gives

$$\tilde{\beta}_h = (\rho - \hat{\rho})\kappa_h, \qquad \kappa_h = \frac{\sigma_\mu^2 J}{(1-\rho\phi)\,\mathrm{Var}(h_t)}\sum_{\ell=0}^{\infty} R_\ell A_{\ell+1} > 0. \tag{A72}$$

because positive hiring variance rules out $R_\ell = 0$ for every ℓ in this common-information case, so the inequality is strict.

A.8 Proof of Proposition 3

Again we have

$$\begin{aligned}\iota_t - \mathbb{F}^k_{t-1}[\iota_t] &= F^n_\iota F^s_h(s^n_{t-1} - \mathbb{F}^k_{t-1}[s^n_{t-1}]) + F^\zeta_\iota(\zeta_t - \mathbb{F}^k_{t-1}[\zeta_t]) + F^s_\iota(s^k_t - \mathbb{F}^k_{t-1}[s^k_t]) \\ &= F^n_\iota F^s_h(s^n_{t-1} - \mathbb{E}^k_{t-1}[s^n_{t-1}]) + F^\zeta_\iota(\zeta_t - \mathbb{E}^k_{t-1}[\zeta_t]) + F^s_\iota(s^k_t - \mathbb{E}^k_{t-1}[s^k_t]) \\ &+ F^n_\iota F^s_h(\mathbb{E}^k_{t-1}[s^n_{t-1}] - \mathbb{F}^k_{t-1}[s^n_{t-1}]) + F^\zeta_\iota(\mathbb{E}^k_{t-1}[\zeta_t] - \mathbb{F}^k_{t-1}[\zeta_t]) + F^s_\iota(\mathbb{E}^k_{t-1}[s^k_t] - \mathbb{F}^k_{t-1}[s^k_t])\end{aligned}$$

We first consider the predictability of behavioral components by past decisions. Since

$$\mathbb{E}^k_{t-1}[s^n_{t-1}] - \mathbb{F}^k_{t-1}[s^n_{t-1}] = -\theta(\hat{a}^k_{t-1} - \rho\hat{a}^k_{t-2})$$

$$\begin{aligned}\mathbb{E}^k_{t-1}[\zeta_t] - \mathbb{F}^k_{t-1}[\zeta_t] &= \mathbb{E}^k_{t-1}[(\Sigma^-)^{-1}\bar{a}_t + \sigma^{-2}_{\epsilon,p}s^p_t] - \mathbb{F}^k_{t-1}[(\Sigma^-)^{-1}\bar{a}_t + \sigma^{-2}_{\epsilon,p}s^p_t] \\ &= (\Sigma^-)^{-1}\left(\mathbb{E}^k_{t-1}[\bar{a}_t] - \mathbb{F}^k_{t-1}[\bar{a}_t]\right) + \sigma^{-2}_{\epsilon,p}\left(\mathbb{E}^k_{t-1}[s^p_t] - \mathbb{F}^k_{t-1}[s^p_t]\right) \\ &= (\Sigma^-)^{-1}\cdot(-\theta)\bar{\Sigma}\sigma^{-2}_{\epsilon,n}(\rho\hat{a}^k_{t-1} - \rho^2\hat{a}^k_{t-2}) + \sigma^{-2}_{\epsilon,p}\cdot(-\theta)(\rho\hat{a}^k_{t-1} - \rho^2\hat{a}^k_{t-2}) \\ &= -\theta\rho\left[(\Sigma^-)^{-1}\bar{\Sigma}\sigma^{-2}_{\epsilon,n} + \sigma^{-2}_{\epsilon,p}\right](\hat{a}^k_{t-1} - \rho\hat{a}^k_{t-2})\end{aligned}$$

$$\mathbb{E}^k_{t-1}[s^k_t] - \mathbb{F}^k_{t-1}[s^k_t] = -\theta\rho(\hat{a}^k_{t-1} - \rho\hat{a}^k_{t-2})$$

And therefore

$$\mathbb{E}^k_{t-1}[\iota_t] - \mathbb{F}^k_{t-1}[\iota_t] = -\theta\left\{F^n_\iota F^s_h + \rho\left[(\Sigma^-)^{-1}\bar{\Sigma}\sigma^{-2}_{\epsilon,n} + \sigma^{-2}_{\epsilon,p}\right]F^\zeta_\iota + \rho F^s_\iota\right\}(\hat{a}^k_{t-1} - \rho\hat{a}^k_{t-2}) \quad \text{(A73)}$$

Consider the theoretical own-decision predictability

$$\tilde{\beta}_\iota = \frac{cov(\iota_t - \mathbb{F}^k_{t-1}[\iota_t], \iota_{i,t-1})}{var(\iota_{i,t-1})} = \frac{cov(\mathbb{E}^k_{t-1}[\iota_t] - \mathbb{F}^k_{t-1}[\iota_t], \iota_{i,t-1})}{var(\iota_{i,t-1})}$$

$$= -\theta \underbrace{\left\{ F_\iota^n F_h^s + \rho \left[(\Sigma^-)^{-1} \bar{\Sigma} \sigma_{\epsilon,n}^{-2} + \sigma_{\epsilon,p}^{-2} \right] F_\iota^\zeta + \rho F_\iota^s \right\} \frac{cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, F_\iota^n n_{t-2} + F_\iota^\zeta \zeta_{t-1} + F_\iota^s s_{t-1}^k)}{var(\iota_{i,t-1})}}_{\equiv \omega_\iota} \tag{A74}$$

The sign of ω_ι depends on the sign of $cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, F_\iota^n n_{t-2} + F_\iota^\zeta \zeta_{t-1} + F_\iota^s s_{t-1}^k)$. We know that

$$\begin{aligned} cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, s_{t-1}^k) &= cov(\hat{a}_{t-1}^k - a_{t-1} + a_{t-1} - \rho \hat{a}_{t-2}^k, s_{t-1}^k) \\ &= cov(a_{t-1} - \rho \hat{a}_{t-2}^k, s_{t-1}^k) = cov(\rho a_{t-2} + \mu_{t-1} - \rho \hat{a}_{t-2}^k, \rho a_{t-2} + \mu_{t-1} + \epsilon_{t-1}^k) \\ &= \rho^2 \hat{\Sigma}^k + \sigma_\mu^2 \end{aligned}$$

$$cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, s_{t-1}^p) = cov(a_{t-1} - \rho \hat{a}_{t-2}^k, s_{t-1}^p) = \rho^2 \hat{\Sigma}^k + \sigma_\mu^2$$

$$\begin{aligned} cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, \bar{a}_{t-1}) &= cov(a_{t-1} - \rho \hat{a}_{t-2}^k, \bar{a}_{t-1}) \\ &= cov(a_{t-1} - \bar{a}_{t-1} + \bar{a}_{t-1} - \rho \hat{a}_{t-2}^k, \bar{a}_{t-1}) \\ &= cov(\rho \bar{\Sigma} \sigma_{\epsilon,n}^{-2} (s_{t-2}^n - \hat{a}_{t-2}^k), \rho \bar{\Sigma} \sigma_{\epsilon,n}^{-2} (s_{t-2}^n - \hat{a}_{t-2}^k) + \rho \hat{a}_{t-2}^k) \\ &= \rho^2 \bar{\Sigma}^2 \sigma_{\epsilon,n}^{-4} (\hat{\Sigma}^k + \sigma_{\epsilon,n}^2) = \rho^2 (\hat{\Sigma}^k - \bar{\Sigma}) \end{aligned}$$

$$\begin{aligned} cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, n_{t-2}) &= cov(\rho a_{t-2} - \rho \hat{a}_{t-2}^k, n_{t-2}) \\ &= cov(\rho (a_{t-2} - \hat{a}_{t-2}^k), n_{t-3} + F_h^0 + F_h^k k_{t-3} + F_h^n n_{t-3} + F_h^\zeta \zeta_{t-2} + F_h^s s_{t-2}^n) \\ &= \rho F_h^s cov(a_{t-2} - \hat{a}_{t-2}^k, s_{t-2}^n) = \rho F_h^s \hat{\Sigma}^k \end{aligned}$$

Hence

$$\begin{aligned} &cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, F_\iota^n n_{t-2} + F_\iota^\zeta \zeta_{t-1} + F_\iota^s s_{t-1}^k) \\ &= F_\iota^s (\rho^2 \hat{\Sigma}^k + \sigma_\mu^2) + F_\iota^\zeta \sigma_{\epsilon,p}^{-2} (\rho^2 \hat{\Sigma}^k + \sigma_\mu^2) + F_\iota^\zeta (\Sigma^-)^{-1} \rho^2 (\hat{\Sigma}^k - \bar{\Sigma}) + \rho F_\iota^n F_h^s \hat{\Sigma}^k > 0 \end{aligned}$$

and therefore $\omega_\iota > 0$.

For the theoretical cross-decision predictability, the behavioral component is similar: we have

$$\tilde{\beta}_h^B = \frac{cov(\mathbb{E}_{t-1}^k[\iota_t] - \mathbb{F}_{t-1}^k[\iota_t], h_{i,t-1})}{var(h_{i,t-1})}$$

$$= \underbrace{-\theta \left\{ F_\iota^n F_h^s + \rho \left[(\Sigma^-)^{-1} \bar{\Sigma} \sigma_{\epsilon,n}^{-2} + \sigma_{\epsilon,p}^{-2} \right] F_\iota^\zeta + \rho F_\iota^s \right\} \frac{cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, F_h^n n_{t-2} + F_h^\zeta \zeta_{t-1} + F_h^s s_{t-1}^n)}{var(h_{i,t-1})}}_{\equiv \omega_h} \tag{A75}$$

and the sign of ω_h is determined by $cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, F_h^n n_{t-2} + F_h^\zeta \zeta_{t-1} + F_h^s s_{t-1}^n)$:

$$\begin{aligned} &cov(\hat{a}_{t-1}^k - \rho \hat{a}_{t-2}^k, F_h^n n_{t-2} + F_h^\zeta \zeta_{t-1} + F_h^s s_{t-1}^n) \\ &= F_h^s (\rho^2 \hat{\Sigma}^k + \sigma_\mu^2 - \hat{\Sigma}^k) + F_h^\zeta \sigma_{\epsilon,p}^{-2} (\rho^2 \hat{\Sigma}^k + \sigma_\mu^2) + F_h^\zeta (\Sigma^-)^{-1} \rho^2 (\hat{\Sigma}^k - \bar{\Sigma}) + \rho F_h^n F_h^s \hat{\Sigma}^k \end{aligned}$$

Under common information, $F_h^s = 0$, and the covariance degenerates to $F_h^\zeta \sigma_{\epsilon,p}^{-2} (\rho^2 \hat{\Sigma}^k + \sigma_\mu^2)$, which makes $\omega_h > 0$.

The rational part of cross-decision predictability $\tilde{\beta}^R \equiv \frac{cov(\iota_t - \mathbb{E}_{t-1}^k[\iota_t], h_{i,t-1})}{var(h_{i,t-1})}$ is positive under communication friction and is equal to 0 under common information. This follows directly from the arguments in the proof of Proposition 1.

A.9 Full Statement and Proof of Proposition 4

We suppress the firm subscript and center all log variables around their frictionless steady-state values. Throughout this subsection, σ denotes the capital–labor elasticity of substitution rather than the standard deviation of a shock.

Full statement of Proposition 4. Suppose there is no adjustment friction on capital and labor, i.e. $\xi_k = \xi_n = 0$, and the firm's gross revenue is given by (35). Suppose the log deviation of factor-augmenting technology $a_{j,t} \equiv \log(A_{j,t}/\bar{A}_j)$ follows an AR(1) process:

$$a_{j,t} = \rho a_{j,t-1} + u_{j,t}, \qquad 0 \leq \rho < 1, \tag{A76}$$

and let the firm observe

$$s_{j,t} = a_{j,t} + \varepsilon_{j,t}. \tag{A77}$$

The innovations $u_{k,t}$ and $u_{n,t}$ and the signal errors $\varepsilon_{k,t}$ and $\varepsilon_{n,t}$ are mutually independent across factors and time, with strictly positive variances. The firm has common information and correctly perceives all innovation and signal-noise variances, but perceives the persistence

of both technologies to be $\hat{\rho}$, where

$$0 \leq \rho < \hat{\rho} < 1. \tag{A78}$$

It uses the corresponding stationary Kalman filters. In particular, define

$$\hat{a}_{j,t} \equiv \mathbb{F}_t[a_{j,t}] = \hat{\rho}\,\hat{a}_{j,t-1} + G_j\left(s_{j,t} - \hat{\rho}\,\hat{a}_{j,t-1}\right), \qquad G_j \in (0,1). \tag{A79}$$

Here the subscript on $\hat{a}_{j,t}$ identifies the factor-specific technology, not the manager forming the belief.

Let $\bar{s}_K$ be the steady-state effective-capital share in (36) and define

$$\psi_{kk} \equiv \bar{s}_K(\epsilon - 1) + (1 - \bar{s}_K)(\sigma - 1), \qquad \psi_{kn} \equiv (1 - \bar{s}_K)(\epsilon - \sigma), \tag{A80}$$

$$\psi_{nk} \equiv \bar{s}_K(\epsilon - \sigma), \qquad \psi_{nn} \equiv (1 - \bar{s}_K)(\epsilon - 1) + \bar{s}_K(\sigma - 1). \tag{A81}$$

Suppose that $\epsilon > 1$, $\sigma > 0$, and

$$\psi_{kk} > 0, \qquad \psi_{nn} > 0. \tag{A82}$$

Then

$$\beta_\iota < 0, \qquad \operatorname{sgn}(\beta_h) = \operatorname{sgn}(\sigma - \epsilon). \tag{A83}$$

Consequently,

$$\beta_\iota < 0 < \beta_h \quad \Longleftrightarrow \quad \sigma > \epsilon. \tag{A84}$$

Proof. The proof proceeds in three steps.

Step 1: Log-linear factor demands. Define

$$\vartheta \equiv 1 - \frac{1}{\epsilon}$$

and the time-t effective-capital share

$$s_{K,t} \equiv \frac{\alpha^{1/\sigma}(A_{k,t}K_t)^{1-1/\sigma}}{\alpha^{1/\sigma}(A_{k,t}K_t)^{1-1/\sigma} + (1-\alpha)^{1/\sigma}(A_{n,t}N_t)^{1-1/\sigma}}. \tag{A85}$$

The revenue elasticities with respect to capital and labor are

$$\frac{\partial \log Y_t}{\partial \log K_t} = \vartheta s_{K,t}, \qquad \frac{\partial \log Y_t}{\partial \log N_t} = \vartheta(1 - s_{K,t}). \tag{A86}$$

Therefore the flexible-input first-order conditions are

$$\vartheta s_{K,t} \frac{Y_t}{K_t} = \bar{R}, \qquad \vartheta(1 - s_{K,t}) \frac{Y_t}{N_t} = \bar{W}. \tag{A87}$$

Certainty equivalence in the first-order approximation means that the firm evaluates these conditions using $\hat{a}_{k,t}$ and $\hat{a}_{n,t}$. Let y_t be the log deviation of revenue and let k_t^* and n_t^* denote the log deviations of the flexible factor choices. Log-linearizing revenue gives

$$y_t = \vartheta \left[\bar{s}_K (\hat{a}_{k,t} + k_t^*) + (1 - \bar{s}_K)(\hat{a}_{n,t} + n_t^*) \right]. \tag{A88}$$

The corresponding log-linearizations of the two CES shares are

$$\log \left(\frac{s_{K,t}}{\bar{s}_K} \right) = (1 - \bar{s}_K) \left(1 - \frac{1}{\sigma} \right) \left[(\hat{a}_{k,t} + k_t^*) - (\hat{a}_{n,t} + n_t^*) \right], \tag{A89}$$

$$\log \left(\frac{1 - s_{K,t}}{1 - \bar{s}_K} \right) = -\bar{s}_K \left(1 - \frac{1}{\sigma} \right) \left[(\hat{a}_{k,t} + k_t^*) - (\hat{a}_{n,t} + n_t^*) \right]. \tag{A90}$$

Since factor prices are fixed, the log-linearized first-order conditions from (A87) are

$$0 = y_t - k_t^* + (1 - \bar{s}_K) \left(1 - \frac{1}{\sigma} \right) \left[(\hat{a}_{k,t} + k_t^*) - (\hat{a}_{n,t} + n_t^*) \right], \tag{A91}$$

$$0 = y_t - n_t^* - \bar{s}_K \left(1 - \frac{1}{\sigma} \right) \left[(\hat{a}_{k,t} + k_t^*) - (\hat{a}_{n,t} + n_t^*) \right]. \tag{A92}$$

Subtracting (A92) from (A91) and rearranging yields

$$k_t^* - n_t^* = (\sigma - 1)(\hat{a}_{k,t} - \hat{a}_{n,t}). \tag{A93}$$

Multiplying (A91) by $\bar{s}_K$, multiplying (A92) by $1 - \bar{s}_K$, and adding cancels the share terms and gives

$$y_t = \bar{s}_K k_t^* + (1 - \bar{s}_K) n_t^*.$$

Combining this equality with (A88) gives

$$\bar{s}_K k_t^* + (1 - \bar{s}_K) n_t^* = (\epsilon - 1) \left[\bar{s}_K \hat{a}_{k,t} + (1 - \bar{s}_K) \hat{a}_{n,t} \right]. \tag{A94}$$

Solving (A93) and (A94) for the two choices gives

$$k_t^* = \psi_{kk}\hat{a}_{k,t} + \psi_{kn}\hat{a}_{n,t}, \tag{A95}$$

$$n_t^* = \psi_{nk}\hat{a}_{k,t} + \psi_{nn}\hat{a}_{n,t}, \tag{A96}$$

with the coefficients in (A80) and (A81).

Step 2: The covariance generated by over-extrapolation. For each $j \in \{k, n\}$, define

$$\delta_{j,t} \equiv \hat{a}_{j,t} - \hat{\rho}\,\hat{a}_{j,t-1}, \qquad d_{j,t-1} \equiv \hat{a}_{j,t-1} - \hat{a}_{j,t-2}, \tag{A97}$$

and let

$$\chi_j \equiv \mathrm{Cov}(\delta_{j,t}, d_{j,t-1}). \tag{A98}$$

We now show that $\chi_j < 0$ whenever $\hat{\rho} > \rho$.

Let $\ell_j \equiv 1 - G_j$ and $\phi_j \equiv \ell_j \hat{\rho}$. Under the true data-generating process, (A79) can be written as

$$\hat{a}_{j,t} = \phi_j \hat{a}_{j,t-1} + G_j a_{j,t} + G_j \varepsilon_{j,t}. \tag{A99}$$

Define

$$V_{a,j} \equiv \mathrm{Var}(a_{j,t}) = \frac{\sigma_{u,j}^2}{1-\rho^2}, \qquad c_j \equiv \mathrm{Cov}(a_{j,t}, \hat{a}_{j,t}), \qquad V_{\hat{a},j} \equiv \mathrm{Var}(\hat{a}_{j,t}).$$

Equation (A99) implies

$$c_j = \frac{G_j V_{a,j}}{1 - \rho\phi_j}, \tag{A100}$$

$$V_{\hat{a},j} = \frac{G_j^2}{1-\phi_j^2}\left[\sigma_{\varepsilon,j}^2 + V_{a,j}\frac{1+\rho\phi_j}{1-\rho\phi_j}\right]. \tag{A101}$$

Let $\gamma_{j,q} \equiv \mathrm{Cov}(\hat{a}_{j,t}, \hat{a}_{j,t-q})$. For $q \geq 1$, (A99) gives

$$\gamma_{j,q} = \phi_j \gamma_{j,q-1} + G_j \rho^q c_j. \tag{A102}$$

In particular,

$$\gamma_{j,1} = \phi_j V_{\hat{a},j} + G_j \rho c_j, \qquad \gamma_{j,2} = \phi_j \gamma_{j,1} + G_j \rho^2 c_j.$$

It follows from (A97) that

$$\chi_j = (1+\hat{\rho})\gamma_{j,1} - \gamma_{j,2} - \hat{\rho}\gamma_{j,0}$$

$$= (\phi_j - \hat{\rho})(1 - \phi_j)V_{\hat{a},j} + G_j \rho c_j (1 + \hat{\rho} - \phi_j - \rho). \tag{A103}$$

Because G_j is the stationary Kalman gain under the firm's perceived model, the perceived Riccati equations are

$$P_j^- = \hat{\rho}^2 P_j + \sigma_{u,j}^2, \qquad G_j = \frac{P_j^-}{P_j^- + \sigma_{\varepsilon,j}^2}, \qquad P_j = (1 - G_j)P_j^-. \tag{A104}$$

Equivalently,

$$\sigma_{\varepsilon,j}^2 = \frac{\sigma_{u,j}^2 \ell_j}{G_j(1 - \hat{\rho}^2 \ell_j)}. \tag{A105}$$

Substituting (A100), (A101), and (A105) into (A103) and collecting terms yields

$$\chi_j = \frac{\sigma_{u,j}^2 (\rho - \hat{\rho}) G_j^2 \left[1 - \hat{\rho}^2 \ell_j + \hat{\rho}^2 \ell_j^2 (1 + \rho)\right]}{(1 + \rho)(1 + \hat{\rho}\ell_j)(1 - \hat{\rho}^2 \ell_j)(1 - \rho\hat{\rho}\ell_j)}. \tag{A106}$$

Every factor in (A106) other than $\rho - \hat{\rho}$ is strictly positive. In particular, $1 - \hat{\rho}^2 \ell_j > 0$ and the bracketed numerator is the sum of this positive term and $\hat{\rho}^2 \ell_j^2 (1 + \rho) > 0$. Consequently,

$$\hat{\rho} > \rho \quad \Longrightarrow \quad \chi_k < 0, \qquad \chi_n < 0. \tag{A107}$$

Step 3: Signs of the projection coefficients. Under the firm's perceived model,

$$\mathbb{F}_{t-1}[\hat{a}_{j,t}] = \hat{\rho}\, \hat{a}_{j,t-1}.$$

Moreover, k_{t-1}^* is known at time $t - 1$. Therefore

$$\begin{aligned} FE_t^{\iota} &= \iota_t - \mathbb{F}_{t-1}[\iota_t] \\ &= k_t^* - \mathbb{F}_{t-1}[k_t^*] \\ &= \psi_{kk} \delta_{k,t} + \psi_{kn} \delta_{n,t}. \end{aligned} \tag{A108}$$

Equations (A95) and (A96) also imply

$$\iota_{t-1} = \psi_{kk} d_{k,t-1} + \psi_{kn} d_{n,t-1}, \tag{A109}$$

$$h_{t-1} = \psi_{nk} d_{k,t-1} + \psi_{nn} d_{n,t-1}. \tag{A110}$$

Let

$$v_j \equiv \text{Var}(d_{j,t-1}) > 0.$$

Independence of the two factor-specific processes implies that all cross-factor covariances in the projection moments are zero. Hence

$$\beta_\iota = \frac{\psi_{kk}^2 \chi_k + \psi_{kn}^2 \chi_n}{\psi_{kk}^2 v_k + \psi_{kn}^2 v_n} < 0, \tag{A111}$$

where the inequality follows from (A107). Similarly,

$$\beta_h = \frac{\psi_{kk}\psi_{nk}\chi_k + \psi_{kn}\psi_{nn}\chi_n}{\psi_{nk}^2 v_k + \psi_{nn}^2 v_n}. \tag{A112}$$

Using (A80) and (A81), the numerator in (A112) factors as

$$\begin{aligned} &\psi_{kk}\psi_{nk}\chi_k + \psi_{kn}\psi_{nn}\chi_n \\ &\quad = (\epsilon - \sigma)\left[\bar{s}_K \psi_{kk}\chi_k + (1 - \bar{s}_K)\psi_{nn}\chi_n\right]. \end{aligned} \tag{A113}$$

The expression in brackets is strictly negative by (A82) and (A107), while the denominator in (A112) is strictly positive. It follows that

$$\text{sgn}(\beta_h) = -\,\text{sgn}(\epsilon - \sigma) = \text{sgn}(\sigma - \epsilon).$$

Together with (A111), this proves (A83) and (A84).

A.10 Robustness Checks

In this section, I consider several robustness checks to Section 3 and Section 5.

A.10.1 Time-to-build

In the capital adjustment literature, it is common to assume that the capital takes time to build (e.g. Cooper and Haltiwanger (2006), David and Venkateswaran (2019)): today's investment transforms into capital stock with a one-period lag. One may wonder whether time-to-build invalidates the power of empirical test: namely, whether incorporating time-to-build in capital alone makes the coefficient of interest $\tilde{\beta}_h \neq 0$.

I verify that time-to-build in capital does not invalidate the regression test. In our framework of Dynamic Narrow Framing, the only change is on the profit function: without time-to-build, we have

$$\Pi(K,N;A,K_{-1},N_{-1}) = \underbrace{AK^{\alpha(1-1/\epsilon)}N^{(1-\alpha)(1-1/\epsilon)}}_{=\mathcal{R}(A,K,N)} - RK - WN - \text{adj. costs}$$

With time-to-build on capital, it seems that the only change is on the profit function:

$$\Pi(K,N;A,K_{-1},N_{-1}) = AK_{-1}{}^{\alpha(1-1/\epsilon)}N^{(1-\alpha)(1-1/\epsilon)} - RK - WN - \text{adj. costs} \tag{A114}$$

In other words, with the time-to-build friction, capital for production today is predetermined, and the cost for tomorrow's capital is paid today. This friction changes the matrices in the log-quadratic profit function (A2): now the matrix $P, Q, R, H_\iota, H_h, H_{\iota h}$ becomes

$$P \equiv \begin{bmatrix} \frac{1}{2\epsilon}\bar{Y} & 0 & 0 & \frac{1}{2}(1-1/\epsilon)\bar{Y} \\ * & \frac{1}{2}\bar{R}\bar{K}(\alpha(1-1/\epsilon)-1) & \frac{1}{2}\alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} & \frac{1}{2}\alpha(1-1/\epsilon)^2\bar{Y} \\ * & * & \frac{1}{2}\bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) & \frac{1}{2}(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ * & * & * & \frac{1}{2}(1-1/\epsilon)^2\bar{Y} \end{bmatrix} \tag{A115}$$

$$Q \equiv \begin{bmatrix} -\bar{R}\bar{K} \\ -\bar{R}\bar{K} \\ 0 \\ 0 \end{bmatrix} \text{ and } R \equiv \begin{bmatrix} 0 \\ \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y} \\ \bar{W}\bar{N}((1-\alpha)(1-1/\epsilon)-1) \\ (1-\alpha)(1-1/\epsilon)^2\bar{Y} \end{bmatrix} \tag{A116}$$

and

$$H_\iota \equiv -\frac{1}{2}(\bar{R}\bar{K} + \xi_k\bar{K}), H_h \equiv -\frac{1}{2}([1-(1-\alpha)(1-1/\epsilon)]\bar{W}\bar{N} + \xi_n\bar{N}) \text{ and } H_{\iota h} = 0 \tag{A117}$$

The learning process of firm managers is not changed by the time-to-build friction, and therefore the intuition we built in Section 3 remains true: lagged hiring cannot forecast the investment forecast error.

A.10.2 Reduced-form Distortions

I show that reduced-form distortions cannot reproduce the sign patterns by themselves. Consider reduced-form distortions as in Hsieh and Klenow (2009): the profit of the firm is

given by

$$\Pi_{it} = (1-\tau_{it}^y)R(K_{it}, N_{it}; A_{it}) - (1+\tau_{it}^k)R_t K_{it} - W_t N_{it} - \frac{\xi_k}{2}\left(\frac{K_{it}}{K_{i,t-1}} - 1\right)^2 K_{i,t-1} - \frac{\xi_n}{2}\left(\frac{N_{it}}{N_{i,t-1}} - 1\right)^2 N_{i,t-1} \tag{A118}$$

where τ_{it}^y is a revenue distortion and τ_{it}^k is a capital-specific distortion. I follow David and Venkateswaran (2019) and assume that these distortions have the following structure:

$$\tau_{it}^y = \gamma^y a_{it} + \delta_i^y + \eta_{it}^y \tag{A119}$$

$$\tau_{it}^k = \gamma^k a_{it} + \delta_i^k + \eta_{it}^k \tag{A120}$$

where $\gamma^y, \gamma^k > 0$ are the "correlated distortions" as in Bartelsman et al. (2013), δ^y, δ^k captures firm characteristics (such as management quality in Bloom et al. (2019)) and η_{it}^y and η_{it}^k are the uncorrelated part of distortion as in Hsieh and Klenow (2009).

Define $\tilde{\eta}_{it}^y \equiv \delta_i^y + \eta_{it}^y$ and $\tilde{\eta}_{it}^k \equiv \delta_i^k + \eta_{it}^k$. The state vector x_{it} is now defined as

$$x_{it} = [1, k_{i,t-1}, n_{i,t-1}, a_{it}, \tilde{\eta}_{it}^k, \tilde{\eta}_{it}^y]'$$

and the profit function can be approximated as

$$\begin{aligned}
\Pi_{it} \approx\ & \frac{1}{\epsilon}\bar{Y} + \left(1-\frac{1}{\epsilon}\right)\bar{Y}a_{it} - \bar{Y}(\gamma^y a_{it} + \tilde{\eta}_{it}^y) - \bar{R}\bar{K}\left(\gamma^k a_{it} + \tilde{\eta}_{it}^k\right) - \bar{R}\bar{K}\left(\gamma^k a_{it} + \tilde{\eta}_{it}^k\right)k_{it} \\
& + \frac{1}{2}\left([\alpha(1-1/\epsilon)-1]\bar{R}\bar{K} - \xi_k\bar{K}\right)k_{it}^2 + \frac{1}{2}\left([(1-\alpha)(1-1/\epsilon)-1]\bar{W}\bar{N} - \xi_n\bar{N}\right)n_{it}^2 \\
& + \frac{1}{2}\left(1-\frac{1}{\epsilon}\right)^2\bar{Y}a_{it}^2 - \frac{\xi_k}{2}\bar{K}k_{i,t-1}^2 - \frac{\xi_n}{2}\bar{N}n_{i,t-1}^2 + \xi_k\bar{K}k_{it}k_{i,t-1} + \xi_n\bar{N}n_{it}n_{i,t-1} \\
& + \alpha(1-\alpha)(1-1/\epsilon)^2\bar{Y}k_{it}n_{it} + \alpha(1-1/\epsilon)^2\bar{Y}k_{it}a_{it} + (1-\alpha)(1-1/\epsilon)^2\bar{Y}n_{it}a_{it} \\
& - \left(1-\frac{1}{\epsilon}\right)\bar{Y}(\gamma^y a_{it} + \tilde{\eta}_{it}^y)a_{it} - \bar{R}\bar{K}(\gamma^y a_{it} + \tilde{\eta}_{it}^y)k_{it} - \bar{W}\bar{N}(\gamma^y a_{it} + \tilde{\eta}_{it}^y)n_{it}
\end{aligned}$$

Everything can be mapped into the primitives:

$$P = \begin{pmatrix}
\frac{1}{\epsilon}\bar{Y} & 0 & 0 & \frac{1}{2}\Big((1-\frac{1}{\epsilon})\bar{Y} - \bar{R}\bar{K}\gamma_k - \bar{Y}\gamma_y\Big) & -\frac{1}{2}\bar{R}\bar{K} & -\frac{1}{2}\bar{Y} \\
* & \frac{1}{2}\bar{R}\bar{K}\Big(\alpha(1-\frac{1}{\epsilon})-1\Big) & \frac{1}{2}\alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} & \frac{\epsilon-1}{2\epsilon}\bar{R}\bar{K} - \frac{1}{2}\bar{R}\bar{K}(\gamma_k+\gamma_y) & -\frac{1}{2}\bar{R}\bar{K} & -\frac{1}{2}\bar{R}\bar{K} \\
* & * & \frac{1}{2}\bar{W}\bar{N}\Big((1-\alpha)(1-\frac{1}{\epsilon})-1\Big) & \frac{\epsilon-1}{2\epsilon}\bar{W}\bar{N} - \frac{1}{2}\bar{W}\bar{N}\gamma_y & 0 & -\frac{1}{2}\bar{W}\bar{N} \\
* & * & * & \frac{1}{2}\frac{(\epsilon-1)^2}{\epsilon^2}\bar{Y} - (1-\frac{1}{\epsilon})\bar{Y}\gamma_y & 0 & -\frac{1}{2}(1-\frac{1}{\epsilon})\bar{Y} \\
* & * & * & * & 0 & 0 \\
* & * & * & * & * & 0
\end{pmatrix}$$

$$Q = \begin{pmatrix} 0 \\ \bar{R}\bar{K}\big(\alpha(1-\frac{1}{\epsilon})-1\big) \\ \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} \\ \frac{\epsilon-1}{\epsilon}\alpha(1-\frac{1}{\epsilon})\bar{Y} - \bar{R}\bar{K}(\gamma_k+\gamma_y) \\ -\bar{R}\bar{K} \\ -\bar{R}\bar{K} \end{pmatrix}, \quad R = \begin{pmatrix} 0 \\ \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y} \\ \bar{W}\bar{N}\big((1-\alpha)(1-\frac{1}{\epsilon})-1\big) \\ \frac{\epsilon-1}{\epsilon}(1-\alpha)(1-\frac{1}{\epsilon})\bar{Y} - \bar{W}\bar{N}\gamma_y \\ 0 \\ -\bar{W}\bar{N} \end{pmatrix}$$

$$H_\iota = \frac{1}{2}\bar{R}\bar{K}\Big(\alpha(1-\frac{1}{\epsilon})-1\Big)-\frac{1}{2}\xi_k\bar{K}, \quad H_{\iota h} = \alpha(1-\alpha)(1-\frac{1}{\epsilon})^2\bar{Y}, \quad H_h = \frac{1}{2}\bar{W}\bar{N}\Big((1-\alpha)(1-\frac{1}{\epsilon})-1\Big)-\frac{1}{2}\xi_n\bar{N}$$

$$A = \mathrm{diag}(1,1,1,\rho,0,0), \quad B = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A.10.3 Private Benefit of Managers

The team production literature (Marschak and Radner (1972)) typically assumes that managers are cooperative and non-strategic: they choose their actions to jointly maximize a common objective, which is the firm's profits. Here I deviate from this assumption and allow capital and labor managers to have private benefits. Following the managerial empire building literature (e.g. Ebsim, Lian, Ma, Ottonello, and Perez (2025)), I model this managerial friction by assuming that the capital manager's payoff function is

$$\Pi_{it} + g(\xi k_{it})$$

where $g(x) = x^\phi$ captures the private benefit of the capital manager by diverting a fraction ξ of the capital stock to their career concerns. After doing log-quadratic approximation to the private benefit term g, the capital manager's payoff can be written as

$$x_t' P^k x_t + x_t' Q^k \iota_t + x_t' R h_t + H_\iota^k \iota_t^2 + H_{\iota h}\iota_t h_t + H_h h_t^2$$

where P^k, Q^k and H_ι^k are altered by the private benefit term. The policy function of the capital and labor manager's problem remains linear, and hence Lemma 2 continue to hold in this case. The decomposition of investment forecast error in Proposition **??** remains valid.

B Data Appendix

This appendix shows the details of sample construction and variable definition for the expectation test and the calibration exercise.

B.1 Construction of IBES-Compustat Dataset

Prepare the CRSP-Compustat Dataset After downloading the CRSP-Compustat Annual Dataset from the WRDS website, I apply the following filters:

- Keep only the firms incorporated in the US (`fic = "USA"`);
- Keep only the firms using USD as accounting units (`curncd = "USD"`);
- Drop the observations with missing NAICS codes;
- Drop the observations with 2-digit NAICS codes = 52 (finance) and = 22 (utilities);
- Drop the observations with non-positive sales;
- Drop the observations with non-positive employment;
- Drop the observations with missing capital expenditure;
- Drop the observations with missing sales.
- Drop the observations with missing employment.
- Remove duplicate (`LPERMNO`, `fyear`) combinations.

I end up with a CRSP-Compustat Annual sample containing 81,937 firm-year observations from 2000 to 2024. Because the IBES capital-expenditure guidance data used below begin in 2003, I restrict the Compustat comparison sample in Table B1 to the regression sample period, 2003–2024. This period-aligned Compustat sample contains 68,591 firm-year observations from 7,567 firms.

Prepare the IBES-Compustat Joint Sample After downloading the IBES Guidance Dataset from WRDS, I apply the following filters and imputations:

- Keep only U.S. firms reporting in U.S. dollars (`usfirm = 1` and `curr = "USD"`);
- Keep only the annual guidance (`pdicity = "ANN"`);
- Keep capital-expenditure guidance (`measure = "CPX"`). Although the initial data-cleaning code also retains sales guidance, sales guidance is not used in the current expectation regression.
- Keep only observations with range guidance (`range_desc = 1`) or point estimates (`range_desc = 2`). For range guidance, drop observations with a missing lower or upper bound and impute the forecast using the midpoint of the lower and upper bounds, `(val_1 + val_2)/2`.
- Merge the IBES-CRSP bridge file with the IBES sample by matching the IBES ticker and requiring the guidance announcement date to fall between the bridge link's start and end dates. This maps the IBES firm identifier to the CRSP-Compustat identifier `PERMNO`.
- Construct the fiscal-year start and end dates and measure the forecast horizon in quarters. Keep guidance with `cpx_horizon_q < 4`.
- Within each firm and fiscal year, order guidance by its announcement and activation timestamps and retain the first capital-expenditure guidance. A firm-year with a single guidance is retained; intermediate and last-only guidance records are dropped.
- Merge the resulting first-guidance sample with CRSP-Compustat by `LPERMNO` and fiscal year, and retain one observation for each (`LPERMNO`, `fyear`) combination.
- Construct hiring and capital-expenditure lags with exact-year self-joins. In particular, $h_{i,t-1}$ requires employment observations in both $t-1$ and $t-2$.
- For the main Q1 regression sample, keep observations with `cpx_horizon_q = 3`, nonmissing capital-expenditure forecast error, nonmissing lagged hiring, and positive lagged capital expenditure.

The raw IBES file contains no annual U.S.-firm capital-expenditure guidance before fiscal year 2003. Consequently, no observations from 2000–2002 can enter the expectation regression. Although 85 observations in 2003 have a nonmissing capital-expenditure forecast error,

only one satisfies the Q1 and lag requirements; the economically substantive sample therefore begins in 2004.

Summary Statistics The final Q1 regression sample contains 14,804 firm-year observations from 2,418 firms and spans 2003–2024. Table B1 compares this sample with the period-aligned CRSP-Compustat sample. Firms in the regression sample are about twice as large in mean sales and employment, while median sales are more than four times larger than in the Compustat sample. The regression sample contains about 21.6 percent as many firm-year observations as the period-aligned Compustat sample. Table B2 reports total sample sales relative to U.S. GDP in selected years. Excluding 2003, which contains only one regression observation, the average annual sales-to-GDP ratio is 29.05 percent over 2004–2024. Because firm sales are gross revenues whereas GDP measures value added, this ratio should not be interpreted as the sample firms' contribution to GDP.

Table B1: Summary Statistics: Regression and Compustat Samples

Variable	Mean	Std. dev.	Median	Observations
Panel A: IBES/Compustat sample				
Sales	7,662.09	24,743.65	1,891.06	14,804
Gross property, plant, and equipment	6,235.37	23,815.35	1,157.65	14,768
Capital expenditures	499.86	1,831.10	88.89	14,804
Employment	21.84	69.21	6.40	14,804
Panel B: Full CRSP/Compustat Sample				
Sales	3,903.91	18,756.34	437.47	68,592
Gross property, plant, and equipment	2,762.06	15,765.27	172.64	65,413
Capital expenditures	230.50	1,382.78	13.20	68,592
Employment	11.43	53.99	1.30	68,592

Notes: Panel A is the IBES/Compustat sample used in the main expectation regressions. It requires a Q1 capital-expenditure forecast, nonmissing forecast error and lagged hiring, and positive lagged capital expenditures (14,804 firm-year observations, 2,418 firms, 2003–2024). Panel B is the CRSP/Compustat annual sample immediately before merging with IBES, restricted to the same year range as Panel A (68,592 firm-year observations, 7,567 firms, 2003–2024). It retains U.S.-incorporated firms reporting in U.S. dollars, excludes utilities (NAICS 22) and finance (NAICS 52), and requires positive sales and employment and nonmissing capital expenditures. Sales, gross property, plant, and equipment, and capital expenditures are in millions of nominal U.S. dollars; employment is in thousands.

Construction of Variables for the Expectation Test The key variables in the expectation tests in Section 3 are constructed as follows. For the capital-expenditure forecast error, I use the arc-percent difference between the forecast and actual values, as in Bloom,

	CRSP-Compustat Sample			
	2005	2010	2015	2020
Total Sales (Billions USD)	8812	10057	11957	13413
Sales/GDP	68%	67%	65%	63%
	IBES/Compustat Sample			
	2005	2010	2015	2020
Total Sales (Billions USD)	2918	5653	5743	5355
Sales/GDP	22%	38%	31%	25%

Table B2: Total Sales Relative to U.S. GDP

Codreanu, and Fletcher (2025):

$$\text{CapEX Forecast Error}_{it} = \frac{\text{CapEX}_{it} - \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}]}{1/2(\text{CapEX}_{it} + \mathbb{E}^k_{i,t-1}[\text{CapEX}_{it}])}$$

The arc-difference forecast error is bounded between $[-2, 2]$, limiting the influence of extreme realized or forecast values. It also avoids the problems that arise when taking logs of values close to zero. I verify that the regression results are robust to alternative definitions of the forecast error, including the log difference between realized and forecast capital expenditure.

The definition of hiring is standard. For firm i in year t, hiring h_{it} is defined as

$$h_{it} \equiv \log(\text{emp}_{it}) - \log(\text{emp}_{i,t-1})$$

where `emp` is the employment of the firm in CRSP-Compustat. Alternatively, an arc-difference measure of hiring is widely adopted in the literature as well (e.g. Davis and Haltiwanger (1992)). The regression results do not change with alternative definitions of hiring.

The main regression uses lagged hiring, $h_{i,t-1}$, which requires nonmissing employment in years $t-1$ and $t-2$. Lagged investment is measured as $\log(\text{CapEX}_{i,t-1})$. Both variables are constructed with exact-year lags, and observations with non-positive lagged capital expenditure are excluded before taking logs.

B.2 Robustness of IBES/Compustat Results

The first robustness exercise addresses the issue of omitted variable bias. The literature has documented that managerial forecast errors are persistent (Ma et al. (2020)). The decomposition in Lemma 3 also shows that lag nowcast error of productivity helps predict the investment forecast error. One may wonder whether the predictability of investment forecast error by past investment and past hiring in Table 1 are merely picking up the effects of these omitted lag forecast or nowcast error terms. To address this concern, I add lagged sales nowcast error as an additional control on the right-hand side of the regression (27) and (26). The sales nowcast error is defined as the arc difference between the actual sales in fiscal year t and the last sales guidance posted for fiscal year t. It serves as a proxy for the lag productivity nowcast error term in the decomposition of Lemma 3. The results are tabulated in Table B3. We can see that the estimates show similar pattern as Table 1: that the investment forecast overreacts to investment and underreacts to hiring. It confirms that the opposite-sign pattern we see in Table 1 is not an artifact of omitted lag forecast or nowcast errors.

The other exercise concerns the robustness of the results to alternative definitions of the investment forecast error. I consider two alternatives: the log difference between actual and expected capital expenditure, and the level difference between actual and expected investment rate, defined as the ratio of capital expenditure in year t to the plants, properties and equipments (PPENT) in period $t-1$. I run the same regression (27) and (26) using these alternative forecast error measures and tabulate the results in Table B4. We can see that the opposite-sign pattern of own- and cross-decision predictability is robust to these alternative measures of forecast errors.

B.3 Construction of the Census Sample

To assemble the Census sample I use for the analysis in Section 3 requires access to microdata of Annual Survey of Manufacturers (ASM) from year 2014 to 2016, Census of Manufacturers (CM) for year 2017, and Management and Organizational Practices Survey (MOPS) for the 2015 wave. The analysis is performed at a Federal Statistical Research Data Center under project 3363. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product.

Table B3: Robustness to Controls with Lag Forecast/Nowcast Errors

	Dependent variable:		
	$\iota_t - \mathbb{F}^k_{t-1}[\iota_t]$		
	(1)	(2)	(3)
ι_{t-1}	-0.033^{***}		-0.037^{***}
	(0.010)		(0.010)
h_{t-1}		0.096^{***}	0.104^{***}
		(0.022)	(0.023)
$y_{t-1} - \hat{y}^k_{t-1}$	0.113^{***}	0.089^{***}	0.095^{***}
	(0.034)	(0.033)	(0.033)
Firm Fixed Effect	Yes	Yes	Yes
Sector-Year Fixed Effect	Yes	Yes	Yes
Observations	8,455	8,455	8,455
R^2	0.562	0.562	0.563

Notes: This table reports regressions of the investment forecast error on lagged investment, lagged hiring, and the lagged sales nowcast error. The investment forecast error is defined as the arc difference between actual and expected capital expenditure, as in Table 1. The lagged sales nowcast error is defined analogously as the arc difference between actual sales in year $t-1$ and the last sales guidance issued before the end of year $t-1$. Firm and sector-year fixed effects are included as indicated. Standard errors clustered by firm are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

The first step of constructing the Census sample is to clean the MOPS 2015 dataset. Since it is a supplement to ASM 2015, each observation is an establishment in ASM 2015, and I merge the ASM 2015 with MOPS 2015 by matching the `SURVU_ID` variable. I keep the observations that do not have missing values for the questions in "Section D - Uncertainty", which elicit the backcast, 2016 forecast and 2017 probabilistic forecast for sales, employment, capital expenditure and material costs. Then, I construct the co-location dummy D_i in the following steps:

- In "Section B - Organization", obtain the answers to the following questions:
  - Question 17: "Was the headquarters for this company at the same location as this establishment? 1-Yes, 2-No" (If the answer is yes, no need to answer the rest of the questions in the section)
  - Question 18: "Where were decisions on hiring permanent full-time employees

Table B4: Robustness to Alternative Definitions of Investment Forecast Errors

	(1) Arc difference	(2) Log difference	(3) Investment-rate difference
h_{t-1}	0.094^{***}	0.079^{***}	0.021^{***}
	(0.0185)	(0.0155)	(0.0040)
ι_{t-1}	-0.022^{**}	-0.027^{***}	-0.008^{***}
	(0.0086)	(0.0073)	(0.0018)
Firm fixed effects	Yes	Yes	Yes
Sector-year fixed effects	Yes	Yes	Yes
Q1 forecasts only	Yes	Yes	Yes
Observations	14,804	14,504	14,519
Firms	2,418	2,379	2,376
R^2	0.5362	0.4759	0.5146

Notes: This table reports joint regressions of alternative investment forecast-error measures on lagged hiring and lagged investment. In column (1), the dependent variable is the arc difference $(\text{CAPX}_{it} - F_{i,t-1}[\text{CAPX}_{it}]) \,/\, [0.5\,(\text{CAPX}_{it} + F_{i,t-1}[\text{CAPX}_{it}])]$. In column (2), it is $\log(\text{CAPX}_{it}) - \log(F_{i,t-1}[\text{CAPX}_{it}])$. In column (3), it is the investment-rate difference $(\text{CAPX}_{it} - F_{i,t-1}[\text{CAPX}_{it}]) \,/\text{PPENT}_{i,t-1}$, where PPENT denotes net property, plant, and equipment. Columns (2) and (3) trim observations below the 1st percentile or above the 99th percentile of their respective forecast-error measures. All specifications use Q1 forecasts and include firm and sector-year fixed effects. Standard errors, clustered by firm, are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

made for this establishment? 1-Only at this establishment, 2-Only at headquarters, 3-Both at this establishment and at headquarters";

- Question 23: "what was the dollar amount that could be used to purchase a fixed/capital asset for this establishment without prior authorization from headquarters? 1-Under \$1,000, 2-\$1,000 to \$9,999, 3-\$10,000 to \$99,999, 4-\$100,000 to \$999,999, 5-\$1 million or more".

- Create a variable `CAPEXDEC` that encodes the location of capital expenditure decisions in 2015. `CAPEXDEC` = 1 if the capital decision is made in the establishment, i.e. the actual capital expenditure in ASM 2015 is below the threshold reported in Question 23 of MOPS 2015; `CAPEXDEC` = 2 if the capital decision is made in the headquarter, i.e. the actual capital expenditure in ASM 2015 is above the threshold reported in Question 23 of MOPS 2015.

- Suppose the answer to Question 18 of MOPS 2015 is encoded by variable `HIREDEC`. Construct the co-location dummy D_i in the following way:

  - $D_i = 1$ if either the answer to Question 17 is "Yes" (so that the headquarter is at

the same location of the plant), or `HIREDEC` = `CAPEXDEC` (so that the capital and labor decisions are strictly made in the same locations);

  - $D_i = 0$ otherwise.

- Keep the observations that has a non-missing value for D_i.

Lastly, keep the establishments in the MOPS/ASM 2015 that also shows up in ASM 2014, ASM 2016 and CM 2017, and append the realized outcome of sales, employment and capital expenditure from ASM 2014, ASM 2016 and CM 2017 to the sample. Table B5 tabulates the summary statistics.

Table B5: Summary statistics: establishment activity levels, 2015

Variable	Units	Mean
Total value of shipments, 2015	$ thousands	105,000
Total capital expenditures, 2015	$ thousands	3,100
Total employment, 2015	headcount	176.4

Notes: Establishment-level unweighted means of 2015 activity levels from the Annual Survey of Manufactures (ASM). The sample is the set of manufacturing establishments in the 2015 MOPS that report complete 2016 and 2017 forecasts and are observed in the ASM in 2014–2016; all statistics are unweighted. Results were cleared for release by the U.S. Census Bureau Disclosure Review Board (project 3363, release request 13405); observation counts are rounded per Census disclosure rules.

B.4 Construction of the Quantitative Sample for Calibration

The quantitative analysis uses a trimmed sample constructed from the CRSP-Compustat/IBES merged dataset described in Section B.1. The CRSP-Compustat observations underlying this dataset are for U.S.-incorporated firms reporting in U.S. dollars. I require a nonmissing NAICS code, exclude utilities (NAICS 22) and finance (NAICS 52), require positive sales and employment, and require nonmissing capital expenditures. The underlying CRSP-Compustat file spans 2000–2024. Because the IBES capital-expenditure guidance begins in 2003, the final quantitative sample spans 2003–2024.

Value Added and Factor Shares I construct value added and factor shares as follows:

1. Define industries using three-digit NAICS codes for manufacturing and two-digit NAICS codes for other sectors. Merge industry-year wage data from the Census County Business Patterns with the CRSP-Compustat/IBES sample.

2. For observations with missing Compustat labor expense, impute the wage bill as employment multiplied by the corresponding industry-year wage.

3. Define material expenditure as

$$\text{Material Expenditure}_{it} = \text{COGS}_{it} + \text{XSGA}_{it} - \text{DP}_{it} - \text{Wage Bill}_{it}.$$

   Drop observations with missing or negative material expenditure.

4. Define value added as sales minus material expenditure and retain observations with positive value added.

5. With $\epsilon = 4$, calculate the aggregate labor and capital shares as

$$s_N = \frac{\sum_{i,t} \text{Wage Bill}_{it}}{\sum_{i,t} \text{Value Added}_{it}}, \qquad \alpha = 1 - \frac{s_N}{1 - 1/\epsilon}, \qquad s_K = \alpha(1 - 1/\epsilon).$$

   The main quantitative analysis uses these aggregate shares. I also construct industry-specific shares as an alternative.

Construction of Quantitative Variables I define log capital and employment as $k_{it} = \log(\text{PPEGT}_{it})$ and $n_{it} = \log(\text{emp}_{it})$. Under the time-to-build timing assumption, current production uses the previous year's capital stock. The value-added productivity residual is therefore

$$a_{it} = \log(\text{Value Added}_{it}) - s_N n_{it} - s_K k_{i,t-1}.$$

The marginal-revenue-product proxies use the same timing convention:

$$mrpk_{it} = \log(\text{Value Added}_{it}) - k_{i,t-1}, \qquad mrpn_{it} = \log(\text{Value Added}_{it}) - n_{it}.$$

I construct investment growth $\Delta k_{it} = k_{it} - k_{i,t-1}$, hiring $h_{it} = n_{it} - n_{i,t-1}$, productivity growth $\Delta a_{it} = a_{it} - a_{i,t-1}$, and the time-to-build log capital–labor ratio $k_{i,t-1} - n_{it}$. All lags are constructed by exact-year firm-level matches, so a missing intermediate year does not generate an incorrect lag.

The IBES component retains the first annual capital-expenditure guidance within the fiscal year with a forecast horizon below four quarters. Unlike the Q1 regression sample, the quantitative sample is not restricted to `cpx_horizon_q = 3` and includes first-guidance horizons from zero to three quarters. Sales guidance and sales-nowcast errors are not used.

Industry-Year Residualization and Trimming I map fiscal years to calendar years by assigning observations with fiscal-year-end months January through May to the previous calendar year and all other observations to the reported fiscal year. I then remove industry-by-calendar-year fixed effects from investment growth, hiring, productivity growth, the productivity level, *mrpk*, *mrpn*, and the capital-labor ratio.

Before trimming, I construct changes in residualized investment growth and hiring using exact one-year lags. I calculate the first and ninety-ninth percentiles in the untrimmed sample and jointly retain observations within these bounds for the following variables:

- the capital-expenditure forecast error, lagged capital expenditure, and lagged hiring;
- changes in residualized investment growth and hiring;
- residualized productivity growth and the residualized productivity level;
- residualized *mrpk*, *mrpn*, and the capital-labor ratio.

This procedure produces a quantitative sample of 13,194 firm-year observations from 2,086 firms over 2003–2024. It is distinct from the Q1 regression sample: 10,261 quantitative-sample observations have a three-quarter forecast horizon, while 2,933 have horizons from zero to two quarters.

Table B6 compares the quantitative sample with the CRSP-Compustat sample before the IBES merge, restricted to the same 2003–2024 period.

Variable	Units	Mean	Std. dev.	Median	Observations
Panel A: Quantitative sample					
Sales	Millions USD	5,346.81	14,803.90	1,755.31	13,194
Gross PPE	Millions USD	3,086.40	6,718.11	894.13	13,194
Capital expenditures	Millions USD	232.89	488.00	69.40	13,194
Employment	Thousands	15.38	32.89	6.00	13,194
Panel B: CRSP-Compustat sample, 2003–2024					
Sales	Millions USD	3,903.97	18,756.48	437.51	68,591
Gross PPE	Millions USD	2,762.10	15,765.38	172.62	65,412
Capital expenditures	Millions USD	230.50	1,382.79	13.20	68,591
Employment	Thousands	11.43	53.99	1.30	68,591

Table B6: Summary Statistics for the Quantitative and Compustat Samples